

The Causal Effects of Heterogeneous Expectation Formation

PRELIMINARY AND INCOMPLETE

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Abstract

This paper studies the causal effect of heterogeneously-formed expectations on aggregate economic outcomes. To do this, I develop a ‘temporary-information equilibrium’ approach that characterizes the causal impact of expectation changes in a broad class of macroeconomic models using simple cross-sectional statistics on how agents form expectations. This approach uncovers a novel ‘narrative heterogeneity channel’ through which heterogeneities in expectation formation can affect the transmission of both expectational and fundamental shocks. This channel arises in a wide range of commonly-used models of expectation formation. In two applications to data on firm and household expectations, I find that narrative heterogeneity accounts for a large but time-varying share of the causal effects of expectations on macroeconomic outcomes.

JEL codes: D83, D84, E31, E71

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1 Introduction

Expectations are fundamental to our understanding of macroeconomics. At least since [Lucas \(1972\)](#) they have been at the core of macroeconomic modelling, and more recently a vast empirical literature has demonstrated their importance for the decisions of firms, households, and investors.¹ Yet despite the seemingly obvious centrality of expectations, it is notoriously difficult to say precisely how a given change in expectations would affect aggregate outcomes.

The reason for this uncertainty is that in the equilibrium conditions that define the behavior of macroeconomic models, expectations are *endogenous*. Take, for example, the linearized New Keynesian Phillips Curve (NKPC), as presented in e.g. [Galí \(2008\)](#):

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t \quad (1)$$

This shows that inflation π_t and expected future inflation $E_t \pi_{t+1}$ are related to one another, but it *does not* reveal anything about the causal effect of inflation expectations. As emphasized recently by [Werning \(2022\)](#), the β coefficient combines the causal effect of $E_t \pi_{t+1}$ on π_t , the causal effect of π_t on $E_t \pi_{t+1}$, and the fact that both $E_t \pi_{t+1}$ and π_t react simultaneously to other shocks.²

The best approach to date has been to study models in *temporary equilibrium* ([Grandmont, 1977](#); [Woodford, 2013](#); [Werning, 2022](#)). However, this approach necessarily abstracts from one of the central findings of recent empirical literature: that expectations are formed *heterogeneously*. Between otherwise similar decision-makers there are large differences in information, and in how people interpret any information they have.³ How, if at all, does this heterogeneity affect the causal effects of aggregate expectation shifts, and macroeconomic dynamics?

In this paper, I develop a new method that can reveal the causal effect of expectations in general equilibrium while allowing for heterogeneous expectation formation. Applying the method to a general class of linear macroeconomic models, I uncover a novel ‘*narrative heterogeneity channel*’, through which heterogeneity influences both the causal effects of expectations and the transmission of fundamental shocks. I show that many popular models of expectation formation implicitly generate such a channel whenever sufficient heterogeneity is allowed for. I then measure the strength of the narrative heterogeneity channel, and demonstrate the use of my approach to studying expectations in general equilibrium models, in two applications.

¹See e.g. the extensive reviews in [Bachmann et al. \(2023\)](#), and the discussion of related literature below.

²Nor would it be sufficient in this case to simply replace $E_t \pi_{t+1}$ with $(E_t \pi_{t+1} + \varepsilon)$, where ε is some exogenous belief shock, and to study the effects of that shock. The NKPC equation itself is derived under the assumption of rational expectations, which imposes a very specific endogeneity on expectations.

³See for example [Link et al. \(2023\)](#) for information, [Andre et al. \(2022\)](#) for subjective models used to interpret information, and [Pfajfar and Santoro \(2010\)](#), [Beutel and Weber \(2024\)](#), and [Macaulay and Moberly \(2025\)](#) for both.

In the first application, I combine the Calvo model underpinning equation (1) with the information-treatment randomized controlled trial (RCT) results of [Baumann et al. \(2024\)](#) to measure the causal effects of firm inflation expectations on realized inflation. In doing so, I demonstrate an auxiliary benefit of the ‘*temporary-information equilibrium*’ method I introduce in this paper, that it can be used to map partial-equilibrium information-treatment RCT results into general equilibrium insights, offering a solution to the ‘missing intercept’ problem in these studies.

In the second application, I use unique features of a UK survey to provide a direct measurement of the narrative heterogeneity channel over time, in the context of household inflation expectations. I then combine this measurement with a New Keynesian model to measure the causal effects of household inflation expectations on inflation, output, and interest rates. In both applications, I find that the narrative heterogeneity channel is substantial. In the household application in particular, the narrative heterogeneity channel accounts for 35% of the average causal effect of household inflation expectations, and two-thirds of the time-series variation in that effect.

Causal effects of expectations. Temporary equilibrium analysis starts by solving the decision problems of agents in a model, but rather than imposing rational expectations, the expectations of all variables are held fixed. The researcher then solves for equilibrium as a function of those fixed and now exogenous expectations. Finally, they compute comparative statics of endogenous variables of interest with respect to each expectation. This reveals the causal effect of an aggregate shift in expectations of a particular variable, holding all other expectations constant.

This abstracts from two plausible features of expectation formation. First, as all expectations are shifted equally, there can be no heterogeneity in any aspect of expectations. Second, there is no general equilibrium feedback from endogenous outcomes back to expectations. These points are related. If, for example, an increase in expectations of inflation in period $t + 2$ causes current inflation to rise, any firms that use current inflation as an input into their forecasts will react by updating their expectations of inflation at all horizons. This second round of updating will have further effects on pricing decisions, and its extent will vary depending on a firm’s ability to observe current inflation realizations, and on how they map from their observations to expectations.

To obtain the general equilibrium effects of a given expectation, we therefore need to allow expectations to respond endogenously and heterogeneously to aggregate conditions. How can we reconcile this with the fact that this very endogeneity is what makes it hard to uncover causal effects in the first place?

I propose a solution, “*temporary-information equilibrium*,” inspired by the information-treatment randomized-controlled-trials (RCTs) that have been successful in estimating the causal effects of

expectations on individual beliefs and actions in the empirical literature (see [Haaland et al., 2023](#), for a review). In those experiments, researchers exogenously vary one part of the information available to participants, and track how that affects expectations of the shocked variable, expectations of other variables (“cross-learning”), and actions. Instrumenting the shocked expectation with the exogenously varied information, they identify the causal effect of that expectation, despite not controlling all aspects of expectation formation.

As with temporary equilibrium, I start from agent decision problems. However, rather than holding expectations constant, I model them as composed of information (a noisy signal) and a residual, which depends on expectations of other variables. While expectations are not therefore fully exogenous, the noise in signals is. Shocking this noise creates exogenous variation in a given expectation in the model, while still allowing that variation to affect other expectations through cross-learning via the ‘residual’ components of other expectations. Any changes to endogenous variables can also affect expectations through the endogenous (non-noise) component of agent signals. Since all effects of the information shift come through the expectation of interest, we can use this to study the causal effect of changes in that expectation while still allowing endogenous general equilibrium responses. Moreover, we can allow the precision of agent signals, and cross-learning, to vary across agents, and still compute the causal effects of a shock causing aggregate expectations to move by a given amount. This mirrors how information-treatment RCTs use exogenous variation in only part of participant information sets to assess the causal effects of interest.

This temporary-information equilibrium approach is the key methodological contribution of the paper. It straightforwardly nests standard temporary equilibrium as a special case, but allows researchers the ability to include both general-equilibrium feedback through endogenous signals and expectations heterogeneity. In the two applications in this paper, both of these features turn out to be of first-order importance.

Linking to the empirical RCT literature comes with a further benefit: the parameters needed to pin down the causal effect of expectations in general equilibrium, and the role of expectations in the transmission of fundamental shocks, are closely related to those estimated in information-treatment RCTs. As a result, temporary-information equilibrium provides a way to map the estimates from a RCT into general equilibrium insights. Typically the empirical literature is unable to do this, because of a “missing intercept” problem: RCTs compare treated and untreated individuals, and so necessarily cannot reveal the general equilibrium effects of aggregate expectation changes. Moreover, since those general equilibrium effects likely also depend on the behavior of expectations, there are no available equivalence results allowing the intercept to be obtained from alternative time-series literature, as is the case for shocks to incomes ([Wolf, 2023](#)).

The Calvo model. For clarity, I begin by demonstrating temporary-information equilibrium in the Calvo model of firm price-setting. This is one way to derive the NKPC (1), and its temporary equilibrium has been extensively analysed by Werning (2022). The *temporary-information equilibrium* nests Werning’s results, while also allowing for heterogeneity in expectation formation.

This turns out to be of first-order importance. The temporary-information equilibrium reveals a previously overlooked channel of shock transmission, which I term the ‘*narrative heterogeneity channel*,’ and which qualitatively alters features of the passthrough from inflation expectations to realized inflation derived in Werning (2022). The novel channel arises whenever the precision of firm signals is correlated with the subjective models they use for cross-learning. Intuitively, if the firms who receive precise signals on current inflation are also those who extrapolate most strongly from current to future inflation, then a given change in inflation will have a greater effect on aggregate pricing than if all firms had the same average information precision and cross-learning.⁴

This narrative heterogeneity channel affects both the partial equilibrium effects of expectations, and the feedback into other expectations in general equilibrium. Moreover, I show that a wide range of common theories of expectation formation naturally imply narrative heterogeneity effects whenever heterogeneity is permitted in both information precision and the cross-learning used to interpret that information. This includes models of rational inattention (Maćkowiak et al., 2023), learning (Evans and McGough, 2020), level- k thinking (Farhi and Werning, 2019) and others.

Having developed these results and procedures in the simple Calvo model, I then go on to show that the same approach can be applied to a broad class of linear general equilibrium models. Many of the same insights apply, notably the potentially critical role of narrative heterogeneity effects in determining partial and general equilibrium effects of expectations.

Application 1: the effect of firm inflation expectations. Because the temporary-information equilibrium framework is conceptually linked to information-provision RCTs, the estimates commonly computed in those experiments correspond closely to the key parameters determining the causal effect of expectations in the temporary-information equilibrium of standard macroeconomic models. I therefore apply the framework to the recent RCT on European firm inflation expectations conducted by Baumann et al. (2024), using the temporary-information equilibrium of the Calvo model to analyse whether the experimental results imply large changes in macroeconomic dynamics relative to rational expectations. Importantly, RCT estimates are informative about both the partial equilibrium effects of expectations and general equilibrium feedback effects, allowing a data-driven solution to the missing intercept problem in these studies.

⁴This echoes the redistribution channel in the broader heterogeneous-agent literature (Auclert, 2019), where shocks are amplified when exposure to shocks correlate with responsiveness to those shocks.

I find that the [Baumann et al. \(2024\)](#) estimates imply that general equilibrium effects of one-year-ahead inflation expectations are larger than that implied by rational expectations, but that the differences are dampened by the narrative heterogeneity channel, and the magnitudes depend strongly on the degree of price stickiness. If the average price duration is 5 quarters, the general equilibrium effect of inflation expectations is just 6% larger than that implied by rational expectations. If, however, the average price duration is 2 years, the difference rises to 21%. Without the narrative heterogeneity channel, these numbers would be 7% and 24% respectively.

In this application the narrative heterogeneity channel is inferred by combining the results of regressions in the RCT that are informative about different statistics in the temporary-information equilibrium of the Calvo model. In the second application, I turn to a context in which the correlation of interest between information and subjective models, can be observed directly.

Application 2: the effect of household inflation expectations. To shed particular light on the role of the novel narrative heterogeneity channel, I end by applying the temporary-information equilibrium approach to a question that has received substantial policy attention: what are the effects of household inflation expectations on inflation and the real economy? In this context, survey data from the UK directly reveals the joint distribution of household inflation information and the subjective models used to interpret it. This dataset has a further advantage of allowing me to study how the causal effect of inflation expectations have changed over time.

I document two key patterns in this data. First, households who believe inflation makes little difference to the strength of the economy use less information about inflation than households with other subjective models. Those who believe inflation has positive or negative effects use similar information sources. Crucially, this means there is a systematic relationship between information and subjective models, implying that the narrative heterogeneity channel will operate.

Second, information that inflation is high is associated with more negative subjective models of the effects of inflation, both in the cross-section and over time. A greater proportion of households report that inflation makes the economy weaker in periods with high realized inflation, and within a period, those who believe inflation is currently higher are more likely to hold such negative subjective models. The joint distribution of information and subjective models therefore varies over time, and is systematically related to the state of the economy.

Combining these survey results with an otherwise-standard New Keynesian model, I find that contrary to a common prevailing view, increased household inflation expectations actually cause output to fall. This occurs because, when inflation expectations rise, the majority of households revise their expectations for future output down, and reduce consumption. However, the time-

variation in the distribution of subjective models means that this general-equilibrium causal effect is time-varying, becoming strongest in periods when realized inflation is high, when the correlation between information precision and cross-learning from inflation to output is at its most negative. The model implies that the causal effects of household inflation expectations were more than three times their 2009 level in the second half of 2016, for example, when annualized inflation rose above 5%. The narrative heterogeneity channel is responsible for approximately 35% of the average causal effect, and almost two-thirds of its time-variation.

Related literature. Principally, this paper contributes to the vast literature on the role of expectations in macroeconomic dynamics. While many papers examine how particular models of expectation formation affect the transmission of fundamental shocks (Maćkowiak and Wiederholt, 2015; Eusepi and Preston, 2018; Bianchi et al., 2024, among many others), the causal effect of changes in expectations themselves are much less clear. Where recent papers have made progress on this issue, they have typically abstracted from cross-sectional heterogeneity in expectations and expectation formation (e.g. Werning, 2022; Ascari et al., 2023), despite the prominent role for these features in the empirical literature (e.g. Mankiw et al., 2004; Andrade et al., 2016; Link et al., 2025, among many others). I contribute to this literature by isolating the causal effects of *heterogeneously-formed* expectations on aggregate economic dynamics.

This *temporary-information equilibrium* framework uncovers a novel mechanism through which heterogeneity in expectation formation affects aggregate dynamics, which I term the *narrative heterogeneity channel*. The channel operates when information precision is correlated in the cross-section with the subjective models agents use to interpret their information. This extends the literature on heterogeneous expectations (e.g. Mankiw et al., 2004; Branch and Evans, 2006; Andrade et al., 2019, and many others), adding a new force through which such heterogeneity affects aggregate outcomes. Conceptually, the mechanism is related to channels in the heterogeneous-household literature that operate through correlations between exposure to shocks and responses to them, particularly via marginal propensities to consume (Auclert, 2019; Bilbiie, 2019).⁵ I show that the relevant cross-sectional correlations for the narrative heterogeneity channel arise naturally in many common models of expectation formation, whenever heterogeneity is permitted in both information and subjective models. This complements recent empirical work documenting these correlations in survey data (Beutel and Weber, 2024; Macaulay and Moberly, 2025).

Methodologically, this paper is most closely related to Werning (2022), who studies the causal effect of inflation expectations in price-setting models using the temporary equilibrium approach

⁵Nord (2022) and Guerreiro (2023) study a further related channel driven by cross-sectional correlations between the sensitivity of expectations to a shock, and the consumption responses to expectations.

of [Grandmont \(1977\)](#). My temporary-information equilibrium nests this approach as a special case, but extends it to settings with rich heterogeneity in expectation formation, as well as general equilibrium feedback from endogenous variables to expectations.⁶ Since the method relies on modelling exogenous changes in agents' information, it is also related to the growing empirical literature on information-provision RCTs (reviewed in [Fuster and Zafar, 2023](#); [Haaland et al., 2023](#)), which aim to create variation in expectations through manipulation of participant information.

In general, translating the results of information-provision RCTs into insights for macroeconomic dynamics is complicated by the 'missing intercept' problem that affects all cross-sectional empirical work in macroeconomics ([Wolf, 2023](#); [Donaldson, 2025](#); [Matthes et al., 2025](#)).⁷ Because the expectation statistics that discipline the temporary-information equilibrium are closely related to the objects estimated in these experiments, the framework I introduce provides a natural way to map their results into general-equilibrium insights, through the lens of standard macroeconomic models. I illustrate this way of addressing the missing intercept problem in an application to firm expectations using the RCT in [Baumann et al. \(2024\)](#). The results speak to the growing literature on firm inflation expectations (see [Candia et al., 2022](#), for a review).

Finally, in applying my framework to household survey data, I relate to the broader literature on inflation expectations, which includes a variety of theoretical and empirical approaches (see reviews in [D'Acunto et al., 2023](#); [Milani, 2023](#)). My contribution in this context is to assess the causal effect of household inflation expectations on aggregate output, uncovering substantial time variation, drive in large part by the narrative heterogeneity channel.

Outline. In Section 2, I introduce the temporary-information equilibrium approach, and the narrative heterogeneity channel, in the simple context of the Calvo model of firm price-setting. Section 3 generalizes the method to a broad class of linear macroeconomic models. In Section 4 I apply this to the RCT in [Baumann et al. \(2024\)](#), interpreting the experimental results through the lens of the Calvo model of Section 2, and decomposing the resulting causal effects of firm inflation expectations into those driven by average firm expectations and the narrative heterogeneity channel. In Sections 5-7 I turn to my second application, to household inflation expectations: in Section 5 I measure the correlations behind the narrative heterogeneity channel directly, and in Sections 6 and 7 I interpret these measures with a New Keynesian model. Section 8 concludes.

⁶Allowing for the reaction of expectations to endogenous variables also connects this method to the literature on learning from prices ([Lucas, 1972](#); [Gaballo, 2018](#); [Chahrour and Gaballo, 2021](#)).

⁷A small number of papers use such RCTs to discipline macroeconomic models (e.g. [Hoffmann et al., 2022](#); [Hajdini et al., 2023](#); [Andre et al., 2025](#)). Related work disciplines models with non-experimental moments of expectations data (e.g. [Leombroni et al., 2020](#); [Ludwig et al., 2024](#)). While this avoids the missing intercept problem, the data used cannot reveal how expectations change in response to shocks or policy changes ([Piazzesi and Schneider, 2016](#)).

2 Calvo model

2.1 Setup

We start, as in [Werning \(2022\)](#), with the log-linearized first order condition of a price-setting firm:

$$p_{it}^* - P_{t-1} = (1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s E_{it} (\pi_{t+s} + mc_{t+s}) + \mu \quad (2)$$

where p_{it}^* is the desired price of price-setting firm i in period t ; P_{t-1} is the aggregate price level in period $t - 1$; $\pi_{t+s} \equiv P_{t+s} - P_{t-1}$ is cumulative inflation between periods $t - 1$ and $t + s$; and mc_{t+s} is real marginal costs in period $t + s$. The parameters β and λ are the discount factor and the probability a firm is unable to reset their price in a given period respectively, and both are $\in [0, 1]$. μ is the desired steady state markup, and the operator E_{it} denotes the expectation of firm i in period t .

In New Keynesian models, the standard approach is to impose rational expectations, and then derive the familiar New Keynesian Phillips Curve, in which the coefficient on $\bar{E}_t \pi_{t+1}$ is β (e.g. [Galí, 2008](#)). However, as discussed extensively in [Werning \(2022\)](#), this coefficient does not reveal the causal effects of inflation expectations, because the use of rational expectations means that the expected inflation term in the Phillips curve is endogenous. High inflation expectations simultaneously cause and are caused by high current inflation, making it impossible to disentangle the strength of each individual mechanism.

Temporary equilibrium. Instead of imposing rational expectations, [Grandmont \(1977\)](#) and others recommend holding expectations fixed, then aggregating and solving for equilibrium in terms of expectations. In the Calvo model this is straightforward:⁸

$$\pi_t = (1 - \lambda)(p_{it}^* - P_{t-1}) = (1 - \lambda)(1 - \beta\lambda) \left(\sum_{s=0}^{\infty} (\beta\lambda)^s \bar{E}_t (\pi_{t+s} + mc_{t+s}) + \mu \right) \quad (3)$$

where $\bar{E}_t(\cdot) = \int_i E_{it}(\cdot) di$ denotes aggregate expectation across firms.

With this, we can then compute comparative statics with respect to any given aggregate expectation. Since we have not imposed rational expectations, the expectations are exogenous in these equations, and thus these comparative statics can be interpreted as the causal effects of

⁸Note we do not need to include a general equilibrium process determining marginal costs, because it is expected marginal costs that enter the firm first order condition, and these (like all other expectations) are held constant in the temporary equilibrium approach.

changes in expectations. Specifically,

$$\frac{d\pi_t}{d\bar{E}_t(\pi_t)} = (1 - \lambda)(1 - \beta\lambda) \quad (4)$$

gives the causal effect of a change in aggregate expectations of current inflation, *holding all other expectations constant*. This latter point arises as a direct consequence of the fact we are removing all endogeneity from expectations: expectations of one variable cannot therefore be endogenous to expectations of something else.

In many settings this is more extreme than is desirable. This Calvo model is a good example, because π_{t+s} is defined as the cumulative inflation between periods $t - 1$ and $t + s$. Increasing $\bar{E}_t(\pi_t)$ but leaving all longer-horizon $\bar{E}_t(\pi_{t+s})$ unchanged, as in equation (4), therefore entails a decrease in expected inflation between periods t and $t + 1$ to exactly offset the expected increase in π_t . That is, it gives the causal effects of a transitory deviation from the expected path of the price level. The majority of evidence on inflation expectations of firms (as well as households) suggests that there is no such reversion of price-level expectations after shocks to short-run expected inflation (Weber et al., 2022), so this may not be the most informative way to study inflation expectations.

Extended temporary equilibrium. The simple way to avoid this problem is to impose some structure to link expectations that we think should co-move after solving for equilibrium (3), but before computing comparative statics. This is the approach taken in Werning (2022), who assumes:

$$\bar{E}_t(\pi_{t+s}) = \pi^e(1 + s) \quad (5)$$

which corresponds to the assumption that per-period expected inflation rates are constant at π^e . While each expectation in equation (3) is no longer fully exogenous – because inflation expectations at each horizon are endogenous to each other – the common factor π^e is still exogenous. The comparative static

$$\frac{d\pi_t}{d\pi^e} = \frac{1 - \lambda}{1 - \beta\lambda} \quad (6)$$

therefore gives the causal effect of a change in aggregate expectations of per-period inflation, assuming that per-period inflation is expected to be equal at all horizons. Werning (2022) gives an extensive analysis of this result. In principle, the researcher can also specify relationships between expected inflation and expected marginal costs, and recompute the causal effects.

While this is a useful advance on studying the equilibrium New Keynesian Phillips Curve, and

on the standard temporary equilibrium approach above, it still leaves several questions unanswered. First, how do heterogeneities in expectation formation affect the transmission of expectation changes? How do general equilibrium feedback effects, that may cause further endogenous changes in expectations, affect the results? And finally, what role do expectations play in the transmission of fundamental shocks? To answer these, I propose an alternative approach, inspired by the success of information-treatment experiments in the empirical study of expectations (see review in [Haaland et al., 2023](#)). This “temporary-information equilibrium” nests the extended temporary equilibrium approach presented here, but allows for a richer analysis of general equilibrium and heterogeneous expectation formation, while maintaining the causal interpretation of temporary equilibrium analysis. Because the approach is inspired by the RCT literature, the key variables in the model correspond to objects of interest in those studies.

Temporary-information equilibrium. When assessing the causal impact of a particular expectation, information-treatment RCTs do not vary that expectation directly. Rather, they exogenously vary the *information* participants have on that variable, and that information is passed through into the expectation of interest. There are also commonly effects on expectations of other variables, which are known in that literature as ‘cross-learning’ ([Haaland et al., 2023](#)). Taking this empirical procedure as our inspiration, we assume that the expectation of any variable z_j is composed of two parts: a signal $s_{it}(z_j)$ and a residual $R_{it}(z_j)$.

$$E_{it}(z_j) = (1 - \tau_{ij})R_{it}(z_j) + \tau_{ij}s_{it}(z_j) \quad (7)$$

This follows the form of a standard signal extraction problem, including that used by [Maćkowiak and Wiederholt \(2024\)](#) to model information acquisition during information-treatment RCTs. Under that signal extraction interpretation, $R_{it}(z_j)$ would reflect the firm’s prior belief, and τ_{ij} would reflect the signal to noise ratio. Following this interpretation, we will refer to τ_{ij} as capturing the precision of firm i ’s information on variable z_j .

Rather than exogenously varying the expectation as in temporary equilibrium approaches, we will analyse the effects of exogenous variation in the signal, allowing the residual component to (potentially) respond endogenously. Specifically, the residual is the expectation firm i would set if they received no information on that variable. The only way to form such an expectation is by using information and beliefs about *other* variables. Since the model here is linear, we impose

$$R_{it}(z_j) = \sum_{z_k \in \mathbf{z}: z_k \neq z_j} M_{ijk} E_{it}(z_k) \quad (8)$$

where $\mathbf{z}$ is the set of all expectations of interest in the model. Stacking equation (7) over all expectations in $\mathbf{z}$, substituting in equation (8), and rearranging gives:

$$E_{it}(\mathbf{z}) = \chi_i \tau_i \mathbf{s}_{it} \quad (9)$$

where τ_i is a square diagonal matrix, with j th diagonal element equal to τ_{ij} ; $\mathbf{s}_{it}$ is a column vector stacking $s_{it}(z_j)$ over z_j , and the square matrix χ_i is defined as

$$\chi_i \equiv (I - (I - \tau_i)M_i)^{-1} \quad (10)$$

where M_i has zeros on the diagonal, and (j, k) th element equal to M_{ijk} in equation (8).

For this section, we will restrict attention to cases in which M_i is upper-triangular. Conceptually, this means we only consider cases where the subjective model of each agent can be expressed as a recursive system of equations, as discussed in detail in the literature on Directed Acyclic Graphs (Spiegler, 2020).⁹ We relax this assumption in Section 3.

With this assumption, χ_i is upper-triangular, with diagonal elements equal to 1. Each element of τ_i therefore represents the degree to which a firm updates their expectations of a particular variable in response to direct information about that variable. In many applications this will be proportional to the precision of the agent's information. Off-diagonal elements of χ_i capture cross-learning, or equivalently the links across expectations considered in extended temporary equilibrium analysis like that in Werning (2022).

The cross-learning matrix χ_i captures both direct and indirect links between expectations of different variables: after a signal on short-run inflation, a firm may update expectations of long-run inflation partly because they believe inflation is persistent. They may also update their expectations of short-run marginal costs, which in turn affects their expectations of long-run marginal costs, which in turn affects their expectations of long-run inflation. It will be convenient from here to work directly with χ_i , rather than considering each of these possible links individually.¹⁰

Substituting equation (9) into equation (3) yields

$$\pi_t = (1 - \lambda)(1 - \beta\lambda) \left(\sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}} \tau_i \mathbf{s}_{it} + \chi_{imc_{t+s}} \tau_i \mathbf{s}_{it}) di + \mu \right) \quad (11)$$

⁹As an example, this means that if the firm uses $E_{it}\pi_t$ to update $R_{it}mc_t$, they cannot simultaneously use $E_{it}mc_t$ to update $R_{it}\pi_t$. $E_{it}mc_t$ may still affect $R_{it}\pi_{t+s}$ for $s > 0$, as these are treated as separate elements of $\mathbf{z}$ from π_t .

¹⁰This has a parallel in the literature on production networks (Carvalho and Tahbaz-Salehi, 2019). The direct links in M_i are analogous to the elements of the input-output matrix, and χ_i is the corresponding Leontief inverse. As with production networks, this Leontief inverse regulates the transmission of shocks.

where $\chi_{i\pi_{t+s}}$ and $\chi_{imc_{t+s}}$ are rows of χ_i corresponding to $E_{it}(\pi_{t+s})$ and $E_{it}(mc_{t+s})$ respectively.

We will now use this equation to analyse the effects of exogenous variation in the signals, rather than varying expectations directly. As in information-treatment RCTs, this exogenous variation in information will create exogenous variation in expectations. This is therefore a simple generalization of the temporary equilibrium approach, and we show in the next subsection that it nests both of the exercises discussed above. Generalizing the approach in this way will allow us to answer several questions which cannot be studied with existing approaches.

2.2 The causal effect of inflation expectations.

I now use the temporary-information equilibrium to study the causal effect of changes in expected π_t and in expected π_{t+1} . The former is the effect studied in [Werning \(2022\)](#), while the latter more closely corresponds to the questions in the empirical information-treatment literature.

Partial equilibrium. To begin with, consider an exogenous shock to the signals firms receive about current inflation $s_{it}(\pi_t)$, holding constant the signals firms receive about all other variables. All proofs for this section and Section 3 are in Appendix A.

Proposition 1 *The causal effect of a shock ε_{π_t} that increases $s_{it}(\pi_t)$ equally for all firms is*

$$\frac{1}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = (1 - \lambda)(1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i \left(\chi_{i\pi_{t+s}\pi_t} \frac{\tau_{i\pi_t}}{\bar{\tau}_{\pi_t}} + \chi_{imc_{t+s}\pi_t} \frac{\tau_{i\pi_t}}{\bar{\tau}_{\pi_t}} \right) di \quad (12)$$

where $\chi_{i\pi_{t+s}\pi_t}$ and $\chi_{imc_{t+s}\pi_t}$ refer to the elements of χ_i that correspond to the cross-learning from a signal about π_t to expectations of π_{t+s} and mc_{t+s} respectively. Scaling the derivative by the average response to π_t signals, $\bar{\tau}_{\pi_t} \equiv \int_i \tau_{i\pi_t} di$, implies that this effect comes through a unit increase in $\bar{E}_t\pi_t$.

Notice that this can still be interpreted as the causal effect of inflation expectations. Firm signals about π_t have been varied exogenously, and the only way those signals affect anything in the firm decision problem is through their effect on $E_{it}(\pi_t)$. This is therefore the theoretical aggregate equivalent of the practise employed in information-treatment RCTs of using exogenous variation in information as an instrument for expectations, and thus to determine the effects of expectations on other (individual-level) outcomes. In-keeping with this IV-like interpretation, we scale the effect such that the result can be interpreted as the causal effect of a unit increase in the relevant aggregate expectation, which in this case is $\bar{E}_t(\pi_t)$.

Given this interpretation, it is unsurprising that this straightforwardly nests the two standard approaches discussed above. Setting $\tau_{i\pi_t} = 1$ for all firms i recovers the case in which one

expectation is shocked and all others are held constant,¹¹ as in standard temporary equilibrium analysis (equation (4)). If instead $\tau_{i\pi_t} < 1$, and $\chi_{imc_{t+s}\pi_t} = 0$, $\chi_{i\pi_{t+s}\pi_t} = 1 + s$ for all i , then we return to the extended temporary equilibrium case in equation (6), in which a rise in expectations of π_t increases expected per-period inflation one-for-one at all horizons, but has no effect on marginal cost expectations. By selecting different values for $\tau_{i\pi_t}$ and the elements of χ_i , we can also solve for the partial-equilibrium effect of current inflation expectations under a range of common assumptions on expectation formation, which have different implications for responses to information and the linkages between expectations across horizons and variables. We return to this point in Section 2.4.

Similarly, the causal effect of π_{t+1} expectations is given by Proposition 2.

Proposition 2 *The causal effect of a shock $\varepsilon_{\pi_{t+1}}$ that increases $s_{it}(\pi_{t+1})$ equally for all firms is*

$$\frac{1}{\bar{\tau}_{\pi_{t+1}}} \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = (1 - \lambda)(1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i \left(\chi_{i\pi_{t+s}\pi_{t+1}} \frac{\tau_{i\pi_{t+1}}}{\bar{\tau}_{\pi_{t+1}}} + \chi_{imc_{t+s}\pi_{t+1}} \frac{\tau_{i\pi_{t+1}}}{\bar{\tau}_{\pi_{t+1}}} \right) di \quad (13)$$

where the scaling here implies that the effect comes through a unit increase in $\bar{E}_t \pi_{t+1}$.

The narrative heterogeneity channel. The key novel insight to notice from equations (12) and (13) concerns the impact of heterogeneity in expectation formation. There is substantial evidence of heterogeneity across households and firms in the precision of information (e.g. Link et al., 2023), and more directly in the responsiveness to signals (Balla-Elliott, 2025). There is also substantial heterogeneity in how agents update from one expectation to another (e.g. Andre et al., 2022; Macaulay and Moberly, 2025). Holding signals, rather than expectations, fixed allows us to study how this heterogeneity affects the causal effect of expectations. Specifically, using the definition of a covariance, we can rewrite equation (12) as

$$\begin{aligned} \frac{1}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} &= (1 - \lambda)(1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s (\bar{\chi}_{\pi_{t+s}\pi_t} + \bar{\chi}_{mc_{t+s}\pi_t}) \\ &+ (1 - \lambda)(1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \left(\frac{1}{\bar{\tau}_{\pi_t}} Cov_I[\chi_{i\pi_{t+s}\pi_t}, \tau_{i\pi_t}] + \frac{1}{\bar{\tau}_{\pi_t}} Cov_I[\chi_{imc_{t+s}\pi_t}, \tau_{i\pi_t}] \right) \end{aligned} \quad (14)$$

where $\bar{\chi}_{z\pi_t}$ is defined as the average of $\chi_{iz\pi_t}$ across firms for any expected variable z , and $Cov_I[\cdot, \cdot]$ denotes a cross-sectional covariance. Equation (13) admits an equivalent decomposition for the effects of $\varepsilon_{\pi_{t+1}}$.

¹¹By implication this sets the corresponding row of χ_i to the equivalent row of the identity matrix (equation (10)).

The first summation reflects representative-agent effects: the causal effect of expected current inflation depends on the average cross-learning to expectations of marginal costs and longer-run inflation across all firms. However, the second summation reflects the effects of heterogeneity. If the firms with the most precise information on π_t (i.e. with the highest $\tau_{i\pi_t}$) are also the firms who extrapolate most strongly from π_t to other variables (high $\chi_{i\pi_t+s\pi_t}$ and $\chi_{imc_t+s\pi_t}$), information is concentrated among those who will extrapolate and react most strongly to it. That amplifies the effect of an aggregate signal relative to what would be observed if all firms formed expectations in the same way.

We will refer to this broad category of effects as the ‘narrative heterogeneity channel,’ reflecting the idea that a narrative combines information with a subjective model to interpret it (Gibbons and Prusak, 2020). These effects were not visible in existing temporary equilibrium analysis, because such exercises implicitly remove all heterogeneity in expectations. Once we allow for heterogeneity, then mathematically the narrative heterogeneity terms are a trivial consequence of Jensen’s inequality. In Section 2.4 we will argue that *economically* they are not trivial at all. A wide range of popular models of expectation formation imply potentially large and time-varying narrative heterogeneity effects, with important qualitative and quantitative consequences.

General equilibrium. All of the exercises so far study the effects of a given exogenous change in the signals firms receive about current or one-period-ahead inflation. This captures the causal effect of a given expectation rise, holding all other signals constant. However, holding signals constant means abstracting from second-round effects: when expectations of π_{t+1} rise, that has effects on current realized π_t (equation (13)), which in turn should affect the information firms receive about π_t , which has its own effects. To assess the full passthrough from inflation expectations to realized inflation in general equilibrium, this feedback should be taken into account.

The temporary-information approach can easily accommodate this. As is standard in models with noisy information (e.g. Coibion and Gorodnichenko, 2015), suppose that firm signals are equal to the variable of interest plus some additively separable noise, which here I allow to consist of a public component and an idiosyncratic component

$$s_{it}(\pi_{t+s}) = \pi_{t+s} + \varepsilon_{\pi t+s} + \tilde{\varepsilon}_{i\pi t+s} \quad (15)$$

where $\varepsilon_{\pi t+s}$ and $\tilde{\varepsilon}_{i\pi t+s}$ are both mean zero, and $\tilde{\varepsilon}_{i\pi t+s}$ is additionally i.i.d. across firms.

This functional form assumption preserves the causal interpretation of the exercises above: exogenous variation in $\varepsilon_{\pi t+s}$ still only affects firm decisions through the relevant inflation expectation, so can be used to assess the causal effects of that expectation on aggregate outcomes.

However, that effect can now also include an endogenous response of signals to the shock, through the dependence of the signal on realized π_{t+s} .

For the causal effects of expectations of π_t , this general equilibrium effect turns out to be irrelevant. The partial equilibrium effect in equation (12) increases π_t , which now causes a second round of changes in aggregate π_t expectations, with associated consequences for realized inflation. However, all of those general equilibrium effects operate through $\bar{E}_t(\pi_t)$. As a result, when we scale the causal effect to ensure a unit impact on $\bar{E}_t(\pi_t)$, we get back exactly to the partial equilibrium effect. We show this formally in Appendix A.

However, for the causal effects of $\bar{E}_t(\pi_{t+1})$, the general equilibrium feedback is not irrelevant, because the feedback occurs through a different expectation than the one being shocked. From here, it will be useful to introduce the following notation: let $\Omega^{PE}(\pi_{t+s})$ denote the partial-equilibrium effect of a change in $s_{it}(\pi_{t+s})$, scaled to ensure a unit impact on $\bar{E}_t(\pi_{t+s})$ as in Propositions 1 and 2. Let $\Omega^{GE}(\pi_{t+s})$ denote the equivalent when we allow for general equilibrium feedback through the endogenous part of signals.

Proposition 3 *The causal effect of $\bar{E}_t(\pi_{t+1})$ on realized π_t , after accounting for general equilibrium effects through current inflation signals, is given by*

$$\Omega^{GE}(\pi_{t+1}) = K \cdot \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = \frac{\Omega^{PE}(\pi_{t+1})}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t) + \Omega^{PE}(\pi_{t+1}) \cdot (\bar{\chi}_{\pi_{t+1}\pi_t} \bar{\tau}_{\pi_t} + Cov_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}])} \quad (16)$$

where K is a scaling coefficient defined in Appendix A which ensures the impact on $\bar{E}_t(\pi_{t+1})$ in general equilibrium is equal to 1.

Whether general equilibrium forces amplify or dampen the effects of a change in one-year-ahead expectations therefore depends positively on the partial-equilibrium effect of *current* expectations ($\Omega^{PE}(\pi_t)$). If this is larger, then a given partial-equilibrium effect of the shock on current realized π_t has larger feedback effects through the resulting change in signals about π_t . Importantly, this feedback effect depends critically on the narrative heterogeneity channel. Even if the desired model experiment is one in which all firms receive an identical signal about π_{t+1} , and so there is no narrative heterogeneity term in the direct partial-equilibrium effects of the shock, heterogeneity will still affect dynamics through general equilibrium.

The second general equilibrium force is that the increase in $E_{it}(\pi_t)$ causes firms to further update $E_{it}(\pi_{t+1})$ through cross-learning, again with a narrative heterogeneity effect present. Since we scale the effect to ensure a unit increase in aggregate $\bar{E}_t(\pi_{t+1})$, this feedback effect acts to reduce

the aggregate effects of the shock. Taken together, general equilibrium effects amplify the causal effect of one-year-ahead expectations if

$$\Omega^{PE}(\pi_t) > \Omega^{PE}(\pi_{t+1}) \cdot \left(\bar{\chi}_{\pi_{t+1}\pi_t} + \frac{1}{\bar{\tau}_{\pi_t}} Cov_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}] \right) \quad (17)$$

Of course, this is one general equilibrium mechanism. If realized inflation also affects real marginal costs, which have not been modelled here, that would further affect these dynamics. The temporary-information equilibrium approach can be deployed in the same way.

2.3 The role of expectations in the transmission of fundamental shocks

So far, we have been considering the causal effects of a shock or change in expectations. However, the other related question of interest to macroeconomists is what role expectations play in the transmission of *fundamental* shocks. It turns out that the temporary-information equilibrium apparatus can also be useful in studying that question.

To see why, go back to equation (11), and allow for mc_t to be perfectly observed, with no cross-learning from it to any other expectations. In this simple setting we have

$$\pi_t = (1 - \lambda)(1 - \beta\lambda) \left[\sum_{s=0}^{\infty} (\beta\lambda)^s (\bar{\chi}_{\pi_{t+s}, \pi_t} \bar{\tau}_{\pi_t} + Cov_I[\chi_{i, \pi_{t+s}, \pi_t}, \tau_{i, \pi_t}]) \right] (\pi_t + \varepsilon_{s\pi t}) + (1 - \lambda)(1 - \beta\lambda)mc_t + a \quad (18)$$

where a collects the steady-state markup μ and all terms involving signals of π_{t+s} and mc_{t+s} for $s \geq 1$. Consider an i.i.d. shock to mc_t . Since the Calvo model is purely forward-looking, all variables return to steady state in period $t + 1$, so the only expectation effects will come through changes in $s_{it}(\pi_t)$, and we can treat a as fixed. The aggregate effects of an i.i.d. mc_t shock are therefore given by

$$\frac{d\pi_t}{dmc_t} = (1 - \lambda)(1 - \beta\lambda) \left(\left[\sum_{s=0}^{\infty} (\beta\lambda)^s (\bar{\chi}_{\pi_{t+s}, \pi_t} \bar{\tau}_{\pi_t} + Cov_I[\chi_{i, \pi_{t+s}, \pi_t}, \tau_{i, \pi_t}]) \right] \frac{d\pi_t}{dmc_t} + 1 \right) \quad (19)$$

which rearranges to

$$\frac{d\pi_t}{dmc_t} = \frac{(1 - \lambda)(1 - \beta\lambda)}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \quad (20)$$

This helps us to see the role of endogenous expectations in the transmission of the fundamental

mc_t shock, as all expectational effects are contained in the $\bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)$ term in the denominator of equation (20). Formally, we can compute a counterfactual shock transmission in which there is no response of expectations by setting $\Omega^{PE}(\pi_t) = 0$

$$\left. \frac{d\pi_t}{dmc_t} \right|_{\sim E} = (1 - \lambda)(1 - \beta\lambda) \quad (21)$$

and then from these last two equations we can compute the share of marginal cost shock transmission which can be attributed to expectations.

Proposition 4 *The fraction of the transmission of an i.i.d. marginal cost shock to realized inflation that is due to inflation expectations is*

$$\frac{\left. \frac{d\pi_t}{dmc_t} - \frac{d\pi_t}{dmc_t} \right|_{\sim E}}{\left. \frac{d\pi_t}{dmc_t} \right|_{\sim E}} = \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t) \quad (22)$$

That is, in this environment the share of the transmission of a fundamental marginal cost shock that is due to inflation expectations is equal to the partial equilibrium effect of an exogenous shock to those inflation expectations, multiplied by the average responsiveness of firms to inflation signals. The same channels determining the causal effects of an information shock also determine the share of other shocks' transmission due to expectations.

This is a particularly simple fundamental shock to analyze. However, the approach continues to work for more complicated interactions with multiple variables, or for alternative counterfactuals.¹² The key point is that studying the shock in the temporary-information equilibrium, rather than in a rational-expectations equilibrium or after imposing any other expectation-formation assumptions, allows us to separate the consequences of expectational shifts from their causes, without abstracting from heterogeneity in expectation formation.

2.4 The Narrative Heterogeneity Channel

The analysis above identifies a covariance between firms' direct response to signals, and their cross-learning, as a potential factor in the effects of expectations in general equilibrium. We now consider this 'narrative heterogeneity channel' in more detail.

¹²For example, if firms are assumed to perfectly observe current inflation, we could allow beliefs about π_t to adjust in the counterfactual, but shut down any transmission to expectations of future π_{t+s} . Rather than simply setting $\Omega^{PE}(\pi_t)$ to 0, the χ_{i,π_{t+s},π_t} and τ_{i,π_t} terms that make up $\Omega^{PE}(\pi_t)$ would just need to be modified accordingly.

An example. We begin with a simple example to give intuition for how the narrative heterogeneity channel operates, and how it can change quantitative and qualitative conclusions about the effects of expectations. Suppose that there is no cross-learning from inflation to marginal costs ($\chi_{imc_t+s\pi_t} = 0$), and $\chi_{i\pi_t+s\pi_t}$ and $\tau_{i\pi_t}$ are each split equally between two values

$$\chi_{i\pi_t+s\pi_t} = \begin{cases} 1 \\ 1 + 2s \end{cases}, \quad \tau_{i\pi_t} = \begin{cases} 0 \\ 1 \end{cases} \quad (23)$$

These have been chosen such that $\bar{\chi}_{\pi_t+s\pi_t} = 1 + s$. If $\chi_{i\pi_t+s\pi_t}$ and $\tau_{i\pi_t}$ are independent of one another, the narrative heterogeneity terms are equal to 0, and the causal effect of current inflation expectations in partial equilibrium is therefore as in equation (6) and in [Werning \(2022\)](#) Proposition 1. However, if there is a correlation, this causal effect can change dramatically. Proposition 5 considers the limit as $\beta \rightarrow 1$, which is a useful benchmark to gain intuition, and which [Werning \(2022\)](#) argues is a good approximation in practise.

Proposition 5 *If $\tau_{i\pi_t}$ and $\chi_{i\pi_t+s\pi_t}$ are split equally between the values in equation (23), the causal effect of expected inflation on realized inflation as $\beta \rightarrow 1$ is given by*

$$\frac{1}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = 1 + \lambda(2q - 1) \quad (24)$$

where $q \equiv \Pr(\chi_{i\pi_t+s\pi_t} = 1 + 2s | \tau_{i\pi_t} = 1)$.

[Werning \(2022\)](#) Proposition 1 states that the transmission from expectations to inflation becomes 1 under no-discounting, irrespective of price stickiness. Equation (24) shows that this is no longer true when expectations are formed heterogeneously, unless information precision is independent of subjective models (i.e. $q = \frac{1}{2}$). Outside of this knife-edge case, the narrative heterogeneity channel becomes stronger when prices are stickier (λ is larger), making the total passthrough from expectations to realized inflation depend on λ even though the representative-agent terms do not.

The reason for this is that $\chi_{i\pi_t+s\pi_t} = 1 + s$, as used in [Werning \(2022\)](#), is a special case that balances two opposing forces. As prices get stickier, each price-setting firm expects to wait longer before they can change prices again, and so they react more strongly to increases in their expectations of inflation today. This increases the causal effect of inflation expectations. At the same time, stickier prices mean fewer firms are resetting prices, decreasing the effect of expectations. When $\chi_{i\pi_t+s\pi_t} = 1 + s$ for all firms, these forces exactly cancel out, leaving the causal effect of expectations independent of price stickiness.

Heterogeneity does not change the second of these forces, but it does change the first. When price stickiness increases, firms with high $\chi_{i\pi_t+s\pi_t}$ increase their responsiveness to expected inflation a great deal, because the extra length of a typical fixed-price spell encompasses periods of greater and greater inflation. By the same logic in reverse, firms with a low $\chi_{i\pi_t+s\pi_t}$ change their behavior much less after an increase in stickiness. If information is concentrated among the former group, they dominate the passthrough from aggregate expectations to inflation, and so that passthrough is increasing in stickiness.¹³ If information is instead concentrated among the latter group (with low $\chi_{i\pi_t+s\pi_t}$), that group dominates, and the passthrough is decreasing in stickiness.

Common models of expectations generate narrative heterogeneity. Previous literature studying departures from full information and rational expectations has typically relaxed just one of those assumptions, not both simultaneously (see related literature discussion above). In the language of the temporary-information equilibrium approach developed here, that means there can only be heterogeneity in either χ_i or τ_i , and not both, which forces the narrative heterogeneity terms to be 0 by definition. However, if heterogeneity is allowed in both components of expectation formation, many standard models of expectations naturally generate narrative heterogeneity effects.

In Appendix B, I consider the simple case in which the only possible signals are on π_t , where there is no public noise in those signals, and where both π_t and $\tilde{\varepsilon}_{i\pi_t}$ are normally distributed.¹⁴ I then introduce a range of common assumptions on expectation formation, and show that they each imply systematic correlations between direct responses to signals ($\tau_{i\pi_t}$) and cross-learning ($\chi_{i\pi_t+s\pi_t}, \chi_{imc_t+s\pi_t}$), whenever heterogeneity is allowed in both. Since these models of expectation formation are well-known, I leave the details and derivations to Appendix B, and focus here on the intuition for why the narrative heterogeneity channel arises under each set of assumptions.

Rational Inattention (Maćkowiak et al., 2023): if firms can pay a cost to increase the precision of their signals, then $\tau_{i\pi_t}$ is endogenous to the firm's expected use of information. Firms who believe there is a stronger pass-through from current inflation to future inflation and marginal costs find information on π_t to be more valuable, and so process more of it. If there is heterogeneity in cross-learning, high $\tau_{i\pi_t}$ is therefore correlated with large magnitudes of $\chi_{i\pi_t+s\pi_t}$ and $\chi_{imc_t+s\pi_t}$.

To give a particular example, consider the case in which all firms have positive $\chi_{i\pi_{t+1}\pi_t}$. Rational inattention implies $Cov_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}] > 0$, because firms with larger $\chi_{i\pi_{t+1}\pi_t}$ condition their prices

¹³Macaulay and Moberly (2025) find evidence that among households greater Kalman gains on inflation signals are correlated with greater perceived inflation persistence, consistent with a positive narrative heterogeneity effect here and thus an amplified causal effect of current inflation expectations.

¹⁴The exception is level- k thinking, where for tractability I allow for signals on mc_t rather than π_t .

more strongly on their signals of π_t , so they pay for more precision in those signals. As discussed in Proposition 5, this amplifies the causal effect of aggregate inflation expectations, as information is concentrated in those who respond most strongly to it. As shown in equation (20), this in turn amplifies the transmission of fundamental shocks.

Learning (Evans and McGough, 2020): if firms update their perceived laws of motion for inflation and marginal costs after observing their signals, then χ_i is endogenous to information precision. Firms who observe more precise signals update their subjective models of the economy more strongly after each observation than those with less precise information. If there is heterogeneity in information precision, this therefore leads to systematic correlations between elements of τ_i and χ_i . Notably, the narrative heterogeneity channel generated in this case may vary over time as firms continually update their perceived laws of motion as they receive new signals.

Sticky information (Mankiw and Reis, 2002): if firms can pay a cost to observe information more frequently (as in e.g. Reis, 2006, and others), then subjective models influence this information choice, with a similar logic to the rational inattention model.

Level- k thinking (Farhi and Werning, 2019): again, a narrative heterogeneity channel arises whenever firms can pay a cost to engage in a further round of reasoning, as in Alaoui and Penta (2016, 2022). In this case, firms with more precise information have more to gain from engaging in costly reasoning about how to use that information. Reasoning causes firms to update their cross-learning parameters, so the elements of χ_i differ between firms with different τ_i .

Information delegation to media outlets (Nimark and Pitschner, 2019): firms with larger $\chi_{i\pi_t+s\pi_t}$ and $\chi_{imc_t+s\pi_t}$ place a greater value on information on π_t . Competitive media outlets specialize, and firms with large values of these cross-learning parameters select into reading those which report π_t more often.

In this case, the resulting narrative heterogeneity channel is state-dependent. When inflation is close to steady state, almost no outlets find inflation sufficiently newsworthy for them to report it, and most firms have $\tau_{i\pi_t} = 0$. Similarly, when inflation is very far from steady state, it is highly newsworthy, and almost all outlets do report it, leading most firms to have $\tau_{i\pi_t} = 1$. It is only in the intermediate region where some outlets report inflation, and others don't, that the precision of firm information becomes heterogeneous, generating a narrative heterogeneity channel.

Data generated through production (Farboodi and Veldkamp, 2021): firm subjective models affect their response to a given shock. That in turn affects how much data each firm has to learn from in subsequent periods, again generating a correlation between subjective models and information.

Heterogeneous price indices: for households, it is well-established that inflation rates vary substantially across the population (Kaplan and Schulhofer-Wohl, 2017; Kiss and Strasser, 2024). If households observe their own idiosyncratic price levels (heterogeneous information), and form expectations using the dynamics of those idiosyncratic price levels (heterogeneous cross-learning), then we could see a narrative heterogeneity effect even if all agents have rational expectations. I take this idea to the Calvo model analysed here by assuming firms belong to sectors which differ in the persistence of their inflation processes. This indeed generates a narrative heterogeneity channel, as greater inflation persistence implies a weaker response to inflation signals but a stronger extrapolation from short to long-horizon inflation expectations, generating a negative covariance between $\tau_{i\pi_t}$ and $\chi_{i\pi_{t+s}\pi_t}$, even though all cross-learning is consistent with rational expectations.

Overall, the narrative heterogeneity channel is a robust feature of many models of expectation formation, either because subjective models influence information acquisition, or because information influences the formation of subjective models. With multiple frictions combined, it is possible for causation to run in both directions. Indeed, the final case (heterogeneous price indices) shows that the mechanism can arise even under rational expectations, if there is heterogeneity in the object to be forecast. Allowing for heterogeneity in both components of expectation formation therefore generates dynamics that are not present if heterogeneity is restricted to either information or subjective models alone. Indeed, the correlation between information and subjective models may provide a novel way to test competing models of expectation formation.

3 General model

We now show how to apply the temporary-information equilibrium approach to a general class of linear models in discrete time, and show that the narrative heterogeneity channel is a general feature of the causal effects of expectations.

Decision rules and expectations. There is a mass 1 of agents, indexed i . We start our analysis with a general linear policy function for agent actions, of the form

$$\mathbf{x}_{it} = A_i \mathbf{x}_{it-1} + \sum_{s=0}^{\infty} (B_{iws} E_{it} \mathbf{w}_{t+s} + B_{ixs} E_{it} \mathbf{x}_{t+s}) \quad (25)$$

where $\mathbf{x}_{it}$ is a $N_x \times 1$ vector of choice variables for agent i in period t ; $\mathbf{w}_t$ is a $N_w \times 1$ vector of state variables that the agent takes as given; and $\mathbf{x}_t \equiv \int_i \mathbf{x}_{it} di$ denotes aggregate actions across agents, which each agent i also takes as given. A_i , B_{iws} , and B_{ixs} are matrices of coefficients, which may in principle vary across agents.

Equation (2) in the Calvo model above is clearly nested in this policy function, as are the equations determining agent choices in a range of other models.

As in Section 2.1, define $\mathbf{z}_t$ as the vector of all variables about which the agent will form expectations. Similarly, let B_i be the matrix that collects all B_{iws} , B_{ixs} coefficients.

$$\mathbf{z}_t \equiv \begin{bmatrix} \mathbf{x}_t \\ \mathbf{w}_t \\ \mathbf{x}_{t+1} \\ \mathbf{w}_{t+1} \\ \vdots \end{bmatrix}, \quad B_i \equiv \begin{bmatrix} B_{ix0}, B_{iw0}, B_{ix1}, B_{iw1}, \dots \end{bmatrix} \quad (26)$$

Following the same steps as in Section 2.1, we can express the expectations of $\mathbf{z}_t$ as in equation (9). In Section 2, we restricted attention to upper-triangular M_i matrices. This implied that χ_i had a unit diagonal. In this more general model, we relax that assumption. Doing this implies that τ_i can no longer be interpreted as the extent of expectations updating after a direct signal.¹⁵ However, it will be helpful for the exercises below to re-scale the χ_i and τ_i such that this original interpretation is restored. As in the first stage of an information-treatment RCT, we would like τ_i to reflect the updating from signals to expectations of the variables being signalled.

To do that, define a new square matrix D_i , where the diagonal elements are the diagonal elements of χ_i , but all off-diagonal elements are equal to 0. Then define $\hat{\chi}_i = \chi_i D_i^{-1}$ and $\hat{\tau}_i = D_i \tau_i$, so

$$E_{it}(\mathbf{z}) = \hat{\chi}_i \hat{\tau}_i \mathbf{s}_{it} \quad (27)$$

¹⁵For example, in the Calvo model a signal about π_{t+1} has a direct effect on $E_{it}\pi_{t+1}$, but also may affect $E_{it}mc_{t+1}$. Relaxing the assumption of acyclic subjective models, this change in $E_{it}mc_{t+1}$ may then also feed back into a further change in $E_{it}\pi_{t+1}$. $\tau_{i\pi_{t+1}}$ only captures the extent of the first of these effects, which is no longer the full update after the signal.

This is of the form in equation (9), but $\hat{\chi}_i$ is now guaranteed to have a unit diagonal, such that τ_i captures the updating from direct signals.

Substituting this into equation (25) yields

$$\mathbf{x}_{it} = A_i \mathbf{x}_{it-1} + B_i \hat{\chi}_i \hat{\tau}_i \mathbf{s}_{it} \quad (28)$$

General equilibrium. The state variables $\mathbf{w}_t$ are such that

$$C\mathbf{x}_t + D\mathbf{w}_t + F\boldsymbol{\xi}_t = 0 \quad (29)$$

where $\boldsymbol{\xi}_t$ is a $N_\xi \times 1$ vector of exogenous shocks, and C, D, F are matrices of coefficients.

The Calvo model in Section 2 featured a very simple general equilibrium constraint: the only element of $\mathbf{w}_t$ was mc_t , and this was assumed to be exogenous to firm actions (i.e. pricing). Thus the constraint was, in effect, that $mc_t + \xi_{mc,t} = 0$. Equation (29) nests this (with $C = 0, D = F = 1$), but also nests a wide range of other equilibrium conditions, which could consist of technological constraints, market clearing conditions, or the decision rules of other categories of agents whose expectations are not the focus of the analysis.

Temporary-information equilibrium. For given distributions of $\hat{\tau}_i, \hat{\chi}_i$, and given exogenous shocks $\boldsymbol{\xi}_t$, the temporary-information equilibrium of this model is defined as a sequence of actions $\mathbf{x}_{it}$, endogenous states $\mathbf{w}_t$, and signals $\mathbf{s}_{it}$ such that

1. $\mathbf{x}_{it}$ is chosen according to agent i 's policy function (28), and aggregate actions are given by $\mathbf{x}_t \equiv \int_i \mathbf{x}_{it} di$;
2. $\mathbf{w}_t$ satisfies the general equilibrium condition (29);
3. a specified process links signals $\mathbf{s}_{it}$ with states $\mathbf{w}_t$ and aggregate actions $\mathbf{x}_t$.

In principle, many different processes could be specified for item (3). For simplicity, however, I restrict attention here to the form used in Section 2.2:

$$s_{it}(z_{jt}) = z_{jt} + \varepsilon_{jt} + \tilde{\varepsilon}_{ijt} \quad (30)$$

where the noise terms $\varepsilon_{jt}, \tilde{\varepsilon}_{ijt}$ are mean-zero white noise. The public component ε_{jt} is the same for all agents i , the idiosyncratic component $\tilde{\varepsilon}_{ijt}$ is independent across agents.

From here, all results follow as in Section 2.

Causal effects of expectations in partial equilibrium. The causal effect of $\bar{E}_t(z_{jt})$ on an element of the aggregate choice vector x_t in partial equilibrium is given by Proposition 6. Note that partial equilibrium here means that all signals aside from $s_{it}(z_{jt})$ are held constant: the change in aggregate actions does not affect signals, and nor does any change in the state variables w_t through the general equilibrium condition (29).

Proposition 6 *The causal effect of a shock ε_{jt} in partial equilibrium is*

$$\Omega_k^{PE}(z_{jt}) = \frac{1}{\bar{\tau}_j} \cdot \frac{dx_{kt}}{d\varepsilon_{jt}} = \sum_{m=0}^{N_z} \left(\bar{B}_{km} \bar{\chi}_{mj} + Cov_I \left[B_{ikm}, \hat{\chi}_{imj} \frac{\hat{\tau}_{ij}}{\bar{\tau}_j} \right] + \bar{B}_{km} Cov_I \left[\hat{\chi}_{imj}, \frac{\hat{\tau}_{ij}}{\bar{\tau}_j} \right] \right) \quad (31)$$

where B_{ikm} denotes the k, m th element of B_i , and $\bar{B}_{km} \equiv \int_i B_{ikm} di$. The scaling by $\bar{\tau}_j^{-1}$ ensures that the information shock is such that $\bar{E}_t(z_{jt})$ increases by 1.

This consists of three sets of effects. The first term inside the summation is the *representative agent channel*: the effects of the average policy function coefficients and average cross-learning across agents. This summarizes all shock transmission channels in models with a representative agent, and indeed in many models with heterogeneity, where average expectation formation is sufficient to capture shock responses to first order.¹⁶

The second term is the *response heterogeneity channel*. Since $\hat{\chi}_{imj} \hat{\tau}_{ij}$ gives the response of $E_{it}(z_{mt})$ to a signal on z_{jt} , this reflects that shocks will be amplified if the agents whose expectations react the most to the shock are the agents whose actions are most sensitive to those expectations (i.e. who have the largest B_{ikm}). Macaulay and Moberly (2025) provide evidence of one such correlation, between the behavior of inflation expectations and liquidity constraints among German households. This channel, for other expectations, is also behind the novel dynamics in Broer et al. (2020), Macaulay (2021), and Guerreiro (2023).

Finally, the third term is the *narrative heterogeneity channel*, as explored for the Calvo model in Section 2.4. Heterogeneous expectations can generate a channel of aggregate shock transmission even if every agent has the same policy function, if information ($\hat{\tau}_{ij}$) is correlated with cross-learning ($\hat{\chi}_{imj}$) across agents. Cross-learning determines an agent's response to a given piece of information, so if information is concentrated among agents with particular non-representative $\hat{\chi}_i$, that distorts the aggregate response away from the representative agent effect.

¹⁶In Andrade et al. (2019), for example, households differ in their interpretation of forward guidance announcements, but the aggregate effects of an announcement depend only on the average over this mix of beliefs. Heterogeneity might affect the average information or cross-learning behavior sustained in equilibrium, but unless one of the other two terms in equation (31) is non-zero, those averages alone drive aggregate shock transmission.

Setting $\hat{\tau}_{ij} = 1$ for all agents i and variables j recovers the effect in standard temporary equilibrium, in which $E_{it}(z_{jt})$ shifts equally for all agents, but no other expectations change. This implicitly abstracts from all effects of heterogeneity, through the response heterogeneity and narrative heterogeneity channels. All that matters is how, on average, expectations of z_{jt} enter the policy function for x_{kt} .

$$\Omega_k^{PE}(z_{jt}|\hat{\tau}_{ij} = 1\forall i, j) = \bar{B}_{kj} \quad (32)$$

Setting $\hat{\tau}_{ij} = \bar{\tau}_j < 1$ and $\hat{\chi}_{ix_{jt+s}x_{jt}} = 1 + s$ for all i, j recovers the effect in extended temporary equilibrium, as in [Werning \(2022\)](#). Again, this abstracts from all effects of heterogeneity, though now a broader range of policy function coefficients determine the aggregate response, as signals on x_{kt} also impact expectations of x_{kt+1} , x_{kt+2} , etc.

$$\Omega_k^{PE}(z_{jt}|\hat{\tau}_{ij} = \bar{\tau}_j < 1, \hat{\chi}_{ix_{jt+s}x_{jt}} = 1 + s, \forall i, j) = \sum_{s=0}^{\infty} \bar{B}_{kjs}(1 + s) \quad (33)$$

where $\bar{B}_{kjs}$ denotes the average coefficient on z_{jt+s} in the row of the policy function (28) corresponding to x_{ikt} .

Causal effects of expectations in general equilibrium. The causal effect of $\bar{E}_t(z_{jt})$ on an element of the aggregate choice vector x_t in general equilibrium is given by Proposition 7.

Proposition 7 *The effect of an exogenous shift in information due to ε_{jt} on aggregate action x_{kt} is*

$$\frac{dx_{kt}}{d\varepsilon_{jt}} = [\Lambda^{GE}\Omega^{PE}]_{k,j} \bar{\tau}_j \quad (34)$$

where Ω^{PE} is a $N_x \times N_z$ matrix collecting the partial-equilibrium effects of each expectation $\bar{E}_t(z_{jt})$ on each aggregate action x_{kt} . Λ^{GE} is a $N_x \times N_x$ matrix capturing general-equilibrium effects, defined in Appendix A.

The causal effect of $\bar{E}_t(z_{jt})$ is obtained by scaling $\frac{dx_{kt}}{d\varepsilon_{jt}}$ such that the response of $\bar{E}_t(z_{jt})$ is equal to 1, which implies

$$\Omega_k^{GE}(z_{jt}) = \frac{[\Lambda^{GE}\Omega^{PE}]_{k,j} \bar{\tau}_j}{\bar{\tau}_j + \Gamma_j} \quad (35)$$

where Γ_j , which captures general-equilibrium effects of the information shock ε_{jt} on $\bar{E}_t(z_{jt})$, is defined in Appendix A.

Importantly, the GE feedback matrix Λ^{GE} will typically not be diagonal. This means that when computing the effects of $\bar{E}_t(z_{jt})$ on a specific aggregate action x_{kt} , the effects of that expectation on *all* aggregate actions and state variables will affect the result, because those effects on other variables in turn cause changes in the signals agents receive on those variables, leading to further changes in x_{kt} beyond those picked up by the partial-equilibrium effect. This is a generalization of the result in Proposition 3, where shocks to $\bar{E}_t(\pi_{t+1})$ in part propagated by influencing π_t , which in turn affected $\bar{E}_t(\pi_t)$.

To showcase more clearly how this differs from the partial-equilibrium case, and from the simple Calvo model of Section 2, Corollary 1 in Appendix A shows Λ^{GE} in the special case where agents only receive signals about current realized $\mathbf{x}_t$ and $\mathbf{w}_t$ (i.e. $\tau_{ij} = 0$ for all z_{jt} realized after period t). In this case, Λ^{GE} simplifies substantially, highlighting that general equilibrium feedback from exogenous variation in expectations can come through two distinct channels. First, as in the Calvo model in Section 2, the shock causes a change in aggregate actions $\mathbf{x}_t$, which feeds back into the information agents receive. Second, this change in $\mathbf{x}_t$ also affects the state variables $\mathbf{w}_t$ through the equilibrium condition (29), which *also* affects agent information. This was not present in Section 2 because we abstracted there from all responses of the state (marginal costs) to firm actions.

The other key insight from this is that all three channels identified in partial equilibrium also affect the strength of this general equilibrium feedback. When exogenous variation in a given expectation affects aggregate actions and states, this affects the information agents receive about those variables, changing their expectations again. The effects of that second round of expectation updating also depend on the representative-agent, response heterogeneity, and narrative heterogeneity channels identified in Proposition 6.

The role of expectations in the transmission of fundamental shocks. This follows exactly as in Section 2.3. The causal effect of an exogenous shock ξ_t on aggregate choices is given by Proposition 8.

Proposition 8 *The causal effect of a fundamental shock ξ_t on aggregate choices is*

$$\frac{d\mathbf{x}_t}{d\xi_t} = -\Lambda^{GE} \Omega_{w0}^{PE} \bar{\tau}_{w0} D^{-1} F \quad (36)$$

where the matrix Ω_{w0}^{PE} collects the partial-equilibrium effects of each element of expectations about current state variables ($\bar{E}_t(\mathbf{w}_t)$), and $\bar{\tau}_{w0}$ is a diagonal matrix collecting the corresponding elements of average τ_i .

Intuitively, the general equilibrium condition (29) implies that $-D^{-1}F$ is the response of state

variables w_t to a ξ_t shock, in the absence of any x_t response. $\Omega_{w0}^{PE} \bar{\tau}_{w0}$ then gives the partial-equilibrium effect of that w_t change on x_t , through changes in expectations of w_t . Finally, Λ^{GE} accounts for general equilibrium feedback through both w_t and x_t .

Next, construct a counterfactual transmission in which there is no response of $E_{it}(z_{jt})$ for some set of variables defined by $j \in J$. To do this, replace $\hat{\tau}_{ij} = \hat{\chi}_{ijk} = 0$ for all i, k , and for all $j \in J$, then re-compute $\Lambda^{GE}, \Omega_{w0}^{PE}, \bar{\tau}_{w0}$. The share of shock transmission due to expectations of z_{jt} for $j \in J$ is then given by $\left(\frac{dx_t}{d\xi_t} - \frac{dx_t}{d\xi_t} \Big|_{\sim E_{j \in J}} \right) \cdot \left(\frac{dx_t}{d\xi_t} \right)^{-1}$.

As in Section 4, the expressions in this section can also be applied to interpret the results of information-treatment RCTs in general equilibrium.

4 Mapping RCT estimates into general equilibrium

Since the temporary-information equilibrium method is inspired by the success of information-treatment RCTs, the expectation statistics that appear in the model framework match closely with parameters estimated in those RCTs. The temporary-information equilibrium approach can therefore be used to translate estimates from information-treatment RCTs into general equilibrium insights, overcoming the typical missing intercept problem in the empirical literature. This helps us answer a key question in that literature: are the departures from full-information and rational-expectations observed in a given RCT large or small, in terms of their macroeconomic implications? Moreover, we can also use the same RCT estimates to separate the implied causal effects of expectations into their representative-agent and narrative-heterogeneity components.

To employ the temporary-information equilibrium method, one needs to choose the appropriate macroeconomic model through which to interpret the RCT. We focus here on applying the method to the recent survey experiment conducted on European firms by [Baumann et al. \(2024\)](#), as this lends itself naturally to interpretation through the Calvo model already studied in Section 2. However, the methods [Baumann et al. \(2024\)](#) employ are common to a broad range of survey experiments, so the same procedure could be carried out for other papers.

The missing intercept in information-treatment RCTs. RCTs like that in [Baumann et al. \(2024\)](#) use information treatments to exogenously vary the expectations of some survey participants, while keeping others as a control group. This cross-sectional variation enables researchers to identify the causal effect of expectations on individual beliefs and actions, relative to a similar individual who was not treated. While this has allowed for great strides in our understanding of the micro-level effects of expectations, and for the testing and development of various theories of expectation

formation, it is often hard to determine whether the results in any given RCT are of a magnitude that is meaningful for macroeconomic outcomes. This is because the individual-level effect cannot simply be aggregated up across a population to give a total macroeconomic effect of a given expectation change. If *all* inflation expectations change, for example, then aggregate realized inflation may change, which in turn may affect firm expectations and decisions. Interest rates may also adjust, as well as the labor market, and other variables, each of which have their own consequences. This general equilibrium effect is the ‘missing intercept’ which stands between the estimates from an information-provision RCT and macroeconomic insights.

Of course, this issue is present in all empirical work using cross-sectional variation to identify partial-equilibrium elasticities of interest to macroeconomists (see e.g. discussions in [Nakamura and Steinsson, 2018](#); [Guren et al., 2020](#); [Wolf, 2023](#), among many others). In information-provision experiments, however, the problem is particularly acute, for two reasons. First, the experiment considered typically occurs only once, or a small number of times. There is therefore limited scope for time-series approaches that could complement the cross-sectional identification (as in e.g. [Donaldson, 2025](#); [Matthes et al., 2025](#)).

Second, the very features of expectations that drive the partial-equilibrium results in the information-provision RCT also affect the missing general equilibrium feedback.¹⁷ To see the difference, consider studies of fiscal policy such as [Nakamura and Steinsson \(2014\)](#). The partial equilibrium effect of interest is the effect of government spending, while the general equilibrium effects are the consequences of subsequent changes in interest rates and other macroeconomic variables. In this way there is hope for equivalence results, such as those in [Wolf \(2023\)](#), that allow researchers to use estimates *from other literatures* to quantify the general-equilibrium intercept. The same approach cannot be taken to study the aggregate effects of changes in expectations, because the features of expectations revealed in the partial-equilibrium RCT estimates also affect that intercept. If, for example, firms tend to over-extrapolate from current to future inflation, that will affect the RCT estimates *and* the response of firms to subsequent general equilibrium changes in endogenous aggregate variables.

The temporary-information equilibrium approach developed here provides a way to cut through this issue: it is straightforward to apply the results in Section 3 to interpret the results from information-treatment RCTs in any model that fits the form introduced there.

¹⁷While not an issue in RCTs, this also suggests that any attempt to use heterogeneous exposure to aggregate expectations shocks to identify the causal effect of expectations will suffer by construction from the issue of heterogeneous exposure to general equilibrium effects discussed in [Donaldson \(2025\)](#).

4.1 Extracting key expectation statistics.

Baumann et al. (2024) elicit prior beliefs from their participants about inflation in 1, 3, and 5 years time. They then treat a random subsample of participants with information about inflation. They study two treatments: one treatment group is shown information on current inflation, and another is shown information on one-year-ahead inflation (from the latest SPF forecast).

Total effects. As is common in the RCT literature, **Baumann et al. (2024)** first estimate the regression:

$$E_{it}^{post} \pi_{t+h} = a_0 + a_1 \cdot E_{it}^{prior} \pi_{t+h} + \sum_j b_j \cdot E_{it}^{prior} \pi_{t+h} \cdot D_{ij} + \sum_j c_j \cdot D_{ij} + \Gamma_{it} + \text{error}_i \quad (37)$$

where D_{ij} is an indicator equal to 1 if firm i is in treatment group j , and Γ_{it} is a vector of controls. The coefficients b_j can be interpreted as -1 times the average degree of updating from the observed signal in treatment group j . For the group shown a signal about π_{t+1} , the b_j estimate when $E_{it} \pi_{t+1}$ is the dependent variable is therefore equal to $-\bar{\tau}_{\pi_{t+1}}$ in our model. It is the (negative of) the average Kalman gain from that signal.

For other combinations of treatment and dependent variable, the signal is no longer precisely about the dependent variable. The coefficient therefore represents the average of $-\frac{\partial E_{it}(z_{jt})}{\partial s_{it}(z_{kt})} = \chi_{ijk} \tau_{ik}$ for treatment variable k and outcome variable j across firms in the sample. From Table 5 in **Baumann et al. (2024)** we therefore obtain $\int_i \chi_{ijk} \tau_{ik} di = \bar{\chi}_{jk} \bar{\tau}_k + Cov_I[\chi_{ijk}, \tau_{ik}]$, i.e. a sum of representative agent and narrative heterogeneity effects.

If all that is desired is an estimate of the total causal effects of inflation expectations on aggregate inflation, these statistics would be sufficient. In Propositions 1-3, those causal effects are derived in terms of precisely the same average effects.¹⁸

Decomposition. The second regression **Baumann et al. (2024)** run is:

$$E_{it}^{post} X_i = \gamma_0 + \gamma_1 \cdot E_{it}^{post} \pi_{t+h} + \gamma_2 \cdot E_{it}^{prior} \pi_{t+h} + \Gamma_{it} + \text{error}_i \quad (38)$$

for a variety of expected dependent variables $E_{it}^{post} X_i$. $E_{it}^{post} \pi_{t+h}$ is instrumented with variation from the information treatment. Specifically, equation (37) is used as a first stage.

¹⁸Note that **Baumann et al. (2024)** do not collect priors for the components of marginal costs in their survey, so in this particular case we can only derive causal effects on inflation assuming no cross-learning from inflation to marginal costs. I discuss the effects of cross-learning to cost expectations further in Appendix C.

Baumann et al. (2024) estimate several versions of equation (38), but for our purposes the relevant specifications are those in which a single treatment group is compared to the control group who received no information. In these cases, the coefficient γ_1 captures the average effect of an exogenous change in expectations of π_{t^*} on expectations of the dependent variable, where π_{t^*} is either current π_t or one-year-ahead π_{t+1} depending on which treatment group is being studied. This is exactly the definition of $\bar{\chi}_{\pi_{t^*}, X}$ in the temporary-information equilibrium of the Calvo model. Combining this with decompositions such as equation (14) (or more generally equation (31)), we obtain an estimate of the representative-agent component of the causal effects of inflation expectations on inflation, which we can combine with the total effect described above to infer the remaining narrative-heterogeneity component.

4.2 The causal effect of expected future inflation on current inflation

I focus here specifically on the causal effect of one-year-ahead inflation expectations on realized current inflation. I also focus only on the effects that operate through the updating of inflation expectations at various horizons, abstracting from any cross-learning from π_{t+1} to marginal cost expectations.

This represents the best showcase for the temporary-information equilibrium approach, as all of the required expectation statistics are reported directly by Baumann et al. (2024) or are easily calculable in their data.¹⁹ Formally, I abstract from marginal-cost cross-learning by setting $\chi_{imc_{t+s}\pi_t} = \chi_{imc_{t+s}\pi_{t+1}} = 0$ (relaxed in Appendix C). I then calibrate the remaining elements of χ_i and τ_i to the estimated coefficients in equations (37) and (38), assuming one period is one year (full details in Appendix C). The only remaining parameters are then β and λ . I set the former to 0.96, and plot the results for varying values of price stickiness λ .

Benchmarks. To interpret the RCT-implied causal effects of inflation expectations, we need benchmarks with which to compare them. I will consider two benchmarks: perfect foresight and an approximation to limited-information rational expectations.

After having solved for the temporary-information equilibrium of the model (equation (11)), imposing perfect foresight is straightforward. We set $\tau_{ij} = 1$ for all variables z_j , which in turn implies χ_i becomes the identity matrix (equation (10)). This was also the strategy used to return to

¹⁹The only statistics not reported by Baumann et al. (2024) are the estimates of γ_1 in equation (38) with posterior inflation expectations at medium- and long-term horizons as the dependent variable. While these are not required for calculating the total general-equilibrium causal effect of expectations on inflation, they allow me to decompose that total effect into representative-agent and narrative-heterogeneity components. I am indebted to Timo Reinelt for kindly providing me with these estimates.

the standard temporary equilibrium case in Section 2.2, as with i.i.d. shocks perfect foresight is equivalent to only varying one expectation at a time.

Rational expectations is harder to specify, because we do not know what the true joint dynamics of inflation and marginal costs are in the Euro Area. I provide a simple approximation here by regressing annual Euro Area HICP inflation on a 12-month lag of itself, giving a persistence coefficient of 0.39. I then assume that $\bar{\tau}_{\pi_{t+1}}$ is as in the survey calibration, but cross-learning is set so that all firms extrapolate to future inflation assuming it has that estimated degree of persistence.²⁰

Finally, note that the survey does not provide an estimate of $\bar{\tau}_{\pi_t}$, the average $E_{it}(\pi_t)$ response to signals about π_t . This will matter for the evaluation of general equilibrium effects (see Proposition 3 and the associated discussion). In the RCT calibration and in the benchmarks I will assume that $\bar{\tau}_{\pi_t} = 1$, as if all firms are perfectly well-informed about current inflation. Importantly, however, the RCT results are informative about cross-learning from observed π_t to longer-horizon π_{t+s} , so the estimates inform general-equilibrium feedback (the ‘intercept’ in the missing intercept problem) as well as partial-equilibrium effects.

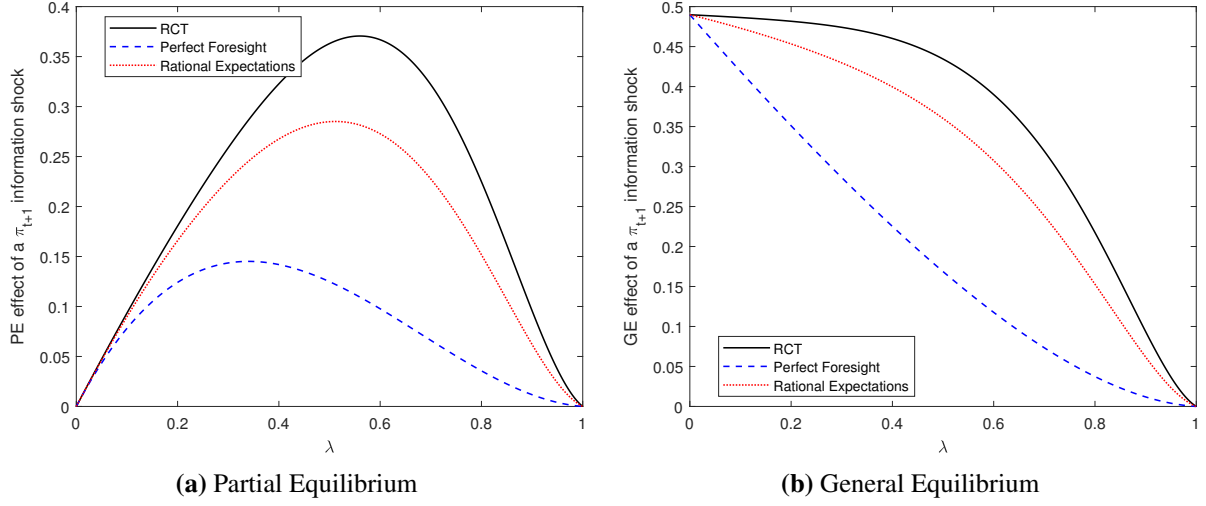
The total causal effect of expectations. Figure 1 plots the causal effect of one-year-ahead inflation expectations on current inflation as price stickiness λ varies. Panel (a) plots the partial equilibrium effect $\Omega^{PE}(\pi_{t+1})$ (i.e. assuming no change in beliefs about current inflation), while panel (b) plots the general equilibrium effect $\Omega^{GE}(\pi_{t+1})$, incorporating the endogenous response of π_t signals.

The key message from these plots is that whether or not the RCT results have large macroeconomic implications relative to common textbook models of expectation formation depends critically on the degree of price stickiness. If we assume a somewhat conservative calibration that targets an average price duration of 5 quarters ($\lambda = 0.2$ since the calibration is annual), the partial equilibrium effect implied by the RCT estimates is 9% larger than under rational expectations, and the general equilibrium effect is 6% larger. However, if prices are sufficiently sticky that the average price duration is two years ($\lambda = 0.5$),²¹ the RCT estimates imply causal effects of expectations that are 28% and 21% larger than rational expectations in partial and general equilibrium respectively. The differences to the perfect foresight benchmark are universally larger, but the same strong dependence on price stickiness is present. With $\lambda = 0.2$, the RCT implies a 46% larger partial

²⁰Note that the model is set up in terms of cumulative inflation, while the survey estimates and the estimated persistence used for this benchmark are in terms of per-period inflation. This rational expectations benchmark therefore entails setting $\chi_{i\pi_{t+s}\pi_{t+1}} = \frac{1-\rho^s}{1-\rho}$ where $\rho = 0.3874$ is the estimated annual persistence of inflation. This gives the correct degree of persistence in cumulative inflation. Appendix C details how the translation to cumulative inflation units is handled for the survey calibration.

²¹This is the calibration used in Werning (2022) footnote 8).

Figure 1: Partial and general equilibrium effects of a shock to $s_{\pi_{t+1}}$, comparing RCT estimates to benchmarks



Note: Figure plots the partial- and general-equilibrium causal effects of an increase in $\bar{E}_t \pi_{t+1}$ in the Calvo model, following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2024) (detail in Appendix C).

equilibrium effect, and a 37% larger general equilibrium effect. With $\lambda = 0.5$, these figures jump to 191% and 158%.

The key reason for this dependence is that the RCT estimates depart most strongly from rational expectations at long horizons, where cross-learning from π_t and π_{t+1} information is found to be much stronger than the benchmark would predict.²² This means that a given change in one-period-ahead expectations is associated with greater updating of long-horizon expectations, and thus a larger inflation response. The difference is even more pronounced when comparing to perfect foresight, where there is no such cross-learning at all.

At very flexible prices (small λ), this does not have a large quantitative effect, because price durations are expected to be short and so long-horizon expectations are relatively unimportant for price-setting. As prices get stickier, long-horizon expectations become more important to current price-setting, and the differences between the RCT-implied model and the benchmarks grow.

Furthermore, price stickiness also determines the importance of the general-equilibrium feedback, i.e. the ‘missing intercept’. At $\lambda = 0.2$, the general equilibrium effect of expected inflation is more than $2.5\times$ larger than the partial equilibrium effect. At $\lambda = 0.5$, it is only 20% larger. The other key determinants are the RCT results themselves, because the general-equilibrium effect of an increase in realized π_t depends on how firms incorporate that signal into their expectations of inflation at all horizons, which is a key object studied in the RCT. In Appendix

²²To get the rational expectations benchmark to deliver the same cross-learning from π_{t+1} to π_{t+5} as implied by the RCT, the annual persistence of inflation would need to be 0.82, more than double its estimated value.

C I show this concretely by plotting the causal effect of expectations using the RCT calibration for the partial-equilibrium effect, but taking the general-equilibrium feedback from either the perfect foresight or rational expectations benchmark. In sharp contrast to the correct results in Figure 1b, this exercise finds that the general-equilibrium effect of inflation expectations is initially *increasing* in price stickiness, with either of these alternative general-equilibrium assumptions. The general-equilibrium effects are also strictly greater than the true effect for all values of λ , with discrepancies exceeding 10% of the true value for some levels of stickiness.

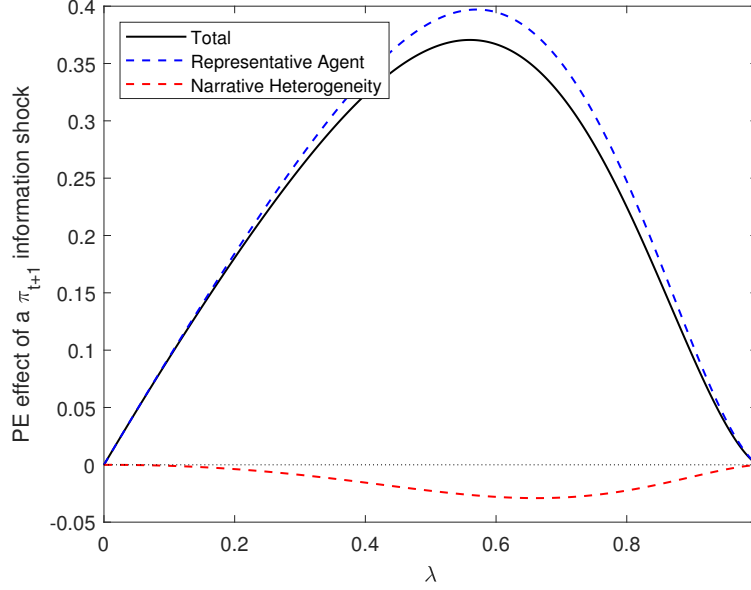
Appendix C also contains a number of other results extending this analysis. In particular, I apply the methods described in Section 2.3 to study the share of transmission of a marginal-cost news shock that is due to inflation expectations, and find that even the sign of the departure from rational expectations implied by the RCT results varies with price stickiness. I also compute alternative benchmarks using two popular models that depart from rational expectations: diagnostic expectations (Bordalo et al., 2020) and inattentiveness (Gabaix, 2020). In each case, adjusting the benchmark simply involves replacing the χ_i and τ_i parameters with those implied by the researcher’s chosen theory of expectations.

The narrative heterogeneity channel. In Figure 2 I decompose the partial-equilibrium effect of expected inflation on realized inflation into its representative-agent and narrative-heterogeneity components. The black solid line is the same as that in Figure 1a, while the dashed lines show its components.

In this experiment, the causal effect of expectations is stronger when we consider the representative-agent effects in isolation. This implies that the narrative heterogeneity effect is negative. In other words, the firms who process the most information about π_{t+1} extrapolate the least to inflation at longer horizons. As a proportion of the total effect, the magnitude of the narrative heterogeneity channel is monotonically increasing in λ . At $\lambda = 0.5$, it reduces the causal effect of expected inflation by 6%, narrowing the gap between the RCT results and the rational expectations benchmark.

This negative narrative heterogeneity channel may seem initially surprising, given that simple models of information acquisition would predict that information is more valuable if the underlying process is believed to be persistent. However, many other forces could generate such a negative correlation between information and cross-learning between inflation horizons. In particular, the RCT results imply that firms on average extrapolate more strongly from π_{t+1} signals to longer horizons than the simple approximation to rational expectations computed above would predict. If this overextrapolation is a reasoning error on the part of firms, then the firms who are better-informed

Figure 2: Partial equilibrium effects of a shock to $s\pi_{t+1}$, decomposition



Note: Figure plots (in black) the partial-equilibrium causal effects of an increase in $\bar{E}_t \pi_{t+1}$ in the Calvo model, following the formula in Proposition 2, and a calibration derived from the estimates in Baumann et al. (2024) (detail in Appendix C). The blue and red lines are the decomposition of this effect into representative-agent and narrative-heterogeneity channels, following Proposition 6.

may use that information to update their beliefs about inflation persistence towards the truth, thus leading to a negative correlation between information precision and cross-learning.

This sharp decomposition of the total causal effect into representative-agent and narrative-heterogeneity effects is only possible when considering the partial-equilibrium causal effects of inflation expectations on inflation. In general equilibrium, the total causal effect is a nonlinear combination of representative-agent and narrative-heterogeneity channels (e.g. equation (16)). The difference between the total general-equilibrium effect and the counterfactual effect if only representative-agent channels were present is therefore indicative of the influence of heterogeneous expectation formation, but does not have the same interpretation as a specific covariance. Moreover, as we assume for these results that all firms observe current inflation precisely ($\tau_{\pi_t} = 1$), there are no further narrative heterogeneity effects in the missing intercept beyond those shown in the partial-equilibrium result in Figure 2. Mechanically, therefore, the dampening of the partial-equilibrium effect due to narrative heterogeneity also results in a dampening of the general-equilibrium effect. This is shown in full in Appendix C.

5 Survey evidence on narrative heterogeneity

In developing the temporary-information equilibrium approach, I have shown above that *narrative heterogeneity* effects have the potential to dramatically alter the general equilibrium effects of expectations. These effects arise from correlations between agent responses to information and the cross-learning they use to map from that information to expectations of other variables. In Section 4, I measured this effect implicitly, backing it out from a decomposition of the Calvo model in Section 2. In this section I measure the correlation behind the narrative heterogeneity channel directly, in the context of household beliefs around inflation. I document three empirical results that reveal a strong and time-varying narrative heterogeneity channel in this context. These results are used to inform the model in Section 6, from which I observe the aggregate consequences of this instance of the narrative heterogeneity channel.

5.1 Data

Most empirical papers on information frictions or subjective models use data on realized expectations. Expectations combine both information and subjective models (as shown in Sections 2 and 3), and so cannot be used to identify the narrative heterogeneity channel. To overcome this, I use data from the Bank of England Inflation Attitudes Survey (IAS), which contains several unique questions which enable me to measure these components of expectation formation separately – and thus to measure their joint distribution.

The IAS is a quarterly survey of a repeated cross-section of UK households, run since 2001 (annual until 2003). After weighting, the sample is representative of the UK adult population. I use the individual-level response data from 2001-2019, omitting the quarters conducted after the onset of the Covid-19 pandemic, as the implementation of the survey had to be changed substantially at this time (see [Bank of England, 2020](#)).

Alongside questions on expectations of inflation, interest rates, and other macroeconomic and personal variables, respondents are asked several questions which are helpful in disentangling information and subjective models about inflation. The first of these asks households about their subjective model of the relationship between inflation and the ‘strength of the economy’.

Question 1 *If prices started to rise faster than they are now, do you think Britain’s economy would end up stronger, or weaker, or would it make little difference?*

This differs from standard questions on expected future economic outcomes because it does not invoke the use of information about the state of the world. The answers are informative about

cross-learning (χ_{ijk} in the models above) only.²³ In the analysis below, I will refer to a respondent answering that inflation would make the economy stronger/little difference/weaker as having a positive/neutral/negative subjective model of inflation respectively.

There are two possible interpretations of this question. Households may view it as asking about the causal effects of inflation on the economy (as in the model of [Spiegler, 2021](#)). Alternatively, they could see it as asking about the most likely source of a rise in inflation, if they believe supply- and demand-driven inflation is associated with different real outcomes (as in e.g. [Kamdar and Ray, 2024](#)). For the purposes of this section, this distinction does not matter. χ_{ijk} is defined above as the degree to which households update their expectations of variable j when their expectation of variable k changes, irrespective of why that link exists. In this case, it is the updating of expectations about the strength of the economy when expected inflation rises. The sign of this updating is captured by the question, whether it occurs because of a perceived direct causal link from inflation to the real economy, or a belief about the type of shocks hitting the economy.²⁴

The next set of novel questions concern the information households use to form their inflation expectations, without asking what those expectations are. This allows us to learn about household information precision (encoded in τ_{ik}) without contamination from cross-learning (χ_{ijk}).

Question 2a *What were the most important factors in getting to your expectation for how prices in the shops would change over the next 12 months?*

Please select up to 4 [from a list reproduced in Appendix [D.1](#)].

Question [2a](#) was only asked in 2016Q1, but very similar questions were asked at other times. In each, the respondent is asked about the information sources they used to arrive at their expected inflation, or that led them to change that expectation over the previous year. In each, some of the options that respondents can choose refer to information specifically on inflation, while others refer to information on other variables, which can only be used to forecast inflation if the household engages in some cross-learning. For each such question I construct a dummy variable equal to 1 if the respondent reports using direct information about inflation, and equal to 0 if they do not. Full details of these questions, and the options representing direct information, are in Appendix [D.1](#). Combining these dummy variables gives an indicator for if the respondent used direct information on inflation in forming their expectations, that is whether $\tau_{i\pi_t} > 0$. This indicator is observed for 8 separate quarters between 2009Q1-2019Q1. I confirm below that the key results of this section do not vary substantially with the changes in question wording over these periods.

²³This is similar to how [Andre et al. \(2022\)](#) interpret the answers to the hypothetical vignettes in their surveys.

²⁴The distinction will matter when using a structural model to analyse counterfactual implications of this data. I therefore return to this issue in Section [6](#).

In Appendix D.2 I confirm that these measures of information and subjective models correlate with questions on planned household consumption, and that the signs of these correlations are consistent with the measures picking up the desired elements of household beliefs. A further possible test of the information indicator would ask if households who obtain direct information about inflation make more accurate forecasts. However, if beliefs about the level of inflation affect subjective models, that may in turn change the incentives to acquire further information, complicating the predicted correlation between information and forecast accuracy.²⁵ Indeed, recent evidence from Link et al. (2025) suggests that greater household information acquisition is often associated with larger forecast errors about inflation.

The other questions used in this section are standard, asking households to give point estimates for “how prices have changed over the last twelve months” and “how much would you expect prices in the shops generally to change over the next twelve months”. For each question, respondents choose from a list of ranges, and follow-up questions may then asked with more precise ranges, until the respondent has selected a 1 percentage-point bin between -5% and +10%, or end ranges $\leq -5\%$, $\geq 10\%$. For the exercises in Section 5.4, I code perceptions and expectations at the midpoint of the selected bin, with the lowest and highest bins coded as -5.5% and 10.5% respectively.

5.2 Information and subjective models in the cross-section

With this data, I first explore the cross-sectional relationship between the distributions of information and subjective models, which is the source of any narrative heterogeneity channel. Table 1 shows the estimated average marginal effects from a probit regression of the information indicator defined in Section 5.1 on the respondent’s subjective model of inflation, represented by their answer to Question 1. The first column shows this with quarter fixed effects only, while the second also includes a range of household controls.²⁶ This gives us our first empirical result.

Empirical Result 1 *Households who believe inflation makes no difference to the economy acquire less information about inflation on average than households who believe inflation does affect the economy (in either direction).*

Those answering that inflation makes no difference to the aggregate economy, or who don’t know the effect of inflation, are significantly less likely to use information about inflation than

²⁵In a previous version of this paper (Macaulay, 2022) I explicitly model this process and show that the data line up qualitatively with the model.

²⁶These are gender, age, class, employment status, income, education, region, and home-ownership status. Age, class, income and education are all reported in bands, and included as categorical variables.

Table 1: Information correlates with subjective models

	(1)	(2)
end up stronger	-0.0102 (0.0191)	-0.00827 (0.0192)
make little difference	-0.0356*** (0.0128)	-0.0315** (0.0129)
dont know	-0.0627*** (0.0172)	-0.0605*** (0.0172)
Controls	None	All
Time FE	Yes	Yes
Observations	8270	8270

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the average marginal effects from estimating a probit regression of the information indicator on the responses to Question 1. The information indicator equals 1 if the household reports using a direct source of information about inflation when forming their expectations, as defined in Appendix D.1. The omitted category is the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

someone who believes inflation makes the economy weaker. There is no significant difference in the probability of using direct inflation information between those holding this view and those with positive subjective models of inflation. The probability of using direct inflation information is 3-3.5 percentage points lower for those with a neutral model of the effects of inflation than those who believe inflation weakens the economy. Over the whole population 23% of respondents use direct inflation information, so this difference is non-trivial. More important than the magnitude, however, is that this shows a systematic cross-sectional relationship between information and subjective models, indicating a role for the narrative heterogeneity channel. Assessing the aggregate relevance of this requires a model, as developed in Section 6 below.

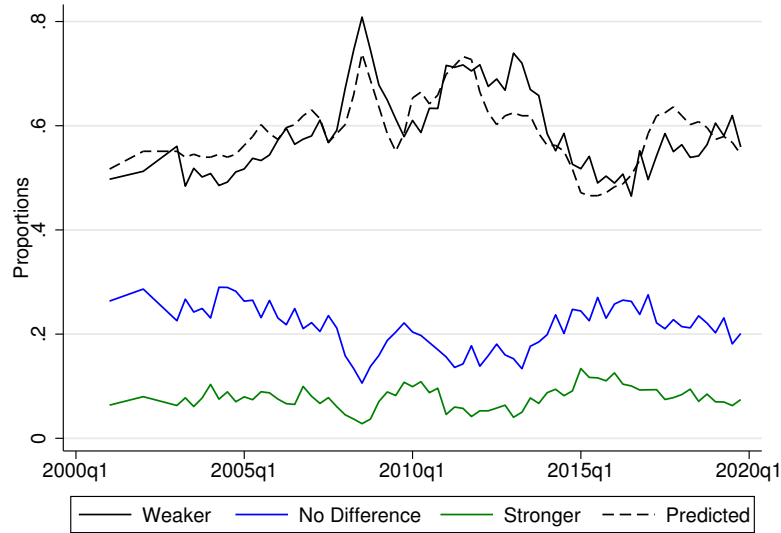
The information indicator is composed of answers to several slightly different questions. In particular, some questions concern information used to arrive at the respondent's point expectation for inflation, while others concern the information they used in changing those expectations over the last year. Most questions concern expected inflation over the next 12 months, but a minority ask about a longer horizon. In Appendix D.3 I repeat the regressions of Table 1 on subsets of the questions, and find the results are robust to these alternatives. As some respondents do not answer the unique survey questions used here, I also account for the concern that there may be selection bias in whose answers are observed, using a selection model as in Heckman (1979). Again, the results are robust.

Result 1 is not consistent with models with exogenous information, as there would be no reason for information to be systematically correlated with household subjective models. It is, however, consistent with models of endogenous information acquisition, as explored in Section 2.4. Intuitively, the value of inflation information is lower for households who believe inflation makes little difference to other variables that matter for their decisions.²⁷

5.3 Subjective models over time

Figure 3 shows the proportions answering Question 1 with each subjective model of inflation over time (‘don’t know’ omitted for figure clarity).

Figure 3: Proportions giving each answer to Question 1: “If prices started to rise faster than they are now, do you think Britain’s economy would end up stronger or weaker or would it make little difference?”



Note: Proportions shown are calculated using the survey weights provided in the IAS. Proportion answering ‘Don’t know’ is omitted for figure clarity. The dashed line is the predicted values from regressing the proportion reporting that inflation makes the economy weaker on annual CPI inflation: $\Pr(\text{weaker})_t = 0.057 \times \text{CPI inflation}_t + 0.466$. The coefficient on inflation is significant at the 1% level.

The majority of households answer that inflation would make the economy weaker in all quarters, in keeping with the findings in Shiller (1997), Andre et al. (2022), and Kamdar and Ray (2024). The second-largest group report that inflation makes no difference. Combined with Empirical Result 1, this suggests that the covariance of information on inflation and cross-learning from inflation to the strength of the economy is *negative*. If households consume more when they believe the economy is strong, the narrative heterogeneity channel will therefore reduce the consumption response to inflationary shocks in partial equilibrium.

²⁷See e.g. Roth and Wohlfart (2020) for evidence that households perceive the state of the aggregate economy as relevant to their own decisions.

The relatively long time series of the IAS also allows us to see that the distribution of answers varies substantially over time, which gives us our second empirical result.

Empirical Result 2 *A greater proportion of households believe inflation weakens the economy when realized inflation is high.*

The correlation between annual CPI inflation and the proportion of respondents with negative models of inflation is extremely high, at 0.8.²⁸ The dashed line in Figure 3 plots the predicted values from regressing this proportion on CPI inflation, showing that this correlation is strong across the whole sample. Tests in Appendix D.4 show that the correlation is robust to the addition of various macroeconomic controls, which themselves explain far less of the variation in the distribution of responses than realized inflation. The correlations are also robust to using inflation measures split by various household characteristics, to get closer to the rate of inflation in each household's own basket of goods. Finally, the proportions giving all other answers are also shown to be significantly negatively correlated with current inflation.

This is not what we would observe if households hold rational expectations. The question is about the effect of an aggregate variable (inflation) on the aggregate performance of the economy. Even if households are differentially exposed to the shock, if they all had model-consistent beliefs they would all give the same answer to this question. The fact that there is heterogeneity at all is evidence that at least some household subjective models are inconsistent with rational expectations.

The time-series patterns also suggest that the majority of households are not using New Keynesian-style models. In a textbook New Keynesian model, a rise in inflation causes the central bank to raise the nominal interest rate. If the Taylor Principle is satisfied, the real interest rate rises, so output falls. If it is not, the real rate falls, and output rises. If most households used this model, they should respond that inflation would make the economy weaker in the periods before interest rates reached the Zero Lower Bound (ZLB), and they should switch to the view that inflation would make the economy stronger once we reach the ZLB in 2009. There is little evidence for this in Figure 3, and indeed statistical tests in Appendix D.4 find no evidence of such a shift.²⁹

²⁸Although I do not use later data due to the structural break induced by the methodology change in 2020, this pattern is robust through the 2021-22 inflation surge. Using data from 2020Q1-2024Q3, the equivalent correlation is 0.65.

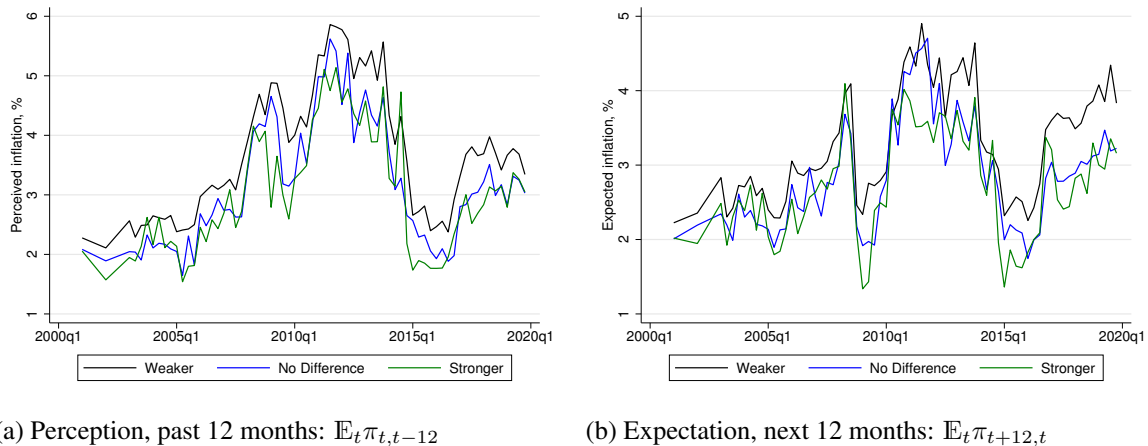
²⁹This argument supposes that at least some households believe cost-push shocks are part of the drivers of inflation. While demand-driven inflation in a New-Keynesian model is associated with higher output whatever the monetary regime, we would still see some shifts in answer distributions at the ZLB if cost-push shocks are perceived to occur with positive probability. In principle, after 2009 a New Keynesian model would predict that a sufficiently large rise in inflation would lift the economy away from the ZLB, implying higher real interest rates and lower output. However, in 2013 the Bank of England began forward guidance committing to maintaining low interest rates, so it is unlikely that households were expecting them to contract in response to small rises in inflation at this time.

5.4 Inflation perceptions, expectations, and subjective models

Finally, I compare perceived and expected inflation across households with different subjective model beliefs. Figure 4 shows the time series of mean perceived and expected inflation by qualitative subjective model of inflation, which gives us our third empirical result.

Empirical Result 3 *Households who believe inflation weakens the economy on average perceive higher current inflation, and expect higher future inflation, than those who believe inflation makes no difference to the economy. They, in turn, perceive and expect higher inflation than those with positive subjective models of inflation.*

Figure 4: Inflation perceptions and expectations by subjective model.



Note: Perceived inflation refers to beliefs about what inflation has been over the past 12 months, and expected inflation refers to expectations for the next 12 months. Averages for each variable are calculated using the survey weights provided in the IAS. Average perceptions and expectations among respondents answering ‘Don’t know’ to the subjective model question (Question 1) are omitted for figure clarity.

The differences are large: Table 2 shows that even after controlling for the full set of available household characteristics, those with a negative model of inflation perceive that inflation has been 54 basis points higher than those with a neutral model, and 70 basis points higher than those with a positive model.³⁰ The gaps are similarly large and strongly significant for expected inflation.³¹ Appendix D.5 shows that these results are not driven by selection bias from missing observations for inflation perceptions and expectations.

³⁰If anything, these gaps are larger in the more recent waves of the survey omitted from the main analysis. At the peak of the UK’s inflation in 2022Q4, the mean difference in inflation perceptions between those with negative and neutral models of inflation was 374 basis points. The mean difference in 1-year ahead inflation expectations was 280 basis points.

³¹Dräger et al. (2022) similarly find for German households that inflation expectations are higher among those reporting that they would prefer inflation to be lower.

Table 2: Perceived and expected inflation are higher for those with more negative subjective models.

	(1)	(2)
	Perceived inflation	Expected inflation
end up stronger	-0.696*** (0.0371)	-0.565*** (0.0353)
make little difference	-0.543*** (0.0226)	-0.466*** (0.0207)
dont know	-0.462*** (0.0315)	-0.413*** (0.0294)
Controls	Yes	Yes
Time FE	Yes	Yes
R-squared	0.179	0.113
Observations	85803	85201

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the results of regressing perceived and expected inflation on respondent subjective models (responses to Question 1) . The omitted category is the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

This is not driven by the households using different kinds of information to arrive at their perceptions and expectations: Table 1 shows that the households with positive subjective models use similar information sources to those with negative models. It is, however, consistent with information about high inflation causing households to update their subjective models towards more negative views. Although the exercises here do not identify the direction of causation, such a mechanism can parsimoniously account for Results 2 and 3. Within a period, those who receive signals that inflation is high shift to more negative subjective models, and when realized inflation rises more households receive such signals. This is explored in detail in Section 6.

6 A model to evaluate the causal effects of household inflation expectations

In this section I build a New Keynesian model in which households have heterogeneous information and subjective models around inflation. This will be used to explore the implications of the empirical observations surrounding the narrative heterogeneity channel in Section 5.1. All proofs for this section are in Appendix E.

6.1 Households

Time is discrete, and the period is denoted by t . The economy is populated by a measure 1 of households. Each period, household i chooses consumption C_{it} to maximize expected discounted utility:

$$E_{it}U_{it} = E_{it} \sum_{s=t}^{\infty} \beta^s \frac{(C_{is})^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} \quad (39)$$

subject to:

$$C_{it} + B_{it} = \frac{R_{t-1}}{\Pi_t} B_{it-1} + Y_t \quad (40)$$

where Y_t is the real income received by all households in period t , R_t is the gross nominal interest rate on one-period bonds B_{it} , and Π_t is gross inflation between periods $t-1$ and t . $\beta \in (0, 1)$ is the discount factor, and $\sigma > 0$ is the elasticity of intertemporal substitution. Income and prices are observed before the consumption choice in period t , but future income and prices are unknown. Household i 's expectations operator E_{it} may not coincide with rational expectations, but given their subjective model for the evolution of R , Π , Y , each household uses their information optimally. Any non-rationality in expectations therefore comes only from misperceptions in these laws of motion.

While households observe the current price level, I assume that they may not perfectly observe the current rate of inflation. This assumption is common in models with noisy information (e.g. [Coibion and Gorodnichenko, 2015](#)), and is consistent with the evidence in [Macaulay and Moberly \(2025\)](#), who find substantial uncertainty about current inflation.³²

To focus on the feedback between subjective models and information choices, I take steady state assets $\bar{B}_i \rightarrow 0$.³³ The log-linearized first order condition is:

$$c_{it} = (1 - \beta) \sum_{s=t}^{\infty} \beta^{s-t} E_{it} y_{is} - \sigma \beta \sum_{s=t}^{\infty} \beta^{s-t} (E_{it} r_s - E_{it} \pi_{s+1}) \quad (41)$$

This is as in e.g. [Eusepi and Preston \(2018\)](#). Imposing rational expectations would restore the standard Euler equation in which consumption is related to expected consumption one period ahead, but as discussed at length above, this is not appropriate for evaluating the causal effects of expectations, and so we must remain with this un-transformed version.

³²One way to microfound this is to assume that households consist of a forecaster, who forms expectations without observing current inflation, and a shopper who uses those forecasts (along with observed current prices) to make consumption decisions. A similar assumption is made in [Pfäuti \(2024\)](#).

³³[Michelacci and Paciello \(2024\)](#) show that with ambiguity aversion, wealth heterogeneity implies heterogeneity in subjective models. Combining this with endogenous information choices, wealth could therefore form an additional reason for a systematic relationship between information and subjective models. This is beyond the scope of this paper.

6.2 Firms, policy, and market clearing

As the focus of this model is the behavior of households, I keep the rest of the model simple. Full details and derivations are in Appendix E.1.

Firms. Monopolistically competitive intermediate goods producers set prices subject to quadratic adjustment costs, and produce using labor as the only input. They supply a perfectly competitive final goods producer, who combines the intermediate goods varieties with a CES production function. Log-linearizing the solution to the intermediate goods firm pricing problem about the zero-inflation steady state yields the Phillips Curve:

$$\pi_t = \beta E_t^{RE} \pi_{t+1} + \kappa mc_t \quad (42)$$

where mc_t is the real marginal cost of intermediate goods producers, and κ is a combination of model parameters. I maintain the textbook assumption of rational expectations for firms, so E_t^{RE} denotes the rational expectations operator.

Labor supply is determined by a continuum of unions, who supply differentiated labor varieties to a perfectly competitive labor packer (as in e.g. Schmitt-Grohé and Uribe, 2005; Auclert et al., 2024). Households supply as much labor as is demanded by the unions, who set nominal wages for their labor variety.

In New Keynesian models with labor markets of this kind, unions are typically subject to frictions in the setting of nominal wages. I depart from this, and instead assume that it is real wages which are sticky, as in e.g. Blanchard and Galí (2007). Furthermore, I take this stickiness to the limit where real wages are perfectly rigid. This simplification ensures that inflation is entirely supply-driven, as the only fluctuations in real marginal costs come from changes in TFP. This substantially aids tractability, particularly as these assumptions eliminate any feedback from inflation expectations back into inflation. The only general equilibrium effect is therefore that inflation expectations affect output, and thus output expectations. The lack of general equilibrium effects on inflation is discussed further in Section 6.3.

All firm profits are transferred back to households as a lump sum, so all revenues are returned to households either through wages or dividends. Costs due to price adjustment costs drop out when the model is log-linearized. To first order, real income is therefore equivalent to real output.

Policy. Nominal interest rates are set according to a simple Taylor rule:

$$r_t = \phi \pi_t + v_{rt} \quad (43)$$

where ϕ is a constant, and $v_{rt} \sim N(0, \sigma_r^2)$ is a monetary policy shock.

Market clearing. All output is consumed by households each period:

$$y_t = c_t \quad (44)$$

where $c_t = \int c_{it} di$ is aggregate consumption across households.

Finally, an exogenous AR(1) process for TFP, combined with equation (42) and fixed real wages, implies:

$$\pi_t = \rho_\pi \pi_{t-1} + v_{\pi t} \quad (45)$$

where the persistence parameter $\rho_\pi \in [0, 1)$, and $v_{\pi t} \sim N(0, \sigma_a^2)$ is driven by an exogenous TFP shock. The full derivation of this is provided in Appendix E.1.

6.3 Expectations

Households form expectations by taking information on each variable and forecasting forward using their subjective models.

Information. Household i 's period- t information set consists of the history to period t of observations of r_t , y_t , and a noisy signal about π_t of the form:

$$s_{it} = \pi_t + \varepsilon_t + \tilde{\varepsilon}_{it}, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2), \quad \tilde{\varepsilon}_{it} \sim N(0, \sigma_{\varepsilon i}^2) \quad (46)$$

Subjective models. Subjective models take the following form:

$$\pi_t = \rho_{\pi i} \pi_{t-1} + u_{\pi t} \quad (47)$$

$$r_t = \phi_i \pi_t + u_{rt} \quad (48)$$

$$y_t = \alpha_i \pi_t + \lambda_i r_t + \rho_{yi} y_{t-1} + u_{yt} \quad (49)$$

where $u_{xt} \sim N(0, \sigma_x^2)$ for $x \in \{\pi, r, y\}$, and $\rho_{\pi i}, \rho_{yi} \in (0, 1)$.

The subjective models for π_t, r_t therefore have the same functional forms as the data generating processes for those variables (equations (45) and (43)). The equilibrium data generating process for y_t is derived in Appendix E.1: the functional form of equation (49) is a tractable approximation to that equilibrium law of motion. Note, however, that the parameters of these subjective models may

differ across households, leading to heterogeneity in cross-learning. Since π_t, r_t, y_t are aggregate variables, this heterogeneity rules out the possibility that all households have rational expectations.

Expectation formation. For simplicity, I assume that households do not infer anything about π_t from observed y_t and r_t . In principle, these are also noisy signals about π_t , but households do not make use of them when forming perceptions of current inflation. This is consistent with the rational inattention microfoundation offered for the household information structure in Section 6.5.³⁴ Households therefore form inflation perceptions using a standard Kalman filter:

$$E_{it}\pi_t = \tau_i s_{it} + (1 - \tau_i)\rho_{\pi i} E_{it-1}\pi_{t-1} \quad (50)$$

As the initial subjective models are constant for each household, so too is $\sigma_{\varepsilon i}^2$. I therefore assume that each household uses the steady-state Kalman gain:

$$\tau_i = \frac{\text{Var}_i(\pi_t | \mathcal{I}_{it-1})}{\sigma_{\varepsilon i}^2 + \text{Var}_i(\pi_t | \mathcal{I}_{it-1})} \quad (51)$$

where $\mathcal{I}_{it-1}$ is the information set of household i in period $t - 1$.

Unlike in Section 5, this specification of expectations does restrict the interpretation of Question 1 in the IAS. The only shock perceived to be driving inflation is $u_{\pi t}$, so there is no room for disagreement about the source of inflation shocks. Heterogeneous cross-learning from inflation to the real economy can only therefore come from heterogeneous beliefs about the causal effects of inflation. In the model, this is consistent with the true law of motion for inflation (equation (45)).

This assumption aids tractability, but also reflects the fact that the distribution of survey answers is very consistent over time, in levels and in how it correlates with realized inflation. If the answers reflected beliefs about the type of shocks driving inflation, this distribution would change across time periods characterized by different types of shocks. Since the distribution of subjective models evolved in the same way with the run-up in inflation before the Great Recession and the currency devaluation-driven spike after the Brexit referendum, it does not appear that the source of inflation shocks plays a key role in the majority of survey answers. Finally, this formulation is also consistent with existing literature finding household inflation expectations are well-described by such simple forecasting rules (e.g. Adam, 2007).

With this setup, the expectations of each relevant variable can be expressed in terms of y_t , and

³⁴Strictly, rational inattention models assume agents choose among all possible signals. So y_t and r_t are available signals, but the household chooses not to pay to process them when forming their inflation perception.

$E_{it}\pi_t$:

$$E_{it}\pi_{t+s} = \rho_{\pi i}^s E_{it}\pi_t \quad (52)$$

$$E_{it}r_{t+s} = \phi_i \rho_{\pi i}^s E_{it}\pi_t \quad (53)$$

$$E_{it}y_{t+s} = \rho_{yi}^s y_t + \frac{(\alpha_i + \lambda_i \phi_i) \rho_{\pi i}}{\rho_{\pi i} - \rho_{yi}^i} (\rho_{\pi i}^s - \rho_{yi}^s) E_{it}\pi_t \quad (54)$$

Substituting these into the consumption function (41), the consumption function of household i is:

$$c_{it} = \frac{1 - \beta}{1 - \beta \rho_{yi}} y_t - \sigma \beta r_t + \tilde{\chi}_i (\tau_i s_{it} + (1 - \tau_i) \rho_{\pi i} E_{it-1} \pi_{t-1}) \quad (55)$$

where

$$\tilde{\chi}_i = \frac{\beta \rho_{\pi i} [(1 - \beta)(\alpha_i + \lambda_i \phi_i) - \sigma(\phi_i \beta - 1)(1 - \beta \rho_{yi})]}{(1 - \beta \rho_{\pi i})(1 - \beta \rho_{yi})} \quad (56)$$

is a composite of cross-learning parameters such that $\tilde{\chi}_i$ gives the elasticity of household i consumption to an exogenous change in their idiosyncratic inflation perception (i.e. to a change in $E_{it}\pi_t$ driven by $\tilde{\varepsilon}_{it}$).

6.4 Temporary-information equilibrium

This model fits into the framework of Section 3. Consumption c_{it} is the choice variable x_{it} , so the consumption function in equation (41) maps into equation (25). The vector of relevant variables $\mathbf{z}_t$ therefore consists of $\{y_{t+s}, r_{t+s}, \pi_{t+s+1}\}_{s=0}^\infty$. Equations (43), (44), (45) form the general equilibrium conditions, mapping into equation (29).

Equilibrium definition. For given distributions of $\tau_i, \rho_{yi}, \tilde{\chi}_i$, and given exogenous shocks $v_{rt}, v_{\pi t}, \varepsilon_t$, the temporary-information equilibrium consists of $\{c_{it}, r_t, \pi_t, y_t\}$ and signals s_{it} such that:

1. *Households*: households choose consumption c_{it} to maximize expected lifetime utility (following (55)).
2. *Firms*: firms set prices to maximize expected lifetime profits, implying inflation π_t follows the process in (45).
3. *Monetary Policy*: policymakers choose the nominal interest rate r_t according to (43).
4. *Market Clearing*: the goods market clears (y_t satisfies equation (44)).
5. *Signals*: inflation signals are determined as a function of realized inflation and idiosyncratic and aggregate noise (equation (46)).

6.5 Fitting the survey results

I now add assumptions on the distributions of information and subjective models across households to match the correlations observed in the survey data in Section 5.

Information. To qualitatively match Result 1, we require that the precision of inflation information $\sigma_{\varepsilon i}^{-2}$ is increasing in $|\tilde{\chi}_i|$. In this way, households who believe greater inflation has strong positive or strong negative effects on consumption both process more inflation information than those who believe inflation is irrelevant.³⁵

Specifically, I impose a functional form derived from a microfoundation in which it is costly for households to process information about current inflation, as in the literature on rational inattention (Maćkowiak et al., 2023). In Appendix E.2 I show that a household facing such costs optimally chooses signals of the form in Equation (46), with noise variances such that:

$$\begin{cases} \tau_i = 0 & \text{if } \tilde{\chi}_i^2 < \delta_i^* \\ \frac{1 - \tau_i}{(1 - \rho_{\pi i}^2(1 - \tau_i))^2} = \frac{\delta_i^*}{(1 - \rho_{\pi i}^2)^2} \tilde{\chi}_i^{-2} & \text{if } \tilde{\chi}_i^2 \geq \delta_i^* \end{cases} \quad (57)$$

where δ_i^* is a threshold defined in Appendix E.2.

Subjective models. To jointly match Results 2 and 3, I assume that increases in perceived inflation $E_{it}\pi_t$ can cause households to update their perceived relationship between inflation and real income (α_i). The timing is therefore as follows: a household starts the period with an initial subjective model, which determines the precision of their inflation signal. After observing the realization of that signal, the household forms a perception of inflation, which they use to update their subjective model. The realized signals and updated subjective model are then combined to form the expectations used to choose consumption. In this decision I use the anticipated utility assumption, common in models with learning, that households act in period t as if they are certain that their subjective model will not change in future periods (see e.g. Bullard and Suda, 2016). For simplicity, the initial subjective models at the start of each period are fixed over time for each household.

As with information, I further impose a functional form derived from a microfoundation contained in Appendix E.2. In this case, the microfoundation is that households are ambiguity averse, and face Knightian uncertainty about α_i . In that environment households distort their

³⁵See Dräger et al. (2022) for evidence that household beliefs about what is good for the economy overall and for them personally are highly correlated.

subjective model towards the ‘worst case’ parameter that would deliver the lowest expected utility.³⁶ When perceived inflation is high, the worst-case model is one in which high inflation is damaging, i.e. in which α_i is strongly negative. Specifically, this microfoundation implies that the updates to subjective models are linear in inflation perceptions:

$$\hat{\alpha}_{it} = \alpha_{i0} + \alpha_{i1} E_{it} \pi_t \quad (58)$$

where $\alpha_{i1} < 0$. In all exercises below I assume $\hat{\alpha}_{it}$ follows this form. This ensures that the average perceived consequences of inflation for real variables are more negative when realized inflation is high (Result 2), and that in the cross-section higher perceived and expected inflation is correlated with more negative subjective models (Result 3).

7 The causal effect of household inflation expectations

I now calibrate the model to UK macro data and the survey results of Section 5, and then use the calibrated model to study the time-varying causal effects of household inflation expectations.

7.1 Calibration

To calibrate the model, I proceed in four steps. First, I set some parameters to standard values in the literature. Second, I assume that all households have the same subjective models for the determination of π_t, r_t , and that these coincide with the true laws of motion of those variables. I therefore set the parameters of those laws of motion, and of equations 47 and 48, to match the output from an OLS estimation of those equations, using UK macroeconomic time-series data from 1993-2019. Third, I assume all households hold the same subjective models for y_t except for α_i , and I use a similar estimation of equation 49 to obtain the values of all parameters except α_i . The estimated coefficient on π_t in that equation is used as the mean α_i across households. The coefficients in the true law of motion for y_t are determined as functions of other calibrated variables, detailed in Appendix E.1.

Fourth, I assume that α_{i0} is normally distributed across households, and that all households share the same α_{i1} . The common subjective model for π_t implies they also share the same δ_i^* . I choose $\text{Var}_I(\alpha_{i0})$, α_1 , and δ^* to target three key moments from the IAS data: the average proportion

³⁶This approach relates to that of Michelacci and Paciello (2024), who note that ambiguity aversion naturally generates the negative correlation between preferences and expectations I observe for inflation. Similarly, in Ilut et al. (2020) firm worst-case beliefs depend on the direction of price changes.

of households who believe inflation makes the economy weaker, the elasticity of this proportion to increases in inflation, and an estimate of the average Kalman gain in inflation perceptions. I use this last moment rather than the estimated marginal effects in Table 1 as it is not clear how to position the boundaries of the ‘makes little difference’ survey answer in the model. Full details of the calibration are in Appendix E.3.

Note that the estimation of equations 47 - 49 used in this calibration, while naïve from the point of view of modern empirical macroeconomics, are not naïve from the households’ point of view. In their subjective models, these regressions uncover the intended underlying parameters. In this it is important that y_t does not appear in the law of motion for π_t .

7.2 The causal effects of inflation expectations in steady state

The first exercise I conduct with the calibrated temporary-information equilibrium model is to find the causal effects of an inflation expectation shock that arrives when the economy is in steady state. I obtain a stationary distribution of inflation perceptions by assuming that $\pi_t = \varepsilon_t = 0$ for many periods, so the only variation in $E_{it}\pi_t$ comes from idiosyncratic noise in household signals. This $E_{it}\pi_t$ distribution implies a stationary distribution of $\hat{\alpha}_{it}$.

Using this stationary distribution, and applying Proposition 7, we find the causal effects of inflation expectations on output in general equilibrium. As in Section 2, I use $\Omega^{GE}(\pi_t)$ to denote the causal effect of $\bar{E}_t\pi_t$, in this case on y_t , scaled such that the change in $\bar{E}_t\pi_t$ equals 1. The results are displayed in Table 3.

Table 3: The causal effect of aggregate inflation perceptions $\bar{E}_t\pi_t$ on output y_t : total and decomposition.

Total $\Omega^{GE}(\pi_t)$	Representative-agent channel	Narrative heterogeneity channel
-0.022	-0.014	-0.008

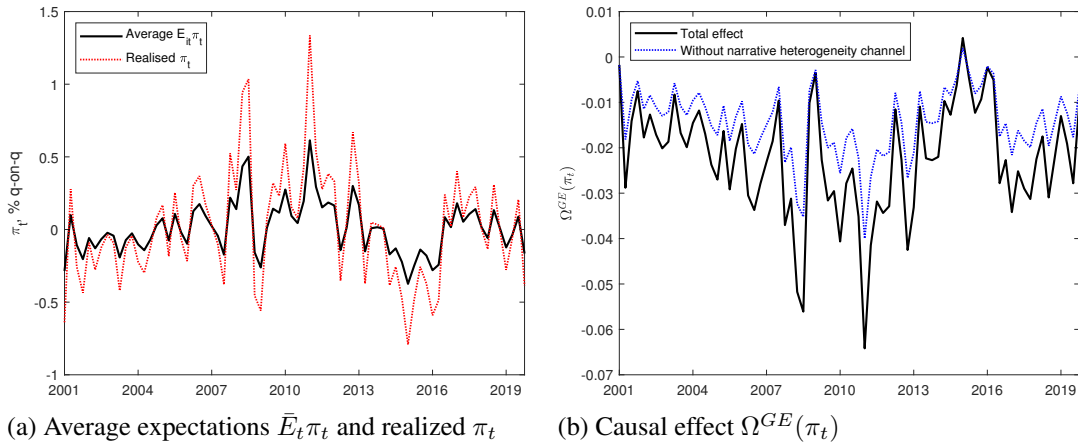
Greater inflation expectations therefore reduce aggregate output, contrary to the intuition popularly put forward for models with rational expectations (see e.g. Eggertsson and Woodford, 2003; Adam and Billi, 2007). This is because the majority of households believe inflation weakens the economy in the survey, in line with e.g. Hajdini et al. (2023). Since all variables here are in log-deviations, a 1% increase in $\bar{E}_t\pi_t$ causes a 0.02% fall in output.

As observed in the IAS data, there is a negative correlation between information τ_i and $\hat{\alpha}_i$, so the narrative heterogeneity channel is also negative. Quantitatively, it accounts for more than a third of the steady state effects of inflation expectations on output.

7.3 The causal effects of inflation expectations over time

Second, I simulate the model for 1000000 households, feeding in the path of de-meaned quarterly CPI inflation observed in the UK over the sample period as realizations of π_t . Figure 5 shows the paths of average perceived inflation and the causal effect of perceived current inflation on aggregate output in general equilibrium. Compared to realized inflation, average perceived inflation is relatively smooth, due to the presence of imperfect information. However, this still implies substantial volatility in the causal effects of expectations.

Figure 5: Simulated inflation perceptions and the causal effect $\bar{E}_t \pi_t$ on y_t . Calibration and simulation details are in Appendix E.3.



During periods of high inflation, the average $\hat{\alpha}_i$ is more negative, and thus the causal effect of expected inflation is more negative. Moreover, the decrease in $\hat{\alpha}_i$ is strongest among those whose expectations of inflation increase the most, that is the households with the most precise information. Increased inflation therefore implies the covariance between $\hat{\alpha}_i$ and τ_i becomes more negative, which means the narrative heterogeneity channel becomes more negative. In this way, the narrative heterogeneity channel is responsible for a large portion of the time-variation in the causal effects of expected inflation: the standard deviation of $\Omega^{GE}(\pi_t)$ over time is 63% larger than it would be in the absence of narrative heterogeneity effects (plotted in blue in Figure 5b).

8 Conclusion

This paper studies the causal effects of heterogeneously-formed expectations in general equilibrium. To do this I develop a new approach, which I label the *temporary-information equilibrium*. This builds on the core insight from the recent literature on information-provision survey experiments

in economics, that is that one can study the causal effect of expectations by creating exogenous variation in a part of agent information sets. I take this idea from the empirical literature and apply it to theoretical macroeconomic modeling. Specifically, this entails an extension of temporary equilibrium, where instead of taking all expectations as fixed and exogenous, we take only a noise term in agent signals as exogenous. Comparative statics with respect to this noise term yield a model equivalent of the first-stage regressions in information-provision experiments. That is, we create exogenous variation in expectations that can be used to identify their causal effects. This therefore retains the precise causal interpretation of temporary equilibrium analysis, without the requirement to hold all aspects of expectation formation fixed. This is how the method allows for cross-learning and general-equilibrium feedback into expectations, and for heterogeneity in the reception and response to information.

I show how to apply this approach to a general class of linear macroeconomic models, and to the specific cases of a Calvo pricing model, and a full New Keynesian model. In doing so, I find that the causal effects of expectations, and by implication the role of expectations in the transmission of fundamental shocks, depend in general on a novel *narrative heterogeneity channel*. This operates whenever information and subjective models covary systematically across agents. Heterogeneous subjective models imply heterogeneous responses to information, so systematic patterns in the distribution of information across agents with different subjective models distort the aggregate response to shocks. Moreover, many common assumptions about expectation formation popular in current literature generate transmission channels of this kind whenever there is non-zero heterogeneity in both information and subjective models.

In the Calvo model, the presence of this channel can have large quantitative and even qualitative effects on the conclusions from temporary-equilibrium analysis that necessarily assumes no expectations heterogeneity. For the New Keynesian model, I use unique features of the Bank of England Inflation Attitudes Survey to show that household subjective models and information about inflation covary systematically with each other, implying that the narrative heterogeneity channel will indeed be present in the causal effects of household inflation expectations. Specifically, when matching the model to the empirical patterns, the narrative heterogeneity channel substantially reduces the effects of inflation expectations on output, and accounts for more than half of the time-variation in that effect.

Finally, an effective use of temporary equilibrium has been to combine “theory with measured expectations” (Piazzesi and Schneider, 2016; Ludwig et al., 2024). The drawback of this approach is that the measured expectations are informative about the level of expectations, but not how they change in response to shocks, or how they affect individual behavior. Since temporary-

information equilibrium is conceptually related to information-provision experiments, it can be used to combine theory with measured expectation *changes*, and expectation *transmission*. I provide one example of this, combining the temporary-information equilibrium of the Calvo model with the RCT results in [Baumann et al. \(2024\)](#). More broadly, this method opens up the possibility of combining information-treatment experiments with macroeconomic theory to answer a wide range of macroeconomic questions.

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A Calvo model proofs

Proposition 1. Differentiating equation (11) with respect to the information shock yields

$$\frac{d\pi_t}{d\varepsilon_{\pi_t}} = (1 - \lambda)(1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_t+s\pi_t} \tau_{i\pi_t} + \chi_{imc_t+s\pi_t} \tau_{i\pi_t}) di \quad (59)$$

Aggregating equation (27) across firms and then differentiating yields

$$\frac{d\bar{E}_t(\pi_t)}{d\varepsilon_{\pi_t}} = \int_i \tau_{i\pi_t} di \equiv \bar{\tau}_{\pi_t} \quad (60)$$

where we have used the fact that the diagonal elements of χ_i are equal to 1 by construction.

Since the model is linear, scaling the shock by $\bar{\tau}_{\pi_t}^{-1}$ implies a unit change in $\bar{E}_t(\pi_t)$, which then generates equation (12).

Proposition 2. The proof follows the same steps as that for Proposition 1. Differentiating equation (11) with respect to the information shock yields

$$\frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = (1 - \lambda)(1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_t+s\pi_{t+1}} \tau_{i\pi_{t+1}} + \chi_{imc_t+s\pi_{t+1}} \tau_{i\pi_{t+1}}) di \quad (61)$$

Aggregating equation (9) across firms and then differentiating yields

$$\frac{d\bar{E}_t(\pi_{t+1})}{d\varepsilon_{\pi_{t+1}}} = \int_i \tau_{i\pi_{t+1}} di \equiv \bar{\tau}_{\pi_{t+1}} \quad (62)$$

Since the model is linear, scaling the shock by $\bar{\tau}_{\pi_{t+1}}^{-1}$ implies a unit change in $\bar{E}_t(\pi_{t+1})$, which then generates equation (13).

GE effects of $\varepsilon_{\pi t}$. With the assumptions on signals in equation (15), equation (11) can be written

$$\pi_t = (1 - \lambda)(1 - \beta\lambda) \left(\sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_t} + \chi_{imc_{t+s}\pi_t}) \tau_{i\pi_t} di \right) (\pi_t + \varepsilon_{\pi_t}) + a \quad (63)$$

where a collects the steady-state markup μ and all terms involving signals of mc_{t+s} , and π_{t+s} for $s \geq 1$. Differentiating with respect to $\varepsilon_{\pi t}$ yields

$$\frac{d\pi_t}{d\varepsilon_{\pi_t}} = (1 - \lambda)(1 - \beta\lambda) \left(\sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_t} + \chi_{imc_{t+s}\pi_t}) \tau_{i\pi_t} di \right) \left(\frac{d\pi_t}{d\varepsilon_{\pi_t}} + 1 \right) \quad (64)$$

which rearranges to

$$\frac{d\pi_t}{d\varepsilon_{\pi_t}} = \frac{\bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \quad (65)$$

where

$$\Omega^{PE}(\pi_t) \equiv (1 - \lambda)(1 - \beta\lambda) \left(\sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_t} + \chi_{imc_{t+s}\pi_t}) \frac{\tau_{i\pi_t}}{\bar{\tau}_{\pi_t}} di \right) \quad (66)$$

denotes the (scaled) partial-equilibrium effect of π_t expectations, as in equation (12).

Aggregating equation (27) across firms and then differentiating yields

$$\frac{d\bar{E}_t(\pi_t)}{d\varepsilon_{\pi_t}} = \bar{\tau}_{\pi_t} \left(\frac{d\pi_t}{d\varepsilon_{\pi_t}} + 1 \right) \quad (67)$$

$$= \bar{\tau}_{\pi_t} \left(\frac{\bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} + 1 \right) \quad (68)$$

$$= \frac{\bar{\tau}_{\pi_t}}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \quad (69)$$

where the second equality uses equation (65). To ensure the impact on $\bar{E}_t(\pi_t) = 1$, we therefore scale the shock by $\frac{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{\bar{\tau}_{\pi_t}}$. This implies

$$\Omega^{GE}(\pi_t) = \frac{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = \Omega^{PE}(\pi_t) \quad (70)$$

After scaling, general equilibrium effects of expected current inflation are irrelevant.

Proposition 3. The proof follows the same steps as for the GE effects of expected π_t above. Equation (11) can be written

$$\begin{aligned} \pi_t = & (1 - \lambda)(1 - \beta\lambda) \left(\sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_t} + \chi_{imc_{t+s}\pi_t}) \tau_{i\pi_t} di \right) (\pi_t + \varepsilon_{\pi_t}) \\ & + (1 - \lambda)(1 - \beta\lambda) \left(\sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_{t+1}} + \chi_{imc_{t+s}\pi_{t+1}}) \tau_{i\pi_{t+1}} di \right) (\pi_{t+1} + \varepsilon_{\pi_{t+1}}) + a' \end{aligned} \quad (71)$$

where a' collects the steady-state markup μ and all terms involving signals of mc_{t+s} , and π_{t+s} for $s \geq 2$. Differentiating with respect to $\varepsilon_{\pi_{t+1}}$ yields

$$\frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t) \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} + \bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1}) \quad (72)$$

where we assume that the change in expectations does not imply any change in realized π_{t+1} . In other words, by the time period $t + 1$ arrives, the firms realize that the change in $s_{it}(\pi_{t+1})$ they observed was noise, so $E_{it+1}(\pi_{t+1})$ return to steady state. Rearranging, we obtain

$$\frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = \frac{\bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1})}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \quad (73)$$

Aggregating equation (27) across firms and then differentiating yields

$$\frac{d\bar{E}_t(\pi_{t+1})}{d\varepsilon_{\pi_{t+1}}} = \bar{\tau}_{\pi_{t+1}} + \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} \int_i \chi_{i\pi_{t+1}\pi_t} \tau_{i\pi_t} di \quad (74)$$

$$= \bar{\tau}_{\pi_{t+1}} + \frac{\bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1})}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \int_i \chi_{i\pi_{t+1}\pi_t} \tau_{i\pi_t} di \quad (75)$$

This is more complicated than the case of a shock to ε_{π_t} because the general equilibrium feedback into aggregate $\bar{E}_t(\pi_{t+1})$ occurs through cross-learning from changes in realized π_t , and thus in $s_{it}(\pi_t)$. Define the constant K as

$$K \equiv \left(\bar{\tau}_{\pi_{t+1}} + \frac{\bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1})}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \int_i \chi_{i\pi_{t+1}\pi_t} \tau_{i\pi_t} di \right)^{-1} \quad (76)$$

$$= \frac{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{\bar{\tau}_{\pi_{t+1}} (1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)) + \bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1}) (\bar{\chi}_{\pi_{t+1}\pi_t} \bar{\tau}_{\pi_t} + Cov_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}])} \quad (77)$$

The scaled aggregate effect of the shock is therefore

$$K \cdot \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = \frac{\bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1})}{\bar{\tau}_{\pi_{t+1}} (1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)) + \bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1}) (\bar{\chi}_{\pi_{t+1}\pi_t} \bar{\tau}_{\pi_t} + Cov_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}])} \quad (78)$$

which after dividing through by $\bar{\tau}_{\pi_{t+1}}$ becomes equation (16).

Proposition 4. This follows from equations (20) and (21). Using these, the fraction of transmission due to expectations is

$$\frac{\left. \frac{d\pi_t}{dmc_t} - \frac{d\pi_t}{dmc_t} \right|_{\sim E}}{\frac{d\pi_t}{dmc_t}} = 1 - \frac{\left. \frac{d\pi_t}{dmc_t} \right|_{\sim E}}{\frac{d\pi_t}{dmc_t}} = 1 - (1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)) \quad (79)$$

which implies equation (22).

Proposition 5. The parameters in equation (23) have been chosen so that the representative-agent effects of inflation expectations (line 1 in equation (14)) are equal to those in equation (6). Since $\chi_{imc_{t+s}\pi_t}$ is set to 0, it only remains to find

$$Cov_I[\chi_{i\pi_{t+s}\pi_t}, \tau_{i\pi_t}] = E_I[\chi_{i\pi_{t+s}\pi_t} \tau_{i\pi_t}] - \bar{\chi}_{\pi_{t+s}\pi_t} \bar{\tau}_{\pi_t} \quad (80)$$

$$= E_I[\chi_{i\pi_{t+s}\pi_t} | \tau_{i\pi_t} = 1] \Pr(\tau_{i\pi_t} = 1) - \frac{1+s}{2} \quad (81)$$

$$= \frac{((1+2s)q + 1 - q)}{2} - \frac{1+s}{2} \quad (82)$$

$$= \frac{s}{2}(2q - 1) \quad (83)$$

where $E_I[\cdot]$ denotes a cross-sectional expectation across firms, and $q \equiv \Pr(\chi_{i\pi_{t+s}\pi_t} = 1 + 2s | \tau_{i\pi_t} = 1)$. With this, the narrative heterogeneity effects (line 2 in equation (14)) are

$$\frac{(1-\lambda)(1-\beta\lambda)}{\frac{1}{2}} \sum_{s=0}^{\infty} (\beta\lambda)^s Cov_I[\chi_{i\pi_{t+s}\pi_t}, \tau_{i\pi_t}] = (1-\lambda)(1-\beta\lambda)(2q-1) \sum_{s=0}^{\infty} (\beta\lambda)^s s \quad (84)$$

$$= \frac{\beta\lambda(1-\lambda)(2q-1)}{1-\beta\lambda} \quad (85)$$

Adding this together with the representative-agent effects from equation (6) yields

$$\frac{1}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = \frac{1-\lambda}{1-\beta\lambda} + \frac{\beta\lambda(1-\lambda)(2q-1)}{1-\beta\lambda} = \frac{1-\lambda}{1-\beta\lambda} \cdot (1 + \beta\lambda(2q-1)) \quad (86)$$

and taking $\beta \rightarrow 1$ then generates equation (24).

Proposition 6. The proof follows the same steps as for Proposition 1. The k th row of equation (28) can be written as

$$x_{ikt} = \sum_{n=0}^{N_z} A_{ikn} x_{int-1} + \sum_{n=0}^{N_z} \sum_{m=0}^{N_z} B_{ikm} \hat{\chi}_{imn} \hat{\tau}_{in} s_{int} \quad (87)$$

where A_{ikn} denotes the k, n th element of A_i , and B_{ikm} denotes the k, m th element of B_i . $\hat{\chi}_{imn}$ and $\hat{\tau}_{in}$ are the m, n th element of $\hat{\chi}_i$ and the n th diagonal element of $\hat{\tau}_i$ respectively.

Aggregating across agents i and differentiating with respect to ε_{jt} we obtain

$$\frac{dx_{kt}}{d\varepsilon_{jt}} = \int_i \left(\sum_{m=0}^{N_z} B_{ikm} \hat{\chi}_{imj} \hat{\tau}_{ij} \right) di \quad (88)$$

$$= \sum_{m=0}^{N_z} (\bar{B}_{km} \bar{\chi}_{mj} \bar{\tau}_j + Cov_I [B_{ikm}, \chi_{imj} \tau_{ij}] + \bar{B}_{km} Cov_I [\chi_{imj}, \tau_{ij}]) \quad (89)$$

where the second equality uses the definition of a covariance.

Similarly, aggregating equation (27) across agents and differentiating yields

$$\frac{d\bar{E}_t(z_{jt})}{d\varepsilon_{jt}} = \int_i \tau_{ij} di \equiv \bar{\tau}_j \quad (90)$$

where we have used the fact that diagonal elements of $\hat{\chi}_i$ are equal to 1 by construction.

Since the model is linear, scaling the shock by $\bar{\tau}_j^{-1}$ implies a unit change in $\bar{E}_t(z_{jt})$, which then implies equation (31).

Proposition 7. Aggregating equation (28) across agents, we can express the aggregate policy function as

$$\mathbf{x}_t = \int_i A_i \mathbf{x}_{it-1} di + \sum_{s=0}^{\infty} [\Omega_{xs}^{PE} \bar{\tau}_{xs} (\mathbf{x}_{t+s} + \varepsilon_{xt+s}) + \Omega_{ws}^{PE} \bar{\tau}_{ws} (\mathbf{w}_{t+s} + \varepsilon_{wt+s})] \quad (91)$$

where Ω_{xs}^{PE} is a $N_x \times N_x$ matrix collecting the partial-equilibrium causal effects of expectations about each element of the future aggregate choice vector $\mathbf{x}_{t+s}$ on each aggregate choice (i.e. each element of $\mathbf{x}_t$), as defined in Proposition 6. Similarly, Ω_{ws}^{PE} is a $N_x \times N_w$ matrix collecting the partial-equilibrium causal effects of expectations about each element of the future state vector $\mathbf{w}_{t+s}$

on each aggregate choice. $\bar{\tau}_{xs}$ and $\bar{\tau}_{ws}$ are square diagonal matrices with diagonals equal to the average $\hat{\tau}_{ij}$ parameters across agents for signals on $\mathbf{x}_{t+s}$ and $\mathbf{w}_{t+s}$ respectively. Finally, $\boldsymbol{\varepsilon}_{xt+s}$ and $\boldsymbol{\varepsilon}_{wt+s}$ are vectors collecting the public component of noise in signals on $\mathbf{x}_{t+s}$ and $\mathbf{w}_{t+s}$. Note that the idiosyncratic noise terms in equation (30) cancel out when we aggregate across agents.

Rearranging equation (29) we obtain

$$\mathbf{w}_t = -D^{-1}C\mathbf{x}_t - D^{-1}F\boldsymbol{\xi}_t \quad (92)$$

Substituting (92) in to (91) and rearranging yields

$$\begin{aligned} \mathbf{x}_t = (I - \Lambda_0)^{-1} & \left(\int_i A_i \mathbf{x}_{it-1} di + \sum_{s=1}^{\infty} [\Lambda_s \mathbf{x}_{t+s}] \right. \\ & \left. + \sum_{s=0}^{\infty} [\Omega_{xs}^{PE} \bar{\tau}_{xs} \boldsymbol{\varepsilon}_{xt+s} + \Omega_{ws}^{PE} \bar{\tau}_{ws} (\boldsymbol{\varepsilon}_{wt+s} - D^{-1}F\boldsymbol{\xi}_{t+s})] \right) \end{aligned} \quad (93)$$

where

$$\Lambda_s \equiv \Omega_{xs}^{PE} \bar{\tau}_{xs} - \Omega_{ws}^{PE} \bar{\tau}_{ws} D^{-1}C \quad (94)$$

Differentiating with respect to $\boldsymbol{\varepsilon}_{xt+h}$ yields

$$\frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} = (I - \Lambda_0)^{-1} \left(\Omega_{xh}^{PE} \bar{\tau}_{xh} + \sum_{s=1}^{\infty} \left[\Lambda_s \frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\varepsilon}_{xt+h}} \right] \right) \quad (95)$$

Next, we turn to $\frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\varepsilon}_{xt+h}}$. For this, note that the terms relating $\mathbf{x}_t$ to $\mathbf{x}_{it-1}$ in equation (91) can be written as

$$\int_i A_i \mathbf{x}_{it-1} di = \tilde{A} \mathbf{x}_{t-1} \quad (96)$$

where

$$\tilde{A} \equiv \int_i A_i di + Cov_I[A_i, \mathbf{x}_{it-1}] \mathbf{x}_{t-1}^+ \quad (97)$$

and $\mathbf{x}_{t-1}^+$ is the Moore-Penrose pseudoinverse of $\mathbf{x}_{t-1}$, such that $\mathbf{x}_{t-1}^+ \mathbf{x}_{t-1} = 1$.

To proceed, we will make the simplifying assumption that $Cov_I[A_i, \mathbf{x}_{it-1}] \mathbf{x}_{t-1}^+$ is constant over time. That is, the correlation between an agent's weight on lagged actions in their policy rule and

their deviation from the aggregate actions may be non-zero, but it is unaffected by shocks. With this simplification, $\tilde{A}$ is constant.

Let the matrix R be defined by the recursion

$$R = (I - \Lambda_0)^{-1} \left(\tilde{A} + \sum_{k=1}^{\infty} \Lambda_k R^{k+1} \right) \quad (98)$$

Given this, we can show the following.

Lemma 1 *If the series $\sum_{k=1}^{\infty} \Lambda_k R^{k+1}$ converges, and if the spectral radius of R is $\rho(R) < 1$, then*

$$\frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\varepsilon}_{xt+h}} = R^s \frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} \quad (99)$$

Proof. We prove equation (99) with a guess-and-verify approach. Rolling equation (93) forwards and differentiating, we have

$$\frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\varepsilon}_{xt+h}} = (I - \Lambda_0)^{-1} \left(\tilde{A} \frac{d\mathbf{x}_{t+s-1}}{d\boldsymbol{\varepsilon}_{xt+h}} + \sum_{k=1}^{\infty} \left[\Lambda_k \frac{d\mathbf{x}_{t+s+k}}{d\boldsymbol{\varepsilon}_{xt+h}} \right] \right) \quad (100)$$

Substituting in equation (99) this becomes

$$\frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\varepsilon}_{xt+h}} = (I - \Lambda_0)^{-1} \left(\tilde{A} R^{s-1} \frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} + \sum_{k=1}^{\infty} \left[\Lambda_k R^{s+k} \frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} \right] \right) \quad (101)$$

$$= (I - \Lambda_0)^{-1} \left(\tilde{A} + \sum_{k=1}^{\infty} [\Lambda_k R^{k+1}] \right) R^{s-1} \frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} \quad (102)$$

which when combined with (98) verifies the guess.

The requirement that $\sum_{k=1}^{\infty} \Lambda_k R^{k+1}$ converges in some consistent matrix norm ensures that R is a well-defined finite matrix. $\rho(R) < 1$ ensures uniqueness of the bounded solution in (99), ruling out explosive alternatives. We assume these conditions hold, in analogy with the determinacy conditions in rational-expectations models (existence and uniqueness of a bounded solution). ■

Substituting equation (99) into equation (95) and rearranging yields

$$\frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} = \left(I - \Lambda_0 - \sum_{s=1}^{\infty} [\Lambda_s R^s] \right)^{-1} \Omega_{xh}^{PE} \bar{\tau}_{xh} \quad (103)$$

Applying the same steps to a shock to information about state variables yields

$$\frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{wt+h}} = \left(I - \boldsymbol{\Lambda}_0 - \sum_{s=1}^{\infty} [\boldsymbol{\Lambda}_s R^s] \right)^{-1} \boldsymbol{\Omega}_{wh}^{PE} \bar{\boldsymbol{\tau}}_{wh} \quad (104)$$

Defining the $N_x \times N_z$ matrix $\boldsymbol{\Omega}^{PE} = (\boldsymbol{\Omega}_{x0}^{PE}, \boldsymbol{\Omega}_{w0}^{PE}, \boldsymbol{\Omega}_{x1}^{PE}, \boldsymbol{\Omega}_{w1}^{PE}, \dots)$, we can combine these into one expression:

$$\frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_t} = \boldsymbol{\Lambda}^{GE} \boldsymbol{\Omega}^{PE} \bar{\boldsymbol{\tau}} \quad (105)$$

where

$$\boldsymbol{\Lambda}^{GE} \equiv \left(I - \boldsymbol{\Lambda}_0 - \sum_{s=1}^{\infty} [\boldsymbol{\Lambda}_s R^s] \right)^{-1} \quad (106)$$

To extract the causal effect of the j th element of $\boldsymbol{\varepsilon}_t$ on the k th aggregate action (x_{kt}), we simply take the k, j th element of the matrix $\frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_t}$. Since $\bar{\boldsymbol{\tau}}$ is diagonal, this implies

$$\frac{dx_{kt}}{d\varepsilon_{jt}} = [\boldsymbol{\Lambda}^{GE} \boldsymbol{\Omega}^{PE}]_{k,j} \bar{\tau}_j \quad (107)$$

as in equation (34).

All that remains is to scale this so the impact on the relevant aggregate expectation is equal to 1. The j th row of equation (27) is

$$E_{it}(z_{jt}) = \sum_{n=0}^{N_z} \hat{\chi}_{ijn} \hat{\tau}_{in} (z_{nt} + \tilde{\varepsilon}_{int} + \varepsilon_{nt}) \quad (108)$$

Aggregating across agents and differentiating with respect to ε_{jt} we find

$$\frac{\bar{E}_t(z_{jt})}{d\varepsilon_{jt}} = \bar{\tau}_j + \Gamma_j \quad (109)$$

where

$$\Gamma_j \equiv \sum_{n=0}^{N_z} [\bar{\chi}_{jn} \bar{\tau}_n + Cov_I[\hat{\chi}_{ijn}, \hat{\tau}_{in}]] \frac{dz_{nt}}{d\varepsilon_{jt}} \quad (110)$$

$$= \sum_{n=0}^{N_z} [\bar{\chi}_{jn} \bar{\tau}_n + Cov_I[\hat{\chi}_{ijn}, \hat{\tau}_{in}]] \cdot \left[\begin{pmatrix} I \\ -D^{-1}C \end{pmatrix} \Lambda^{GE} \Omega^{PE} \right]_{j,n} \bar{\tau}_n \quad (111)$$

The second equality here comes from using equations (105) and (92) to obtain $\frac{dw_t}{d\varepsilon_{jt}}$ in terms of $\frac{dx_t}{d\varepsilon_{jt}}$, and using this to substitute out for $\frac{dz_{nt}}{d\varepsilon_{jt}}$.

To ensure a unit change in $\bar{E}_t(z_{jt})$, the un-scaled effect $\frac{dx_{kt}}{d\varepsilon_{jt}}$ must be multiplied by $K = (\bar{\tau}_j + \Gamma_j)^{-1}$. Combined with equation (107) this implies equation (35).

Corollary 1 *If there are no signals about any variables realized after the current period, the general equilibrium feedback from an exogenous change in expectations of a contemporaneous variable is determined by the matrix*

$$\Lambda^{GE} = (I - \Omega_{x0}^{PE} \bar{\tau}_{x0} + \Omega_{w0}^{PE} \bar{\tau}_{w0} D^{-1} C)^{-1} \quad (112)$$

where Ω_{x0}^{PE} and Ω_{w0}^{PE} collect the partial-equilibrium effects of each expectation $\bar{E}_t(z_{jt})$ for $z_{jt} \in \{x_t, w_t\}$, and $\bar{\tau}_{x0}, \bar{\tau}_{w0}$ are matrices collecting the corresponding elements of average τ_i .

Proof. If there is no forward information, then $\bar{\tau}_{xs}$ and $\bar{\tau}_{ws}$ are zero matrices for all $s \geq 1$. Equation (94) therefore implies Λ_s is a zero matrix for all $s \geq 1$. Equation (106) then becomes

$$\Lambda^{GE} = (I - \Lambda_0)^{-1} = (I - \Omega_{x0}^{PE} \bar{\tau}_{x0} + \Omega_{w0}^{PE} \bar{\tau}_{w0} D^{-1} C)^{-1} \quad (113)$$

where the second equality uses equation (94). ■

From this, we can see that an exogenous change in $\bar{E}_t(z_{jt})$ causes two related general equilibrium feedback effects. First, as in the Calvo model in Section 2, the shock causes a change in aggregate actions x_t , which feeds back into the information agents receive.

Second, this change in x_t also affects the state variables w_t through the equilibrium condition (29), which *also* affects agent information. This was not present in Section 2 because we abstracted there from all responses of the state (marginal costs) to firm actions.

Finally, note that outside of the special case in Corollary 1, there is a further category of effects, as changes in $\bar{E}_t(z_{jt})$ affect future actions and states through the lagged-action term $A_i x_{it-1}$ in the policy function (28). If agents get signals about future aggregate actions or states, those signals will also therefore react to the change in $\bar{E}_t(z_{jt})$.

Proposition 8. Differentiating equation (93) with respect to the fundamental shock vector ξ_t yields

$$\frac{d\mathbf{x}_t}{d\xi_t} = (I - \Lambda_0)^{-1} \left(-\Omega_{w0}^{PE} \bar{\tau}_{w0} D^{-1} F + \sum_{s=1}^{\infty} \left[\Lambda_s \frac{d\mathbf{x}_{t+s}}{d\xi_t} \right] \right) \quad (114)$$

Following analogous reasoning to that in Lemma 1 we can substitute out for $\frac{d\mathbf{x}_{t+s}}{d\xi_t} = R^s \frac{d\mathbf{x}_t}{d\xi_t}$, where R is defined in (98). Rearranging the resulting expression implies equation (36).

B Narrative heterogeneity in common models of expectation formation

I focus here on the model in Section 2 under the simplifying assumptions that $\tau_{ij} = 0$ for all variables except π_t , and $\varepsilon_{\pi_t} = 0$, $\tilde{\varepsilon}_{i\pi_t} \sim N(0, \sigma_i^2)$, $\pi_t \sim N(0, \sigma_\pi^2)$. The reset price of firm i is then given by

$$p_{it}^* - P_{t-1} = (1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s [\chi_{i\pi_{t+s}\pi_t} + \chi_{imc_{t+s}\pi_t}] \tau_{i\pi_t} (\pi_t + \tilde{\varepsilon}_{i\pi_t}) \quad (115)$$

It will be helpful here to define $\tilde{\chi}_i$ as the sum of cross-learning parameters as they enter the reset price first order condition:

$$\tilde{\chi}_i \equiv \sum_{s=0}^{\infty} (\beta\lambda)^s [\chi_{i\pi_{t+s}\pi_t} + \chi_{imc_{t+s}\pi_t}] \quad (116)$$

I also assume that firms incorporate their signals into expectations of π_t optimally. Standard Bayesian updating therefore implies

$$\tau_{i\pi_t} = \frac{\sigma_\pi^2}{\sigma_\pi^2 + \sigma_i^2} \quad (117)$$

Rational inattention. Following Sims (2003), models of rational inattention assume that agents choose the precision of their signals, subject to a cost (see Maćkowiak et al., 2023, for a review).

To study this in our Calvo model, we make two assumptions. First, a resetting firm i 's expected payoff is proportional to $-E_{it-1}(p_{it}^* - p_{it}^{FI})^2$, where p_{it}^{FI} is the reset price that they would choose if

they precisely observed π_t .³⁷ Second, before the signal $s_{it}(\pi_t)$ is observed, each firm can choose the variance of $\tilde{\varepsilon}_{i\pi_t}$, subject to a cost that is proportional to the expected entropy reduction between their prior and the resulting posterior beliefs about π_t .³⁸

Since there is a one-to-one mapping between σ_i^2 and $\tau_{i\pi_t}$ (equation (117)), we can formulate the information-choice problem as

$$\arg \max_{\tau_{i\pi_t} \in [0,1]} -E_{it-1}(p_{it}^* - p_{it}^{FI})^2 - \mu \log \left(\frac{1}{1 - \tau_{i\pi_t}} \right) \quad (118)$$

where $\mu > 0$ is the marginal cost of information.

Substituting out for p_{it}^* and p_{it}^{FI} using (115) and evaluating the expectations, the problem becomes

$$\arg \max_{\tau_{i\pi_t} \in [0,1]} -(1 - \beta\lambda)\tilde{\chi}_i^2 ((\tau_{i\pi_t} - 1)^2 \sigma_\pi^2 + \tau_{i\pi_t}^2 \sigma_i^2) - \mu \log \left(\frac{1}{1 - \tau_{i\pi_t}} \right) \quad (119)$$

which after substituting out for σ_i^2 using (117) and differentiating gives the first order condition:

$$\tau_{i\pi_t} = \max \left\{ 0, 1 - \frac{\mu}{(1 - \beta\lambda)\sigma_{\pi_t}^2 \tilde{\chi}_i^2} \right\} \quad (120)$$

This clearly generates correlations between $\tau_{i\pi_t}$ and $\chi_{i\pi_t+s\pi_t}$, $\chi_{imc_t+s\pi_t}$, as long as some firms choose to process strictly positive amounts of information ($\tau_{i\pi_t} > 0$). Intuitively, firms who believe there is a stronger pass-through from current inflation to future inflation and to marginal costs (larger $\tilde{\chi}_i^2 = [\sum_{s=0}^{\infty} (\beta\lambda)^s [\chi_{i\pi_t+s\pi_t} + \chi_{imc_t+s\pi_t}]]^2$) find information more valuable, so process more of it – implying a $\tau_{i\pi_t}$ closer to 1.

Learning. Models of rational inattention endogenize τ_i . In contrast, models of learning make subjective models endogenous: agents estimate the coefficients of aggregate laws of motion from their observations (Evans and McGough, 2020). This also generates a narrative heterogeneity effect, because agents learning from systematically different information estimate different laws of motion.

To incorporate this in a simple way into the Calvo framework used here, assume that all firms have perceived laws of motion that specify there is no link between π_t and future π_{t+s} or mc_{t+s} for

³⁷An objective of this form can be obtained by taking a log-quadratic approximation to the profit function (Maćkowiak and Wiederholt, 2009).

³⁸Since the firm's objective function is quadratic, and priors are Gaussian, the signal structure in equation (??) is in fact optimal (Maćkowiak and Wiederholt, 2009).

$s \geq 1$, that $\pi_t \sim N(\mu_\pi, \sigma_\pi^2)$, and that current mc_t is perfectly correlated with π_t :

$$\pi_t = \gamma mc_t \quad (121)$$

These assumptions imply that $\tilde{\chi}_i$ is equal to the firm's belief about the parameter γ .

Furthermore, suppose that at the end of period t (i.e. after resetting firms have chosen p_{it}) firms observe the realized mc_t . They can therefore use the combination of mc_t and $s_{it}(\pi_t)$ to learn about the true slope γ .

There are a range of possible learning technologies present in the literature. I proceed here with a tractable application of Bayesian learning (reviewed in ?). Using their observation of mc_t and their inflation signal from period t , firm i can construct an unbiased signal about γ^{-1} :

$$\frac{s_{it}(\pi_t)}{mc_t} = \frac{1}{\gamma} + \frac{\epsilon_{i\pi_t}}{mc_t} \quad (122)$$

To keep the learning problem simple for this example, suppose that firms start period t with a Gaussian prior belief about γ^{-1} , with mean $\mu_{\gamma it}$ and variance $\sigma_{\gamma it}^2$. Learning about the inverse coefficient γ^{-1} is then straightforward: Bayesian updating implies that posterior beliefs about γ^{-1} at the end of period t are:

$$E(\gamma^{-1} | s_{it}(\pi_t), mc_t) = \mu_{\gamma it} + K_{it} \left(\frac{1}{\gamma} + \frac{\epsilon_{i\pi_t}}{mc_t} - \mu_{\gamma it} \right) \quad (123)$$

where the gain K_{it} is given by:

$$K_{it} = \frac{mc_t^2 \sigma_{\gamma it}^2}{mc_t^2 \sigma_{\gamma it}^2 + \sigma_\pi^2 \left(\frac{1 - \tau_{i\pi_t}}{\tau_{i\pi_t}} \right)} \quad (124)$$

This gain is strictly increasing in $\tau_{i\pi_t}$. Firms with more precise information update their subjective models more strongly in response to it. At the extremes, a firm with $\tau_{i\pi_t} = 1$ is able to exactly back out the realized γ , so $K_{it} = 1$ and their posterior belief matches the true correlation between π_t and mc_t . A firm with $\tau_{i\pi_t} \rightarrow 0$ has $K_{it} \rightarrow 0$: their signal is uninformative, so they never update their subjective model. If period- t posteriors persist to influence the prior beliefs used in pricing decisions in period $t + 1$, this difference in learning speeds will create a narrative heterogeneity channel.

We can see the effects of this more concretely if we impose further structure. Assume that in an initial period 0, all firms share the same priors, with mean μ_0 and variance $\sigma_{\gamma 0}^2$. Firms are split

equally between $\tau_{i\pi_t} = \{0, 0.5, 1\}$. Finally, assume all firms believe γ is constant, so prior beliefs in period 1 are centered on their period-0 posteriors.

In period 0, resetting firms choose prices before observing mc_t , so use their prior beliefs on γ_0^{-1} .³⁹ Since there is no heterogeneity in these beliefs, the narrative heterogeneity term is equal to 0.

After mc_0 is realized, firms update their subjective models using equation (123), which yields:

$$E_i(\gamma^{-1}|s_{i0}(\pi_t), mc_0) = \begin{cases} \mu_0 & \text{if } \tau_{i\pi_t} = 0 \\ \mu_0 + \frac{mc_t^2 \sigma_{\gamma_0}^2}{mc_t^2 \sigma_{\gamma_0}^2 + \sigma_{\pi}^2} \left(\frac{1}{\gamma} + \frac{\epsilon_{i\pi_t}}{mc_0} - \mu_0 \right) & \text{if } \tau_{i\pi_t} = 0.5 \\ \gamma^{-1} & \text{if } \tau_{i\pi_t} = 1 \end{cases} \quad (125)$$

In period 1, resetting firms choose their price using $\tilde{\chi}_{i1} = [E_i(\gamma^{-1}|s_{i0}(\pi_t), mc_0)]^{-1}$ as their subjective model. The narrative heterogeneity channel is then:⁴⁰

$$Cov_I[\tilde{\chi}_{i1}, \tau_{i\pi_t}] = \frac{1}{6}(\gamma - \mu_0^{-1}) \quad (126)$$

If prior beliefs in period 0 are below the true γ , then in period 1 the narrative heterogeneity channel is positive. Firms with precise information update their beliefs about γ upwards, while those with imprecise information do not, so more precise information is associated with higher $\tilde{\chi}_{i1}$. As discussed in the rational inattention model, this amplifies the effect of productivity shocks on aggregate investment relative to a model with no heterogeneity. If period-0 priors are above the true γ , the narrative heterogeneity channel has the opposite sign. The only way for the narrative heterogeneity channel to disappear is if period-0 priors are centred on γ .

Sticky information. Following [Mankiw and Reis \(2002\)](#), firms observe information only sporadically. With probability $q_i \in (0, 1)$, firm i observes π_t precisely, which through equation (117) implies $\tau_{i\pi_t} = 1$. With probability $1 - q_i$, firm i observes no information at all ($\tau_i = 0$). As in [Reis \(2006\)](#) and others, firms can pay a cost to increase q_i .⁴¹

Applying the same steps as used to obtain equation (119), expected utility net of the increasing,

³⁹In all periods, I assume that firms make pricing decisions as if they were certain of their estimate of γ , following the standard ‘anticipated utility’ assumption in the learning literature ([Bullard and Suda, 2016](#)).

⁴⁰Calculating this requires taking an expectation over the inverse of the normally distributed random variable $E_i(\gamma^{-1}|s_{i0}(\pi_t), mc_0, \tau_{i\pi_t} = 0.5)$. Since the expectation of the inverse of a normally distributed variable does not exist in the conventional sense, I use the corresponding Cauchy principal value ([Kanwal, 1996](#)). This term cancels out of the resulting expression, but the step is required to ensure that $Cov_I[\tilde{\chi}_{i1}, \tau_{i\pi_t}]$ is well-defined.

⁴¹Specifically, [Reis \(2006\)](#) assumes firms can pay a cost to observe information, which is equivalent to paying to increase q_i to 1.

convex cost of increasing q_i ($C(q_i)$) is:

$$-(1 - \beta\lambda)\tilde{\chi}_i^2\sigma_\pi^2(1 - q_i) - C(q_i) \quad (127)$$

The firm chooses q_i to maximize this objective. The optimal q_i satisfies the first order condition:

$$(1 - \beta\lambda)\tilde{\chi}_i^2\sigma_\pi^2 = C'(q_i) \quad (128)$$

which implies q_i is strictly increasing in $\tilde{\chi}_i^2 = [\sum_{s=0}^{\infty}(\beta\lambda)^s[\chi_{i\pi_t+s\pi_t} + \chi_{imc_t+s\pi_t}]]^2$. The intuition is the same as the rational inattention model: information is more valuable to firms who believe that current inflation has larger effects on future inflation and marginal costs.

By the law of iterated expectations, each narrative heterogeneity covariance can be expressed as:

$$Cov(\chi_{ij\pi_t}, \tau_{i\pi_t}) = E(E(\chi_{ij\pi_t}\tau_{i\pi_t}|q_i)) - \bar{\chi}_{j\pi_t}E(E(\tau_{i\pi_t}|q_i)) \quad (129)$$

$$= E(\chi_{ij\pi_t}q_i) - \bar{\chi}_{j\pi_t}E(q_i) \quad (130)$$

$$= Cov(\chi_{ij\pi_t}, q_i) \quad (131)$$

That is, the narrative heterogeneity channel appears whenever the probability of observing information is correlated with a firm's subjective model. Equation (128) implies that this is indeed the case. In the case where $\chi_{ij\pi_t} > 0$ for all i , then $Cov_I(\chi_{ij\pi_t}, q_i) > 0$.

Information delegation to media outlets. As in [Nimark \(2014\)](#), [Chahrour et al. \(2021\)](#), and others, assume that firms receive their signals from media outlets, who perform an editorial role. Media outlet m observes π_t precisely, and chooses whether or not to report it to their readers depending on its realization.

As is common in the literature, I focus on editorial decision rules in which large shocks are deemed more newsworthy. Media outlet m reports on π_t if $|\pi_t| > \theta_m$, where $\theta_m > 0$ is an outlet-specific threshold. If outlet m reports on π_t , any firm that reads outlet m observes π_t precisely ($\tau_{i\pi_t} = 1$). Using the same quadratic objective as in the rational inattention case, the firm has a payoff of 0. If m does not report π_t , readers do not observe any information about π_t ($\tau_{i\pi_t} = 0$). Instead of π_t , the firms read about a non-economic news story, from which they gain utility x . Each firm only reads the news from one outlet. The expected utility from reading outlet m is:

$$E_{imt-1}U_{it} = \Pr(|\pi_t| > \theta_m) \cdot 0 + \Pr(|\pi_t| < \theta_m) \cdot (x - (1 - \beta\lambda)\tilde{\chi}_i^2 Var(\pi_t| |\pi_t| < \theta_m)) \quad (132)$$

Importantly, the variance of π_t conditional on π_t not being reported is not equal to the

unconditional variance σ_π^2 . This is the key insight of [Nimark \(2014\)](#): the fact that π_t is not reported allows firms to rule out extreme realizations of π_t .

As θ_m rises, $\Pr(|\pi_t| < \theta_m)$ rises, but $Var(\pi_t | |\pi_t| < \theta_m)$ also rises as fewer extreme realizations can be ruled out. This means that as π_t becomes less likely to be reported, the expected utility costs of it going unreported also rise. These costs are proportional to $\tilde{\chi}_i^2$, so firms with larger cross-learning parameters prefer π_t to be reported more frequently. This is the key mechanism that generates a narrative heterogeneity effect, as firms with greater $\chi_{ij\pi_t}$ parameters demand news with a lower threshold θ_m .

The exact value of the narrative heterogeneity channel depends on the nature of the media market. For simplicity, here I follow [Nimark and Pitschner \(2019\)](#) and assume that there is a continuum of media outlets, one for each firm, and that each outlet maximizes the expected utility of their associated firm. The optimal θ_i for firm i is such that firms are indifferent between reported and unreported states. That is:

$$Var(\pi_t | |\pi_t| < \theta_i) = \frac{x}{(1 - \beta\lambda)\tilde{\chi}_i^2} \quad (133)$$

Since $Var(\pi_t | |\pi_t| < \theta_i)$ is strictly increasing in θ_i , greater $\tilde{\chi}_i^2$ is associated with lower θ_i . For any given realization of π_t , we therefore have $\tau_{i\pi_t} = 1$ if $\tilde{\chi}_i^2$ is above a certain threshold, and $\tau_{i\pi_t} = 0$ if not. This generates a narrative heterogeneity effect, as information is concentrated among those with large cross-learning parameters, as in the rational inattention and sticky information models. Unlike those models, the extent of this covariance is now state-dependent: at π_t close to 0, almost no firms receive news about it, so there is little heterogeneity in information and the covariance is small. At very large (positive or negative) realizations of π_t , almost all firms receive the news, and again there is little heterogeneity. The narrative heterogeneity effect is therefore strongest for intermediate realizations of π_t , when news coverage is more divided.

$$Cov_I[\tilde{\chi}_i, \tau_{i\pi_t}] = [E_I(\tilde{\chi}_i | \tilde{\chi}_i^2 > \chi^*(\pi_t)) - \tilde{\chi}] \cdot \Pr(\tilde{\chi}_i^2 > \chi^*(\pi_t)) \quad (134)$$

where $\chi^*(\pi_t)$ is the threshold such that firm i receives news on π_t if $\tilde{\chi}_i^2 > \chi^*(\pi_t)$. The results above imply that $\chi^*(\pi_t)$ is decreasing in $|\pi_t|$. For small $|\pi_t|$, the first component is close to 0. For large $|\pi_t|$, the second component is close to 0.

This analysis abstracts from two further channels through which media provision of information may affect the joint distribution of information and subjective models. First, I considered only news reporting functions which were symmetric about $\pi_t = 0$. This implied that $E_{it\pi_t} = 0$ whenever the firm's media outlet did not report π_t . If there was a bias (e.g. if negative shocks are reported

more often than positive shocks, as in [Chahrour et al. \(2021\)](#)), then a firm observing their outlet not reporting π_t will adjust their posterior expectation of π_t , as well as their posterior variance. Two firms may form different expectations even if neither of their outlets report on π_t , if their outlets have different reporting functions. This creates further information heterogeneity between firms based on their media consumption. Second, recent work has suggested that media may also transmit subjective models as well as information ([Macaulay and Song, 2022](#); [Eliaz and Spiegler, 2024](#)). While not considered here, it provides a further reason why the media sector may induce correlations between information and subjective models.

Data generated through production. [Farboodi and Veldkamp \(2021\)](#) have recently introduced models in which firms generate information (data) through production. To model this in our framework, note that log-linearizing the firm-level demand equation used to derive the Calvo first order condition (115) implies (relative) production is a decreasing function of relative price:

$$y_{it} - Y_t = -\gamma(p_{it} - P_t) \quad (135)$$

where y_{it} and Y_t denote firm i and aggregate output respectively, and γ is the elasticity of substitution between firms' products.

If a firm has a low relative price in period t , they therefore produce more, and thus receive a more precise signal about inflation in the following period. As in the model with media, this introduces time-variation in signal-to-noise ratios. Specifically, I follow [Farboodi and Veldkamp \(2021\)](#) and assume that signal precision is linearly increasing in production (i.e. is linearly decreasing in the relative price), which implies firm i 's signal-to-noise ratio in period t is:

$$\tau_{i\pi_t,t} = \frac{\sigma_\pi^2}{\sigma_\pi^2 + \frac{\sigma_\epsilon^2}{z(p_{it}-P_t)}}, \quad z(p_{it} - P_t) = \max\{\delta, z^* - (p_{it} - P_t)\} \quad (136)$$

where σ_ϵ^2 is constant across firms, $z^* > 0$ is a constant, and $\delta > 0$ is arbitrarily close to 0. I focus here on the tractable case where firms are myopic, so resetting firms do not account for the value of future signals when choosing prices in period t , and the pricing decision is as in equation (115). The precision of a firm's signal in period t^* is therefore:

$$\sigma_\epsilon^{-2} z(p_{it^*} - P_{t^*}) = \sigma^{-2} \cdot \max(\delta, z^* - (p_{it^*} - P_{t^*-1}) + (P_{t^*} - P_{t^*-1})) \quad (137)$$

$$= \sigma^{-2} \cdot \max(\delta, z^* + (P_{t^*} - P_{t^*-1}) - (1 - \beta\lambda)\tilde{\chi}_i \tau_{i\pi_t,t^*-1} s_{it^*-1}(\pi_{t^*-1})) \quad (138)$$

If $s_{it^*-1}(\pi_{t^*-1}) > 0$, this is strictly decreasing in $\tilde{\chi}_i$. If $s_{it^*-1}(\pi_{t^*-1}) < 0$, it is strictly increasing.

The signal-to-noise ratio $\tau_{i\pi_t, t^*}$ is strictly increasing in this precision.

Consider the case where $\pi_{t^*-1} > 0$, and all $\tilde{\chi}_i > 0$. The majority of resetting firms will receive $s_{it^*-1}(\pi_{t^*-1}) > 0$, and will increase prices. Firms with larger $\tilde{\chi}_i$ increase prices by more than those with small $\tilde{\chi}_i$. As a result, large $\tilde{\chi}_i$ firms have less precise information in period t^* than those with small $\tilde{\chi}_i$: there is a negative $Cov_I[\tilde{\chi}_i, \tau_{i, t^*}]$. The reverse would be true if $\pi_{t^*-1} < 0$. Overall, the narrative heterogeneity channel arises because subjective models affect pricing decisions, which in turn affects information production. As in the model with media, the narrative heterogeneity channel produced here is therefore state-dependent.

If firms were not myopic, there would also be a second channel here: firms would set prices below the within-period optimum to increase the precision of signals in the next period. As discussed extensively above, the value of precise information is larger for firms with large $\tilde{\chi}_i^2$, so this is also likely to induce a correlation between $\tilde{\chi}_i$ and information precision.

Heterogeneous price indices. Recent literature (e.g. [Kaplan and Schulhofer-Wohl, 2017](#)) has found that households in particular face highly heterogeneous inflation rates, and so some heterogeneity in survey expectations may come because households are forecasting their own idiosyncratic inflation rates. We now adapt that idea to our simple firm model, and show that it can also generate a narrative heterogeneity channel, even if all agents have rational expectations.

Suppose that each firm i belongs to a sector k , and that each sector faces its own inflation process, which is a weighted average of past sector-specific inflation and an aggregate i.i.d. shock $u_t \sim N(0, \sigma_u^2)$:

$$\pi_{kt} = \rho_k \pi_{kt-1} + (1 - \rho_k) u_t \quad (139)$$

where $\rho_k \in (0, 1)$ is the persistence of that sector's inflation process.

Firms observe their own sector's inflation at the end of each period, and receive signals of the form:

$$s_{i\pi_{kt}} = \pi_{kt} + \epsilon_{i\pi_{kt}}, \quad \epsilon_{i\pi_{kt}} \sim N(0, \sigma_\epsilon^2) \quad (140)$$

The posterior expectation of π_{kt} after observing this signal is:

$$E_{it}\pi_{kt} = (1 - \tau_{k\pi_t})\rho_k \pi_{kt-1} + \tau_{k\pi_t}(\pi_{kt} + \epsilon_{i\pi_{kt}}) \quad (141)$$

where

$$\tau_{k\pi_t} = \frac{(1 - \rho_k)^2 \sigma_u^2}{(1 - \rho_k)^2 \sigma_u^2 + \sigma_\epsilon^2} \quad (142)$$

From this, we see that τ_k is strictly decreasing in ρ_k for all $\rho_k < 1$. This is because greater

persistence in inflation means that in any given period, prior beliefs are more informative, leading to less reaction to current signals. This heterogeneity in $\tau_{k\pi_t}$ arises even though the precision of inflation signals is assumed to be equal across sectors.

Moreover, inflation persistence also affects how firms cross-learn from current to future inflation. It is straightforward to see that

$$E_{it}\pi_{kt+s} = \rho_k^s E_{it}\pi_{kt} \quad (143)$$

and thus that $\chi_{i\pi_{t+s}\pi_t} = \rho_k^s$, which is increasing in ρ_k . The covariance between $\chi_{i\pi_{t+s}\pi_t}$ and $\tau_{i\pi_t}$ is therefore negative. Firms in sectors with less persistent inflation react more to inflation information, but extrapolate less strongly from current inflation expectations to the future. Responsiveness to information is therefore concentrated among those who cross-learn the least from it, which is a narrative heterogeneity effect that weakens the aggregate causal effect of inflation expectations.

Level-k thinking. For this, we switch the setup slightly to facilitate tractability. Specifically, now assume that the only signals available to firms are on current marginal costs mc_t , not inflation. There is cross-learning from mc_t to π_t , but no cross-learning to longer horizons. That is, $\chi_{i\pi_{t+s}mc_t} = \chi_{imc_{t+s}mc_t} = 0$ for all $s > 0$. As in the other cases above, assume there is no public noise in signals. In this case, the optimal price of a resetting firm is

$$p_{it}^* - P_{t-1} = (1 - \beta\lambda)[1 + \chi_{i\pi_t mc_t}] \tau_{imc_t}(mc_t + \varepsilon_{imc_t}) \quad (144)$$

This can alternatively be written as a static beauty contest

$$p_{it}^* - P_{t-1} = (1 - \beta\lambda)[E_{it}mc_t + E_{it}\pi_t] = (1 - \beta\lambda) \left[E_{it}mc_t + (1 - \lambda)E_{it} \int (p_{jt}^* - P_{t-1})dj \right] \quad (145)$$

Following [Farhi and Werning \(2019\)](#) and [Iovino and Sergeyev \(2023\)](#), a firm that reasons to level 0 observes their signal about mc_t , and updates their expectations of the shock accordingly, but fails to update their beliefs about the prices of others. That is, they ignore general equilibrium effects of the shock. The price choice of such a level-0 firm is therefore:

$$p_{it}^{*0} - P_{t-1} = (1 - \beta\lambda)E_{it}mc_t = (1 - \beta\lambda)\tau_{imc_t}s_{it}(mc_t) \quad (146)$$

A firm that reasons to level 1 realizes that others may react to the shock as well, but do not

reason all the way to general equilibrium. Rather, they assume that all other firms are level 0. In this case, the optimal reset price is:

$$p_{it}^{*1} - P_{t-1} = (1 - \beta\lambda) \left[E_{it}mc_t + (1 - \lambda)E_{it} \int (p_{jt}^{*0} - P_{t-1})dj \right] \quad (147)$$

$$= (1 - \beta\lambda)[1 + (1 - \lambda)(1 - \beta\lambda)\bar{\tau}_{mc_t}]\tau_{imc_t}s_{it}(mc_t) \quad (148)$$

For the purpose of this example, I stop at $k = 1$. Letting χ^k be the $\chi_{i\pi_t mc_t}$ of a firm engaged in k levels of reasoning, we have:

$$\chi^0 = 0, \quad \chi^1 = (1 - \lambda)(1 - \beta\lambda)\bar{\tau}_{mc_t} \quad (149)$$

Level-1 reasoners have a strictly greater $\chi_{i\pi_t mc_t}$ than level-0 reasoners. If the depth of reasoning is correlated with the precision of information τ_i , then there will be a narrative heterogeneity effect. I now show that such a correlation arises naturally if firms are able to choose their depth of reasoning, as in [Alaoui and Penta \(2016\)](#). In the model they develop, agents decide whether to complete an extra level of reasoning by weighing up the perceived costs and benefits. Specifically, in their model, two games are said to be cognitively equivalent if they entail the same sequence of actions as reasoning develops (k increases). To fit the firm problem into this framework, denote the choice of $\chi_{i\pi_t mc_t}$ as the firm's action, with pricing following mechanically using equation (144). χ^0 and χ^1 are independent of all firm-level variables, so framed in this way the problem facing firm i is cognitively equivalent to the one facing firm j . As a result, the costs of reasoning are the same for all firms, and the depth of reasoning only differs according to the value of doing so.

[Alaoui and Penta \(2016\)](#) propose a number of functions for the value of reasoning. I demonstrate the narrative heterogeneity channel for the general 'expected gain' formulation from their Example 2, which is a generalization of the form they use for their examples, and is given an axiomatic foundation in [Alaoui and Penta \(2022\)](#). Adapting it to the firm problem, the value of reasoning to level 1 is given by:

$$v_i^1 = \int p(\chi') \left[\mathbb{E}_{it-1} U_{i,t}(BR(\chi'), \chi') - \mathbb{E}_{it} U_{i,t-1}(\chi^0, \chi') \right] d\chi' \quad (150)$$

where $p(\chi')$ are weights such that $p(\chi') \in [0, 1]$ and $\int p(\chi')d\chi' = 1$, $U_{i,t}(\chi_i, \chi')$ is the payoff to choosing χ_i when all other firms choose χ' , and $BR(\chi')$ is the χ_i that maximizes $\mathbb{E}_{it-1} U_{i,t}(\chi_i, \chi')$ for a given χ' . That is, the value of reasoning is the expected utility gain from switching from χ^0 to the best-response χ_i , weighted by the firm's perceived probability that reasoning will reveal $\chi_j = \chi'$. Reasoning is undertaken before the observation of $s_{i,t}$, so the expectations are indexed $t - 1$.

If $\chi_j = \chi'$ for all firms $j \neq i$, the optimal price is (from equation (145)):

$$p_{it}^* - P_{t-1} = (1 - \beta\lambda) [1 + (1 - \lambda)(1 - \beta\lambda)(1 + \chi')\bar{\tau}_{mc_t}] \tau_{imc_t} s_{it}(mc_t) \quad (151)$$

The best-response χ_i is therefore:

$$BR(\chi') = (1 - \lambda)(1 - \beta\lambda)(1 + \chi')\bar{\tau}_{mc_t} \quad (152)$$

Following the steps used to compute indirect utility in the rational inattention case above, the value of reasoning to level 1 is:

$$v_i^1 = (1 - \beta\lambda)^4 (1 - \lambda)^2 \sigma_{mc}^2 \bar{\tau}_{mc_t}^2 \tau_{imc_t} \int p(\chi') (1 + \chi')^2 d\chi' \quad (153)$$

The value of reasoning is therefore linearly increasing in τ_{imc_t} , whatever perceptions $p(\chi')$ the firm holds. If we further assume that all firms use the same $p(\chi')$ when evaluating the benefits of reasoning (as in [Alaoui and Penta, 2016](#)), then there exists a threshold τ^* above which firms decide to reason to level 1, and below which they do not. This implies that high τ_{imc_t} is associated with high $\chi_{i\pi_t mc_t}$, so the narrative heterogeneity channel is positive. Formally:

$$Cov_I[\chi_{i\pi_t mc_t}, \tau_{imc_t}] = q\chi^0\tau^- + (1 - q)\chi^1\tau^+ - \bar{\tau}(q\chi^0 + (1 - q)\chi^1) \quad (154)$$

where q is the proportion with $\tau_{imc_t} > \tau^*$, and τ^-, τ^+ are the average τ_{imc_t} conditional on $\tau_{imc_t} < \tau^*$ and $\tau_{imc_t} > \tau^*$ respectively. Simplifying, this becomes:

$$Cov_I[\chi_{i\pi_t mc_t}, \tau_{imc_t}] = -q(1 - q)(\tau^+ - \tau^-)(\chi^0 - \chi^1) > 0 \quad (155)$$

Intuitively, firms with more precise information on shocks have more to gain from understanding how to use that information, and so they will choose a higher k than firms with less information. Information is therefore concentrated among firms with large $\chi_{i\pi_t mc_t}$.

C Mapping RCT estimates to the Calvo model: details

RCT-implied calibration. To adapt the [Baumann et al. \(2024\)](#) estimates to the particular model set out in Section 2, we need to take two steps: first, interpolate between estimates at different horizons to get the total effect of a given signal on inflation at horizons $t + 2$ and $t + 4$, where there are no estimates available. I use a simple linear interpolation here. Second, the estimates refer to

annual inflation, while the model is expressed in terms of cumulative inflation since period $t - 1$. We therefore sum them up to get the relevant coefficients for aggregate expectations updating.

To see why we need to sum, notice that inflation in the survey is given as net annual inflation:

$$\pi_{t+s}^a = \frac{P_{t+s} - P_{t+s-1}}{P_{t+s-1}} \approx \log \left(\frac{P_{t+s}}{P_{t+s-1}} \right) \quad (156)$$

Cumulative log inflation (as in the model) is therefore given by:

$$\pi_{t+s} = \log \left(\frac{P_{t+s}}{P_{t-1}} \right) = \log \left(\prod_{m=0}^s \frac{P_{t+m}}{P_{t+m-1}} \right) \approx \sum_{m=0}^s \pi_{t+m}^a \quad (157)$$

Any change in cumulative inflation is therefore equal to the sum of changes of annual inflation at all periods between now and the relevant horizon. With these adjustments, the coefficients implied by Table 5 of [Baumann et al. \(2024\)](#) are presented in the left panel of Table 4.

Table 4: Mapping estimates from [Baumann et al. \(2024\)](#) to the Calvo model.

Model object	π_t signal	π_{t+1} signal	Model object	π_{t+1} signal
$E_I[\chi_{\text{signal period}, \pi_t} \tau_{\text{signal period}}]$	1	N/A	$E_I[\chi_{\text{signal period}, \pi_t}]$	N/A
$E_I[\chi_{\text{signal period}, \pi_{t+1}} \tau_{\text{signal period}}]$	1.21	0.53	$E_I[\chi_{\text{signal period}, \pi_{t+1}}]$	1
$E_I[\chi_{\text{signal period}, \pi_{t+2}} \tau_{\text{signal period}}]$	1.51	0.945	$E_I[\chi_{\text{signal period}, \pi_{t+2}}]$	1.87
$E_I[\chi_{\text{signal period}, \pi_{t+3}} \tau_{\text{signal period}}]$	1.9	1.245	$E_I[\chi_{\text{signal period}, \pi_{t+3}}]$	2.61
$E_I[\chi_{\text{signal period}, \pi_{t+4}} \tau_{\text{signal period}}]$	2.235	1.56	$E_I[\chi_{\text{signal period}, \pi_{t+4}}]$	3.365
$E_I[\chi_{\text{signal period}, \pi_{t+5}} \tau_{\text{signal period}}]$	2.515	1.86	$E_I[\chi_{\text{signal period}, \pi_{t+5}}]$	4.135

In principle, we could extrapolate to parameters further into the future, but I will cut off firm planning horizons at 5 years here, for this and the two benchmarks.

For the decomposition in Figure 2, we proceed in a similar fashion. Estimating equation (38) with $E_{it}\pi_{t+3}$ as the dependent variable, and instrumenting $E_{it}^{post}\pi_{t+1}$ with the π_{t+1} treatment only, gives an estimated γ_1 of 0.74. Doing the same with $E_{it}\pi_{t+5}$ as the dependent variable gives an estimated γ_1 of 0.77. Both of these coefficients are significantly different from 0 at the 1% level. Applying the same procedure as described above gives the right panel of Table 4. Note that since we assume $\bar{\tau}_{\pi_t} = 1$, and thus $\tau_{i\pi_t} = 1$ for all i , there is no narrative heterogeneity channel from π_t signals, so we do not need to compute $\bar{\chi}_{\pi_t, \pi_{t+h}}$.

Including cross-learning to marginal costs. [Baumann et al. \(2024\)](#) Table 8 contains results from estimating equation (38), which (as argued in Section 4.1) reflect average cross-learning from the inflation period in the signal to each X_i variable. They report results for $X_i \in \{1\text{-yr-ahead}$

expected wage, non-wage costs, employment}. To go from those variables to the marginal cost that enters the model, we need to specify the firm's production function, before then applying the temporary-information equilibrium approach. For the exercise here, I will use the simple production function

$$y_{it} = \min \left(\frac{n_{it}}{\alpha_n}, \frac{x_{it}}{\alpha_x} \right) \quad (158)$$

where labor n_{it} and other inputs x_{it} are perfect complements, so the optimal factor ratio is

$$\frac{n_{it}}{x_{it}} = \frac{\alpha_n}{\alpha_x} \quad (159)$$

In this case, real marginal costs are given by

$$mc_t = \alpha_n \frac{w_t}{P_t} + \alpha_x \frac{p_{xt}}{P_t} \quad (160)$$

where w_t is the nominal wage and p_{xt} is the price of x_{it} . Log-linearizing we obtain

$$mc_t = \tilde{\alpha}_n w_t + \tilde{\alpha}_x p_{xt} - (\tilde{\alpha}_n + \tilde{\alpha}_x) P_t \quad (161)$$

where

$$\tilde{\alpha}_n \equiv \frac{\alpha_n w^{ss}}{\alpha_n w^{ss} + \alpha_x p_x^{ss}}, \quad \tilde{\alpha}_x \equiv \frac{\alpha_x p_x^{ss}}{\alpha_n w^{ss} + \alpha_x p_x^{ss}} \quad (162)$$

are the steady-state shares of labor and other costs in marginal costs. Since this model features a constant marginal cost, these are also the steady state shares of labor costs and other costs in total costs.

The average update to expected marginal costs after a π_{t+1} signal is therefore

$$\frac{dE_{it} mc_{t+1}}{d\varepsilon_{\pi_{t+1}}} = \tilde{\alpha}_n \frac{dE_{it} w_{t+1}}{d\varepsilon_{\pi_{t+1}}} + \tilde{\alpha}_x \frac{dE_{it} p_{xt+1}}{d\varepsilon_{\pi_{t+1}}} - \frac{dE_{it} \pi_{t+1}}{d\varepsilon_{\pi_{t+1}}} \quad (163)$$

$$= \underbrace{(\tilde{\alpha}_n \chi_{iw_{t+1}\pi_{t+1}} + \tilde{\alpha}_x \chi_{ip_{xt+1}\pi_{t+1}} - 1)}_{\equiv \chi_{imc_{t+1}\pi_{t+1}}} \tau_{i\pi_{t+1}} \quad (164)$$

where I have used the observation that $\tilde{\alpha}_n + \tilde{\alpha}_x = 1$, the assumption that $\tau_{i\pi_t} = 1$, the definition $P_{t+1} = \pi_{t+1} + P_{t-1}$,⁴² and the fact that P_{t-1} is fixed as it was determined before the shock. I have

⁴²Recall that π_{t+1} is defined as the cumulative inflation $P_{t+1} - P_{t-1}$ in this model.

also assumed that there is no cross-learning from π_t to $E_{it}w_{t+1}$ or $E_{it}p_{xt+1}$, because all estimates for cross-learning from the current-inflation treatment in [Baumann et al. \(2024\)](#) are not significantly different from 0.

The average cross-learning from π_{t+1} to mc_{t+1} can therefore be computed from [Baumann et al. \(2024\)](#) Table 8 as

$$\bar{\chi}_{mc_{t+1}\pi_{t+1}} = \tilde{\alpha}_n \bar{\chi}_{w_{t+1}\pi_{t+1}} + \tilde{\alpha}_x \bar{\chi}_{p_{xt+1}\pi_{t+1}} - 1 \quad (165)$$

$$= \tilde{\alpha}_n \times 0.29 + (1 - \tilde{\alpha}_n) \times 0.90 - 1 \quad (166)$$

$$= -0.4538 \quad (167)$$

where for the final equality I have assumed that the labor share of total costs is $\tilde{\alpha}_n = 0.58$. As this data is not available for the Euro area, this is taken from the 2023 measurement for the US.⁴³ On average firms therefore decrease their real marginal cost expectations when they learn of higher future inflation. This is because the positive updating found by [Baumann et al. \(2024\)](#) is over the components of nominal marginal costs, and those updates are dominated by the increase in inflation, causing real marginal cost expectations to fall.

As in the exercises in Section 4, one could in principle extrapolate from this to cross-learning about marginal costs at longer horizons, but here I assume for simplicity that all cross-learning beyond the horizon we observe (here one year) is zero. Allowing for the cross-learning from π_{t+1} to mc_{t+1} , the partial equilibrium effects of $\bar{E}_t(\pi_{t+1})$ are therefore given by

$$\begin{aligned} \Omega^{PE}(\pi_{t+1}) &= \Omega^{PE}(\pi_{t+1} : \chi_{imc_{t+1}\pi_{t+1}} = 0) \\ &\quad + (1 - \lambda)(1 - \beta\lambda)\beta\lambda \times \left(\bar{\chi}_{mc_{t+1}\pi_{t+1}} + \frac{1}{\bar{\tau}_{\pi_{t+1}}} Cov_I[\chi_{imc_{t+1}\pi_{t+1}}, \tau_{i\pi_{t+1}}] \right) \end{aligned} \quad (168)$$

where $\Omega^{PE}(\pi_{t+1} : \chi_{imc_{t+1}\pi_{t+1}} = 0)$ denotes the partial equilibrium effect calculated in Figure 1a under the assumption of no cross-learning to real marginal costs. The general equilibrium effects can then be calculated from this using equation (16). Since I am assuming no cross-learning from π_t to marginal costs, the $\Omega^{PE}(\pi_t)$ term in equation (16) is unaffected.

Note that while we can calibrate $\bar{\chi}_{mc_{t+1}\pi_{t+1}}$ from the data in [Baumann et al. \(2024\)](#), we cannot calibrate the narrative heterogeneity term $Cov_I[\chi_{imc_{t+1}\pi_{t+1}}, \tau_{i\pi_{t+1}}]$. For Figure 6 below I will assume that the narrative heterogeneity term is 0, so the results show what would happen if all firms updated marginal cost expectations in the same way.

The choice of benchmark models is difficult here, because we haven't modelled how wages

⁴³Available at <https://fred.stlouisfed.org/series/MPU4900141>.

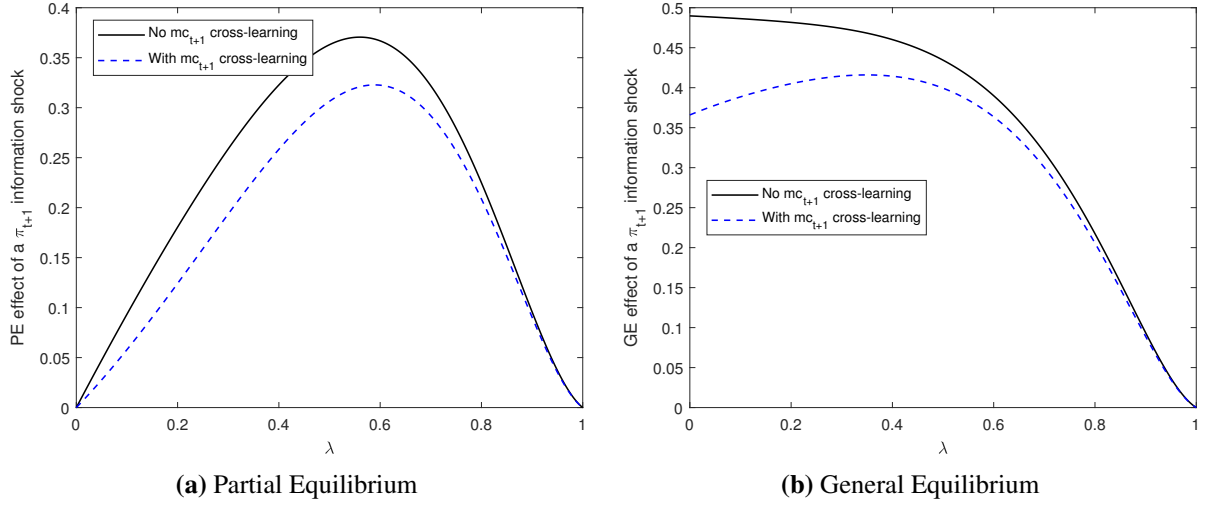
and costs are determined. Without specifying a full model for those processes, we cannot write down the perfect foresight or rational expectations benchmarks. Instead, I therefore compare the results to the case calibrated to the RCT, but with no cross-learning to future marginal costs. That is, the benchmark I will use is the black line in Figures 1a and 1b. This gives a sense of whether the updates to cost expectations found in the RCT are quantitatively important relative to only looking at the effects on inflation expectations.

Figure 6 therefore plots the partial and general equilibrium effects of $\bar{E}_t(\pi_{t+1})$ with and without cross-learning to marginal costs. Again the quantitative significance of that cross-learning depends on price stickiness. In partial equilibrium (panel a), the effects of inflation expectations are smaller than they are without marginal cost learning at all values of price stickiness λ , because higher expected inflation causes firms to reduce their real marginal cost expectations, which tempers their incentives to raise prices. Proportionally, this effect is stronger when prices are more flexible (λ is small): at the plausible calibration of $\lambda = 0.2$, the PE effect including marginal cost cross-learning is 69% as large as the effect without such cross-learning. At $\lambda = 0.5$, it is 84% as large. This is again because stickier prices imply longer-horizon expectations matter relatively more for pricing decisions. Since the marginal-cost cross-learning is assumed to only affect marginal costs one period in the future, its effect on inflation dynamics declines as λ rises.

In general equilibrium, again the effect is smaller once we allow for marginal cost cross-learning, and the magnitude of the difference is large at low values of λ . However, the differences to the baseline case are smaller than in partial equilibrium: at $\lambda = 0.2$ and 0.5 respectively the GE effect with marginal cost cross-learning is 84% and 92% as large as the baseline case without that cross-learning. This is because of an effect discussed in Section 4: a larger partial equilibrium effect $\Omega^{PE}(\pi_{t+1})$ actually dampens general equilibrium feedback, because it causes a greater scaling of the original shock to ensure that $\bar{E}_t(\pi_{t+1})$ increases by 1 unit. When we introduce cross-learning to marginal costs, that partial equilibrium effect falls, but there is no change on the other component of the general equilibrium effect through $\Omega^{PE}(\pi_t)$. General equilibrium feedback is therefore stronger in the presence of marginal cost cross-learning, but this is purely due to the way we are scaling the shock in computing these causal effects.

This mechanism is also why the general equilibrium effect of expected inflation initially rises with λ once we allow for marginal cost cross-learning. As λ increases the general equilibrium feedback through $\Omega^{PE}(\pi_t)$ becomes larger, while the counter-effect on general equilibrium feedback through $\Omega^{PE}(\pi_{t+1})$ and scaling makes the general equilibrium effect smaller. Marginal cost cross-learning weakens the second of these mechanisms, so that overall the causal effect in general equilibrium is initially increasing in λ .

Figure 6: Partial and general equilibrium effects of a shock to $s_{\pi_{t+1}}$, comparing RCT estimates with and without cross-learning to mc_{t+1} .



Note: Figure plots the partial- and general-equilibrium causal effects of an increase in $\bar{E}_t \pi_{t+1}$, following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2024) (detail in Appendix C), with and without allowing cross-learning from π_{t+1} to mc_{t+1} .

The share of fundamental shock transmission due to expectations. Following the strategy in Proposition 4, we now compute the share of the general equilibrium transmission of a fundamental shock that can be attributed to inflation expectations, for the RCT data and both benchmarks (perfect foresight and rational expectations). Since we are assuming $\bar{\tau}_{\pi_t} = 1$, these shares for an i.i.d. mc_t shock would just equal the partial equilibrium effect of a $\bar{E}_t(\pi_t)$ shock (equation (22)). We therefore opt instead to consider a news shock: firms learn in period t of an exogenous increase in mc_{t+1} . This demonstrates the flexibility of the approach developed here.

With this shock the transmission channels are more numerous. Once period $t + 1$ arrives, the increased marginal costs will imply inflation above steady state, as described in equation (20). In period t , firms therefore not only learn that mc_{t+1} will be above steady state, their signals about π_{t+1} also endogenously increase. Even without cross-learning from marginal cost expectations to inflation expectations, there is therefore an increase in expected future inflation, which has cross-learning effects on inflation expectations at other horizons. All of these forces contribute to the increase in π_t , which in turn has general equilibrium feedback effects through $s_{it}(\pi_t)$.

Formally, we model the news shock by assuming that $\tau_{imc_{t+1}} = 1$. We also assume that marginal costs are i.i.d., so there is no cross-learning from the news about future marginal cost to expected marginal costs at any other horizon, or to expected inflation. Differentiating equation (11) with

respect to mc_{t+1} with these assumptions implies

$$\frac{d\pi_t}{dmc_{t+1}} = (1 - \lambda)(1 - \beta\lambda)\beta\lambda + \bar{\tau}_{\pi_{t+1}}\Omega^{PE}(\pi_{t+1})\frac{d\pi_{t+1}}{dmc_{t+1}} + \Omega^{PE}(\pi_t)\frac{d\pi_t}{dmc_{t+1}} \quad (169)$$

where we have also used our recurring assumption that $\bar{\tau}_{\pi_t} = 1$.

Next, from equation (20) we have

$$\frac{d\pi_{t+1}}{dmc_{t+1}} = \frac{d\pi_t}{dmc_t} = \frac{(1 - \lambda)(1 - \beta\lambda)}{1 - \Omega^{PE}(\pi_t)} \quad (170)$$

where it is $\Omega^{PE}(\pi_t)$, not $\Omega^{PE}(\pi_{t+1})$ that matters because by the time period $t + 1$ arrives, the shock will be contemporaneous, so it is the effect of a contemporaneous shock that determines the contemporaneous inflation rate – i.e. π_{t+1} .

Substituting this in to equation (169) and rearranging, we find

$$\frac{d\pi_t}{dmc_{t+1}} = \frac{(1 - \lambda)(1 - \beta\lambda)\beta\lambda}{1 - \Omega^{PE}(\pi_t)} + \frac{(1 - \lambda)(1 - \beta\lambda)\bar{\tau}_{\pi_{t+1}}\Omega^{PE}(\pi_{t+1})}{(1 - \Omega^{PE}(\pi_t))^2} \quad (171)$$

We define the share of this overall transmission that is due to expectations as in equation (22), where the counterfactual shock transmission is computed assuming firms only observe the mc_{t+1} signal, with no updating of any other expectations. Thus this calculation gives the share of shock transmission due to expectations of inflation specifically, as marginal cost expectations are the same in the overall effect and the counterfactual.

This counterfactual with no response of inflation expectations is found by setting $\Omega^{PE}(\pi_t) = \Omega^{PE}(\pi_{t+1}) = 0$, at which point

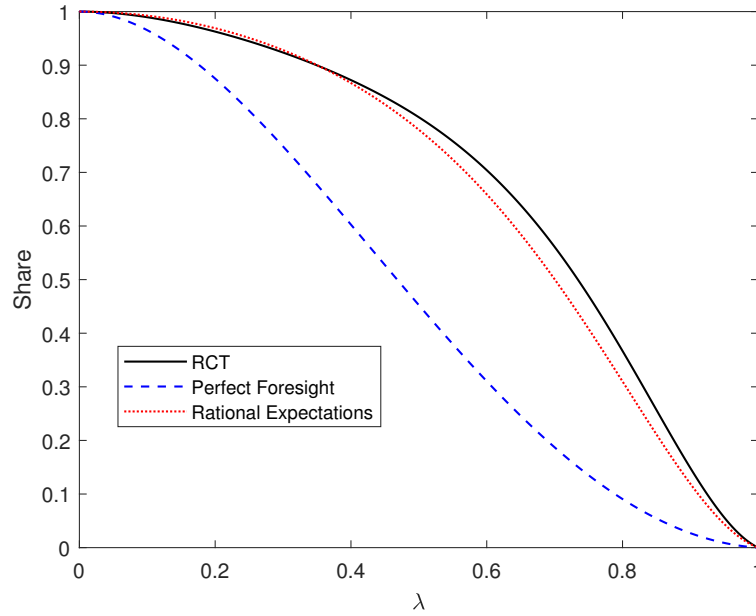
$$\left. \frac{d\pi_t}{dmc_{t+1}} \right|_{\sim E} = (1 - \lambda)(1 - \beta\lambda)\beta\lambda \quad (172)$$

Using these expressions, we calculate the share of the shock transmission to π_t that is due to the response of inflation expectations as

$$\frac{\left. \frac{d\pi_t}{dmc_{t+1}} - \frac{d\pi_t}{dmc_{t+1}} \right|_{\sim E}}{\left. \frac{d\pi_t}{dmc_{t+1}} \right|_{\sim E}} = 1 - \frac{\beta\lambda(1 - \Omega^{PE}(\pi_t))^2}{\beta\lambda(1 - \Omega^{PE}(\pi_t)) + \bar{\tau}_{\pi_{t+1}}\Omega^{PE}(\pi_{t+1})} \quad (173)$$

To construct each line in Figure 7, we replace $\Omega^{PE}(\pi_t)$ and $\Omega^{PE}(\pi_{t+1})$ with values calculated using equations (12) and (13) and the relevant calibrations of χ_i and τ_i .

Figure 7: Share of transmission from a mc_{t+1} news shock to π_t due to the adjustment of inflation expectations, comparing RCT estimates to benchmarks



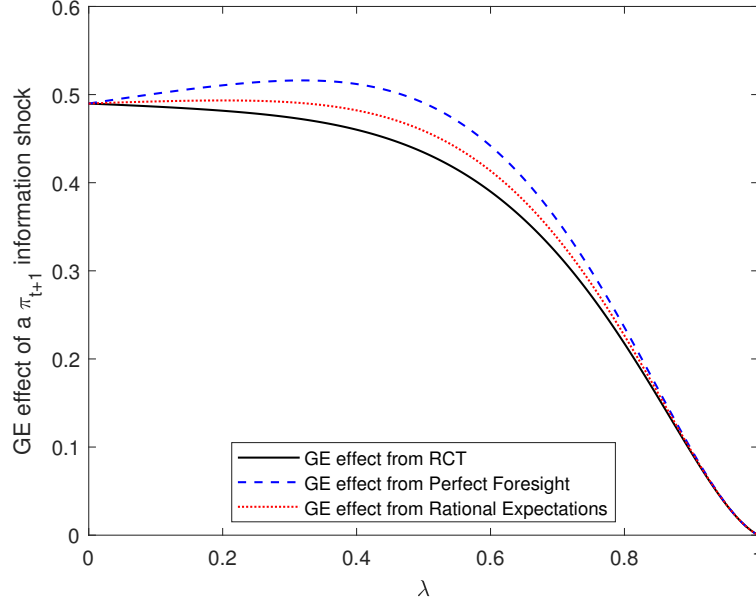
Note: Figure plots the share of the transmission of a mc_{t+1} news shock that is due to the endogenous response of inflation expectations in the Calvo model of Section 2, for calibrations based on perfect foresight, rational expectations, and the RCT results of Baumann et al. (2024) (detail in Appendix C).

In this case, the extent of price stickiness can change the sign of the RCT-implied departure from standard benchmarks, although the quantitative differences remain small. At $\lambda = 0.2$, the share of the shock transmission due to the dynamics of inflation expectations is just under 1% smaller in the RCT calibration than in the rational expectations benchmark. At $\lambda = 0.5$, the RCT-implied share is 3% larger than the rational expectations benchmark. This sign change occurs because the RCT results imply somewhat smaller cross-learning from π_t to π_{t+1} than implied by rational expectations, even if long-horizon cross-learning is stronger (as discussed above). This means that when prices are rather flexible, and short-horizon expectations drive the vast majority of pricing decisions, the mc_{t+1} shock entails a smaller reaction in the most important expectation horizons than is generated with rational expectations. Of course, even the limited short-horizon cross-learning in the RCT calibration is stronger than seen under perfect foresight, so the transmission due to expectations is larger than the perfect foresight case at all levels of price stickiness.

Alternative general equilibrium assumptions. Figure 9b reproduces Figure 1b, except that in all lines the partial equilibrium effect is the one computed using the RCT results, but the general equilibrium feedback is taken from different assumptions on expectations. This incorrect way to calculate the general-equilibrium causal effects of expected inflation highlights how RCT results are

important for the intercept (i.e. general equilibrium feedback) as well as the direct effect of interest.

Figure 8: General equilibrium effects of a shock to $s_{\pi_{t+1}}$, comparing RCT estimates to alternatives with general equilibrium effects from other benchmarks



Note: Figure plots (in black) the general-equilibrium causal effects of an increase in $\bar{E}_t \pi_{t+1}$ in the Calvo model, following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2024). The red and blue lines replace the true calibrated general-equilibrium feedback effect with the equivalent calculated using the perfect foresight and rational expectations calibrations (detail in Appendix C).

Other models as benchmarks. Recall that the equilibrium New Keynesian Phillips Curve cannot reveal the causal effects of inflation expectations because expectations there simultaneously cause and are caused by dynamics of other variables. The same is true if we attempt to compare different models of expectation formation by studying equilibrium conditions. If, for example, a particular model implies a reduced link between inflation expectations and realized inflation, we cannot tell if that is because expectations have been made less responsive to inflation, or if their causal effect on inflation has been reduced.

The temporary-information equilibrium approach therefore offers a structured way to disentangle why certain models have different implications from one another, and to assess whether they match the equilibrium implications of RCT evidence. I provide an example here, re-computing Figure 1 for diagnostic expectations (Bordalo et al., 2020) and inattentiveness (Gabaix, 2020).

In Gabaix (2020), expectations of a variable x_{t+h} are given by

$$E_{it} x_{t+h} = m^h E_t^* x_{t+h} \quad (174)$$

where $E_t^* x_{t+h}$ is the rational expectation of x_{t+h} formed with information on all variables realized up to period t , and $m \in [0, 1]$ is a parameter governing the degree of inattentiveness. The first implication of this setup is that when $h = 0$, inattentiveness has no effect on expectations. That is, inattentiveness does not affect the mapping from observed information on a variable x_t into beliefs about x_t . It only affects how that information is used to form beliefs about x_{t+h} . Inattentiveness therefore affects cross-learning from x_t to x_{t+h} , but not the information matrix τ_i .

Specifically, we will restrict ourselves here to cases with no cross-learning to or from marginal costs. The inattentiveness case can therefore be summarized by

$$\chi_{i\pi_{t+s}^a \pi_t^a}^{\text{IA}} = m^h \chi_{i\pi_{t+s}^a \pi_t^a}^{\text{RE}} \quad (175)$$

where $\chi_{i\pi_{t+s}^a \pi_t^a}^{\text{RE}}$ is the cross-learning from annual inflation π_t^a to annual inflation π_{t+s}^a . We specify this in terms of per-period inflation to fit with the specification in [Gabaix \(2020\)](#). Translating this to the cumulative inflation used in our model, assuming that $\chi_{i\pi_{t+s}^a \pi_t^a}^{\text{RE}} = \rho^s$, is achieved by setting

$$\chi_{i\pi_{t+s} \pi_t}^{\text{IA}} = \frac{1 - (m\rho)^s}{1 - m\rho} \quad (176)$$

The second model we consider here is diagnostic expectations, as in [Bordalo et al. \(2020\)](#). In this model, if x_t is an AR(1) process with persistence ρ , we have

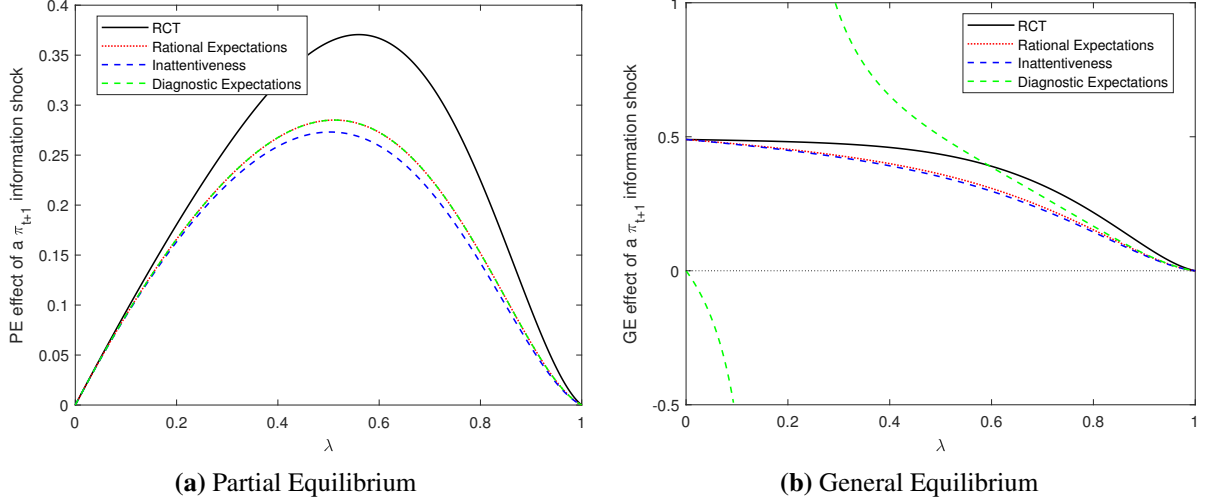
$$E_{it} x_{t+h} = \rho^h E_{it} x_t = \rho^h \left((1 + \theta) \frac{\sigma_x^2}{\sigma_x^2 + \sigma_{\varepsilon x}^2} s_{it}(x_t) \right) \quad (177)$$

where θ is the degree of diagnosticity, $s_{it}(x_t)$ is the (noisy) signal observed about x_t , σ_x^2 is the variance of x_t , and $\sigma_{\varepsilon x}^2$ is the variance of the noise in $s_{it}(x_t)$. In this case, the departure from rational expectations operates solely on the way in which agents map from direct signals on x_t into expectations of x_t . It therefore affects τ_i and not χ_i . Specifically, diagnostic expectations imply τ_i is multiplied by $1 + \theta$, so for $\theta > 0$ agents overreact to news.

Figure 9 plots the partial and general equilibrium effects of $\bar{E}_t(\pi_{t+1})$ in each of these models, alongside the effects under rational expectations and the RCT benchmark. For the calibration, I set $m = 0.8^4$, as advised by [Gabaix \(2020\)](#) for firm expectations. I set $\theta = 0.37$, the average of the results from the two methods employed by [Bordalo et al. \(2020\)](#) to estimate θ in the context of CPI inflation expectations (taken from their Table 4).

Inattentiveness generates similar results to rational expectations, with a small degree of dampening relative to that benchmark in both partial and general equilibrium. The dampening is proportionately larger at higher λ , when longer-horizon expectations are more important for current

Figure 9: Partial and general equilibrium effects of a shock to $s_{\pi_{t+1}}$, comparing RCT estimates to rational expectations, inattentiveness, and diagnostic expectations.



Note: Figure plots the partial- and general-equilibrium causal effects of an increase in $\bar{E}_t \pi_{t+1}$ in the Calvo model, following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2024) (detail in Appendix C). This RCT calibration is compared to calibrations based on perfect foresight, rational expectations, inattentiveness as in (Gabaix, 2020), and diagnostic expectations as in (Bordalo et al., 2020).

pricing.

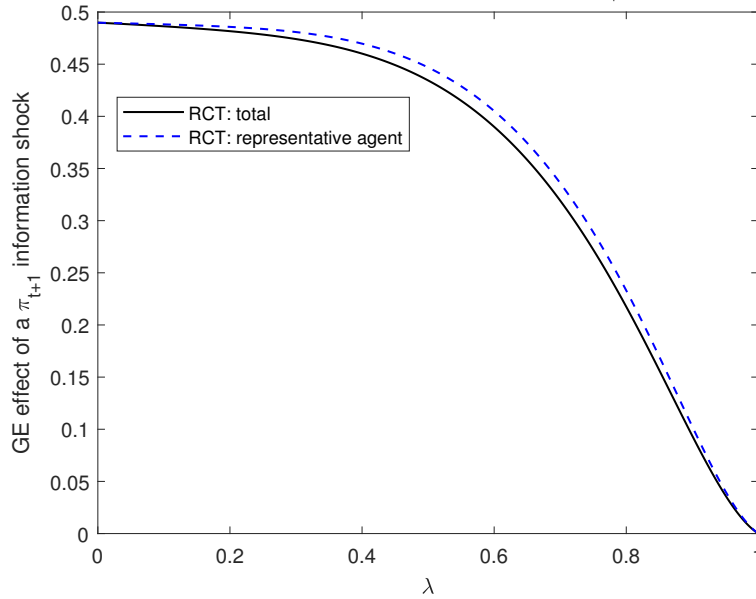
Diagnostic expectations are identical to rational expectations in partial equilibrium, because the only difference to rational expectations is in $\bar{\tau}_{\pi_{t+1}}$. When we scale the shock to ensure a unit response of $\bar{E}_t(\pi_{t+1})$, we therefore cancel out any effects of diagnosticity.

However, in general equilibrium diagnostic expectations deliver radically different results from the other models and the from the RCT results. The over-reaction to news generated by diagnostic expectations makes general equilibrium feedback effects more powerful, as any increase in π_t is met with a further round of over-reaction, entailing even more π_t response, creating powerful strategic complementarities in price-setting. As price stickiness falls from $\lambda = 1$, general equilibrium effects grow in all models. The extra complementarity induced by diagnostic expectations means that as λ falls the causal effects of expectations grow more rapidly with diagnostic expectations than they do with rational expectations. At sufficiently low λ , the strategic complementarities become so strong that an increase in $\bar{E}_t(\pi_t)$ generates a greater than one-for-one response in π_t , implying that in equilibrium greater $\bar{E}_t(\pi_t)$ must in fact be met with a decrease in π_t .

General equilibrium causal effect with and without heterogeneity. Figure 10 repeats the black (RCT-calibration) line from Figure 1b, showing the overall general-equilibrium effect of expected one-year-ahead inflation on realized current inflation. The blue dashed line alongside it shows

the same effect if there is no narrative heterogeneity channel in the model, computed using the parameters from the right panel of Table 4.

Figure 10: General equilibrium effects of a shock to $s\pi_{t+1}$, decomposition



Note: Figure plots (in black) the general-equilibrium causal effects of an increase in $\bar{E}_t\pi_{t+1}$ in the Calvo model, following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2024). The blue line plots the equivalent effect if we assume there is no correlation between χ_i and τ_i (detail in Appendix C).

D Empirical Appendix for Section 5.1

D.1 Defining the direct information indicator in the IAS

The full set of questions used to construct the information dummy is set out below, along with the dates at which each was asked and how the answers are mapped into the information indicator used above. Note that my question numbering differs from the labels in the IAS microdata, to aid the logical organization of the paper. All of the questions were only asked in the first quarter of the year(s) indicated. In the main exercises I exclude questions 2e and 2g from the total information variable, to ensure that there are no periods in which two questions are asked. I remove these rather than the short run questions in those periods to keep the majority of questions as short run expectations. The results are robust to including these extra questions. See Appendix D.3 for this, and robustness checks with other variations in the definition of the information indicator.

Question 2a *What were the most important factors in getting to your expectation for how prices in the shops would change over the next 12 months?*

Please select up to 4.

- 1. How prices have changed in the shops recently, over the last 12 months*
- 2. How prices have changed in the shops, on average, over the longer term i.e the last few years*
- 3. Reports of current inflation in the media*
- 4. Discussion of the prospects for inflation in the media*
- 5. The level of interest rates*
- 6. The inflation target set by the government*
- 7. The current strength of the UK economy*
- 8. Expectations about how economic conditions in the UK are likely to evolve*
- 9. Other factors*
- 10. None*

We can divide the possible answers into four categories. First, options 1 and 2 concern past experienced price rises. Options 3 and 4 are direct information about inflation. Options 5-8 concern other macroeconomic variables, either current or expected, and options 9 and 10 are extras. A rational household may well use the information sources in options 1,2 and 5-9 to forecast inflation, but in the decomposition in Proposition 6 this would represent cross-learning from information about other variables. To use the level of interest rates (5) to forecast inflation, for example, a household must employ a model of how interest rates relate to inflation. Similarly, to use past experienced price changes (1-2), households need a model of the persistence of inflation.⁴⁴ The only answers that represent the use of direct information about inflation are options 3 and 4.

Question 2b *What were the most important factors that led you [to change (insert their response to how expectation has changed)] your expectation of prices in the shops over the next 12 months?*

Please select up to 4:

- 1. How prices have changed in the shops recently, over the last 12 months*
- 2. How prices have changed in the shops, on average, over the longer term i.e the last few years*
- 3. Reports of current inflation in the media*
- 4. Discussion of the prospects for inflation in the media*
- 5. The level of interest rates*

⁴⁴Macaulay and Moberly (2025) find this perceived persistence is very heterogeneous across households. Note that strictly, option 3 also concerns past price changes, so the assumption here is that media reports of inflation tend to discuss both current and future inflation simultaneously. Appendix D.3 shows that the results below are robust to various small changes to this definition of the information indicator.

6. *The inflation target set by the government*
7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *The level of the exchange rate (the value of sterling)*
10. *Other factors*
11. *None*

Asked: 2017

Information indicator: =1 if items 3 or 4 selected, =0 otherwise.

Question 2c *What were the most important factors that led you to change/not change your expectation of prices in the shops in the longer term?*

1. *How prices have changed in the shops recently, over the last 12 months*
2. *How prices have changed in the shops, on average, over the longer term i.e the last few years*
3. *Reports of current inflation in the media*
4. *Discussion of the prospects for inflation in the media*
5. *The level of interest rates*
6. *The inflation target set by the government*
7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *The level of the exchange rate (the value of sterling)*
10. *Other factors*
11. *None*

Asked: 2018, 2019

Information indicator: =1 if items 3 or 4 selected, =0 otherwise.

Question 2d *When you said prices would go up in the next 12 months, how important were the following things in getting to that answer?*

For each option, possible answers are:

- *Very important*
- *Fairly important*
- *Not very important*
- *Not at all important*
- *Don't know*
- *Refused*

Options:

1. *How prices have changed in the shops in your most recent visits (i.e. the last 1 to 6 months).*
2. *How prices have changed in the shops over the longer term (i.e. the last 12 months or more)*
3. *The current level of interest rates.*
4. *The current strength of the British Economy.*
5. *The inflation target set by the government.*
6. *Reports on inflation outlook in the media.*
7. *Reports of VAT changes in the media.*
8. *Other factor(s).*

Asked: 2009, 2010, 2011, 2013

Information indicator: =1 if 'very important' selected for option 6, =0 otherwise.

Question 2e *And which, if any, of the same factors were important in getting to your expectation of how prices will change over the longer term (say in 5 years time)?*

1. *How prices have changed in the shops in your most recent visits (i.e. the last 1 to 6 months).*
2. *How prices have changed in the shops over the longer term (i.e. the last 12 months or more)*
3. *The current level of interest rates.*
4. *The current strength of the British Economy.*
5. *The inflation target set by the government.*
6. *Reports on inflation outlook in the media.*
7. *Reports of VAT changes in the media.*
8. *Other factor(s).*

Asked: 2011, immediately after Question 2d

Information indicator: =1 if item 6 selected, =0 otherwise.

Question 2f *What were the most important factors in getting to your expectation for how prices in the shops would change over the next 12 months?*

Please select up to 4:

1. *How prices have changed in the shops recently, over the last 12 months*
2. *How prices have changed in the shops, on average, over the longer term i.e the last few years*
3. *Reports of current inflation in the media*
4. *Discussion of the prospects for inflation in the media*
5. *The level of interest rates*
6. *The inflation target set by the government*

7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *Other factors*
10. *None*

Asked: 2016

Information indicator: =1 if items 3 or 4 selected, =0 otherwise.

Question 2g *And what were the most important factors in getting to your expectation for how prices in the shops would change over the longer term (say in 5 years' time)?*

Please select up to 4:

1. *How prices have changed in the shops recently, over the last 12 months*
2. *How prices have changed in the shops, on average, over the longer term i.e the last few years*
3. *Reports of current inflation in the media*
4. *Discussion of the prospects for inflation in the media*
5. *The level of interest rates*
6. *The inflation target set by the government*
7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *Other factors*
10. *None*

Asked: 2016

Information indicator: =1 if items 3 or 4 selected, =0 otherwise.

D.2 The relationship of planned consumption with measured information and subjective models

To confirm that the survey measures of information and subjective models uncover meaningful aspects of household beliefs, I consider how they correlate with planned consumption behavior. To this end, I use the following survey question:

Question 3 *Which, if any, of the following actions are you taking, or planning to take, in the light of your expectations of price changes over the next twelve months?*

- *Cut back spending and save more.*

Crucially, this asks about consumption choices which are explicitly driven by expected inflation.⁴⁵ A household answering ‘yes’ to this question, and who reports elsewhere in the survey that they expect prices to rise in the next year, is therefore indicating that $dc_{it}/d\mathbb{E}_{it}p_{t+1} < 0$. A question that only asked about consumption or consumption changes, without reference to the cause of the behavior, would conflate this with reactions to expectations of other variables, which might also be influenced by the same shocks as expected inflation, either directly or through cross-learning. Question 3 is therefore informative about the sign of $\frac{dc_{it}}{d\mathbb{E}_{it}p_{t+1}}$. If current prices are taken as given by the household, then this is the same as the sign of $\tilde{\chi}_i = \frac{dc_{it}}{d\mathbb{E}_{it}\pi_{t+1}}$.

The vast majority of respondents (98%) expect positive inflation over the next 12 months.⁴⁶ For these households, yes and no responses to Question 3 respectively indicate that:

$$\frac{dc_{it}}{d\mathbb{E}_{it}p_{t+1}} \begin{cases} < 0 & \text{if answer yes} \\ \geq 0 & \text{if answer no} \end{cases} \quad (178)$$

For the minority who expect deflation, these inequalities are reversed: responding with ‘yes’ indicates consumption is being cut because of an expected fall in prices. I therefore define the following indicator:

$$\widetilde{\frac{dc_{it}}{d\mathbb{E}_{it}p_{t+1}}} = \begin{cases} 1 & \text{if Q3=‘no’ and } \mathbb{E}_{it}\pi_{t+1} > 0 \\ 0 & \text{if Q3=‘yes’ and } \mathbb{E}_{it}\pi_{t+1} > 0 \\ 1 & \text{if Q3=‘yes’ and } \mathbb{E}_{it}\pi_{t+1} < 0 \\ 0 & \text{if Q3=‘no’ and } \mathbb{E}_{it}\pi_{t+1} < 0 \end{cases} \quad (179)$$

For the large majority who expect inflation, this is equal to 1 if $\frac{dc_{it}}{d\mathbb{E}_{it}p_{t+1}} \geq 0$, and equal to 0 if the reaction to expected price rises is strictly negative. The same is true of the minority who expect deflation, except that any household with $\frac{dc_{it}}{d\mathbb{E}_{it}p_{t+1}} = 0$ would respond ‘no’ to Question 3, and so is counted as if their response to expected price rises is strictly negative. The mislabeling is not a large

⁴⁵Another question in the survey asks if the respondent will “bring forward major purchases such as furniture or electrical goods” as a result of expected inflation. I do not use this for two reasons. First, as Nunes and Park (2020) note, the question refers specifically to durable goods, which may not respond to prices in the same way as aggregate consumption, the object of interest. Second, it is very rarely chosen: just 6% of respondents said they would bring forward major purchases. In contrast, 40% report that they will cut back spending and save more. Any estimation on this variable will therefore be heavily influenced by a small subset of agents.

⁴⁶The analysis in this section excludes any households who report expecting zero inflation over the next 12 months, or who do not answer the inflation expectation question, as Question 3 is difficult to interpret for these households. I discuss the appropriate counterfactual implicit in the question below. Including these people, 79% of respondents to Question 3 expect positive inflation, 7% expect zero inflation, 2% expect deflation, and 12% do not answer.

issue, as less than 1% of respondents to Question 3 both expect deflation and answer ‘no’. The results below are robust to removing the few households who expect deflation (see Table 5 column 2).

Table 5 shows how this is related to the information indicator and the subjective models (responses to Question 1). Column 1 shows the results from estimating a probit regression of $\frac{\tilde{dc}_{it}}{dE_{+1}}$ on the information indicator interacted with subjective models (Question 1), plus the standard household controls and time fixed effects used above. The coefficient on information is significantly negative for those with negative subjective models of inflation, despite the fact that substitution effects imply $\frac{dc_{it}}{dE_{it}p_{t+1}} \geq 0$ in many standard models. Being informed is therefore associated with a *lower* probability of responding positively to expected inflation for these households.

However, for those who believe inflation makes the economy stronger, being informed is associated with a significantly higher $\Pr(\frac{dc_{it}}{dE_{it}\pi_{t+1}} \geq 0)$. For those who believe inflation makes no difference, the average value of $\Pr(\frac{dc_{it}}{dE_{it}\pi_{t+1}} \geq 0)$ with and without information, which is also consistent with the interpretation of these variables as $\frac{\tilde{dc}_{it}}{dE_{+1}} = 1$ includes the case where $\frac{dc_{it}}{dE_{it}\pi_{t+1}} = 0$.

Table 5: Consumption response to inflation correlates with information, by subjective model

	(1)	(2)
	c response to $E\pi$	c response to $E\pi$
information indicator=1	-0.213*** (0.0611)	-0.224*** (0.0613)
end up stronger	0.0108 (0.0891)	0.0392 (0.0906)
information indicator=1 $\times$ end up stronger	0.348* (0.185)	0.313* (0.186)
make little difference	0.130** (0.0594)	0.157*** (0.0600)
information indicator=1 $\times$ make little difference	0.0240 (0.126)	-0.0149 (0.128)
dont know	0.0958 (0.0833)	0.0978 (0.0846)
information indicator=1 $\times$ dont know	-0.0158 (0.186)	-0.0342 (0.187)
Expected Inflation	All	Exclude Deflation
Controls	All	All
Time FE	Yes	Yes
Observations	4940	4871

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the results of probit regressions of the $\frac{dc_{it}}{dE_{it}\pi_{t+1}}$ indicator on the information indicator, interacted with responses to Question 1. The omitted category is a household with information indicator=0 who holds the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

This is consistent with individuals filtering information through their subjective models of the economy. If a household who believes inflation weakens the economy gets more information about future positive inflation, their subjective model implies that they should cut consumption, because bad times lie ahead. If instead a household believes inflation strengthens the economy, then they will react in the opposite way to the same inflation. The overall correlation of information and consumption response is negative because the majority of households believe inflation makes the economy weaker. This therefore supports the claim that the information indicator and answers to Question 1 reflect the information and subjective models used by households in making their consumption decisions.

The analysis here assumes that when asked whether they will cut back consumption and save more, households are comparing their actions to a counterfactual in which there are no price rises over the next 12 months. An alternative possibility is that they are comparing with a consumption plan made in the past, in which case the relevant counterfactual is where expected inflation is unchanged from the level expected when the plan was made. I consider this in two ways, and find that the qualitative patterns in reported consumption responses to inflation are the same for households expecting inflation to increase or decrease relative to the previous year. It does not therefore appear that past inflation is the relevant counterfactual for most respondents.

First, column 2 of Table 5 re-runs the regression in column 1, excluding any respondent who reports expecting prices to fall over the next year. All results are qualitatively the same as over the full sample, showing that the few respondents expecting deflation are not driving the results.

Second, I split the sample by the sign of the respondent's expected change in inflation, computed as the sign of the difference between 12-month ahead inflation forecast and their perception of inflation over the previous 12 months. The results are in Table 6. The sample sizes in each group are substantially smaller than over the full sample, so some significance is lost, but importantly the signs of the key coefficients remain the same. In each group, households who believe inflation makes the economy weaker are less likely to have $\frac{dc_{it}}{dE_{it}\pi_{t+1}} \geq 0$ when they get inflation information. For households who believe inflation makes the economy stronger, this effect is reversed. The similarity of these patterns suggests that most respondents use 'no price change' as the counterfactual when answering Question 3, not 'no inflation change'. If the latter was used, we would expect to see changes of sign across the columns in Table 6, as a household expecting a fall in inflation would be reporting $-1 \times \frac{dc_{it}}{dE_{it}\pi_{t+1}}$, while one expecting a rise in inflation would report $\frac{dc_{it}}{dE_{it}\pi_{t+1}}$.

Table 6: Consumption response to inflation correlates with information, by subjective model and sign of perceived $E\pi$ change.

	(1)	(2)	(3)
	$E\Delta\pi < 0$	$E\Delta\pi = 0$	$E\Delta\pi > 0$
Dc_Dpi			
Information=1	-0.140 (0.116)	-0.305*** (0.101)	-0.257** (0.107)
end up stronger	0.0668 (0.164)	-0.178 (0.151)	0.195 (0.165)
Information=1 × end up stronger	0.586 (0.441)	0.349 (0.293)	0.397 (0.307)
make little difference	0.165 (0.111)	0.136 (0.0957)	0.181 (0.112)
Information=1 × make little difference	0.129 (0.241)	-0.300 (0.211)	0.113 (0.216)
dont know	0.156 (0.176)	0.0293 (0.128)	0.0264 (0.167)
Information=1 × dont know	-0.141 (0.354)	0.469 (0.359)	0.117 (0.325)
Controls	All	All	All
Time FE	Yes	Yes	Yes
Observations	1384	1876	1463

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the results of probit regressions of the $\frac{\widetilde{dc_{it}}}{dE_{it}\pi_{t+1}}$ indicator on the information indicator, interacted with responses to Question 1, split by the sign of the respondent's inflation expectations. The omitted category in all cases is a household with information indicator=0 who holds the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

D.3 Cross-sectional patterns in information on inflation

Columns 1-3 of Table 7 show the results of probit regressions of the information indicator on subjective models, controls, and period fixed-effects, for three subsamples. The first only uses questions about the information used to arrive at the respondent's *change* in expected inflation, and the second uses only questions about information used to form point forecasts. The third column excludes questions relating to forecast horizons longer than 12 months. The signs of the marginal effects are the same as in the main exercise in Table 1, though they are not significant in the case of

the revisions questions, as the sample size is small.

Table 7: Information correlates with subjective models, split by information question type

	(1) Revision	(2) Point	(3) Short horz.	(4) Extra Qs	(5) Q2d wider	(6) +Other
end up stronger	0.0575 (0.0380)	-0.0335 (0.0218)	-0.0123 (0.0206)	0.00114 (0.0196)	-0.00126 (0.0196)	-0.00779 (0.0205)
make little difference	-0.0191 (0.0233)	-0.0331** (0.0155)	-0.0392*** (0.0141)	-0.0310** (0.0132)	-0.0312** (0.0131)	-0.0429*** (0.0139)
dont know	-0.0408 (0.0297)	-0.0715*** (0.0206)	-0.0622*** (0.0192)	-0.0663*** (0.0174)	-0.0472*** (0.0180)	-0.0663*** (0.0191)
Controls	All	All	All	All	All	All
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2364	5906	6848	8306	8270	8270

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the average marginal effects from estimating probit regressions of the information indicators constructed from subsets of the questions listed in Appendix D.1 on the responses to Question 1. The omitted category is the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

The remaining columns of Table 7 repeat the regression for broader definitions of the information dummy than that used in Table 1. In the fourth column, the information indicator includes Questions 2e and 2g. In the fifth column, I extend the criteria for setting the information indicator equal to 1 in Question 2d to account for the fact that some people may be unwilling to select the highest importance box for any information source. I therefore set the information indicator to 1 if in answer to Question 2d, the respondent selects ‘very important’ for direct inflation information (as before), or if they do not select ‘very important’ for any option, but do respond that four or fewer options were ‘fairly important’, and direct inflation information is among them. In the final column, I set the information indicator =1 if the household chooses a direct information source or ‘Other’, in case this includes direct information sources (e.g. checking the Bank of England published forecasts). In all of these, the results are robust.

To account for possible selection bias from missing observations, I estimate a version of Table 1 amended for selection as in Heckman (1979). As in Michelacci and Paciello (2024),

Table 8: Information correlates with subjective models, with selection correction

	(1)	(2)
	Information	Information
end up stronger	-0.0178 (0.0172)	-0.0183 (0.0172)
make little difference	-0.0306** (0.0120)	-0.0315*** (0.0121)
dont know	-0.0575*** (0.0172)	-0.0579*** (0.0173)
Inverse Mills ratio	-0.282*** (0.0820)	-0.0696** (0.0355)
<i>Selection stage</i>		
Economic Literacy	0.205*** (0.0226)	
HH does not know past π		-0.876*** (0.0365)
r affects π in 1-2 months		0.0334 (0.0236)
r affects π in 1-2 yrs		0.0882*** (0.0231)
Pseudo- R^2 (selection)	0.103	0.127
Controls	All	All
Time FE	Yes	Yes
Observations	18026	18026
Standard errors in parentheses		
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$		

Note: The table reports the coefficients from estimating a linear regression of the information indicator defined in Section 5.1 on the responses to Question 1, augmented with the inverse Mills ratio from a first-stage probit regression of whether the information indicator is observed on measures of economic literacy defined above. The omitted category is the belief that inflation makes the economy weaker. The selection stage is only run for quarters in which the information questions were asked. The model is run using the 2-step limited information method in Heckman (1979). Time fixed effects and controls as in footnote 26 are included in both stages.

I predict observing the relevant survey response using a measure of economic literacy. Here the relevant response is only the information indicator, as there are no missing values for Question 1. Following Michelacci and Paciello (2024), economic literacy is measured with three indicators: the household reports a value for perceived current inflation, and they answer ‘agree’ or ‘strongly agree’ to the statements “a rise in interest rates makes prices in the high street rise more slowly in the short term (say a month or two)” and “a rise in interest rates makes prices rise more slowly in the medium term (say a year or two)”. I estimate versions of the model with this as an aggregate index (=1 if

and only if the household scores on all components), and with the components disaggregated. The results are in Table 8. The predictors used in the first stage are strongly significant. Qualitatively the second-stage results are unchanged from Table 1, and the quantitative differences are small.

D.4 Time series patterns in subjective models of inflation

Bhandari et al. (2024) also study the time series of responses to Question 1, and conclude that households are more pessimistic about inflation when output growth is low. To explore this, I regress the proportion of households responding ‘end up weaker’ on realized annual CPI inflation and quarterly GDP growth. The results are in column 2 of Table 9. Consistent with Bhandari et al. (2024), the coefficient on GDP growth is significantly negative. However, the R^2 is only slightly higher than that of a regression on inflation only (column 1), so GDP growth does not account for much of the variation in survey answers. Indeed, GDP growth does not have any significant relationship with the proportion of households with a negative view of inflation outside of the four worst months of the Great Recession (column 3).

Table 9: Regressions of the proportion of households answering weaker to Question 1 on aggregate variables.

	(1)	(2)	(3)
	Proportion weaker	Proportion weaker	Proportion weaker
Inflation	0.0568*** (0.00489)	0.0517*** (0.00479)	0.0501*** (0.00469)
GDP growth		-0.0261*** (0.00869)	-0.0110 (0.0180)
Constant	0.466*** (0.0109)	0.487*** (0.0123)	0.482*** (0.0152)
Omitted quarters	None	None	2008Q2-2009Q1
R-squared	0.615	0.647	0.554
Observations	70	70	66

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the results of regressing the proportion of households answering Question 1 that inflation makes the economy weaker on annual CPI inflation and quarter-on-quarter real GDP growth. Proportions are computed using survey weights.

To explore which measure of inflation affects subjective models, Table 10 reports the results of regressing an indicator variable for if the respondent reports a negative subjective model on a variety of inflation measures. The first column uses CPI inflation, so is very similar to the time-series

regression in Table 9. Columns 2-4 use more granular measures of the inflation rate experienced by different households, split by whether they are above retirement age (65), above median income, and by their housing tenure. Inflation rates split by these characteristics are provided by the ONS.⁴⁷ Column 5 uses perceived current inflation. The different realized inflation measures are strongly correlated, so cannot be included jointly. Although the coefficient sizes vary as the different inflation rates have different levels of volatility, in all cases higher inflation is associated with a significantly greater probability of reporting a negative subjective model. The R^2 is highest for perceived inflation, supporting the choice of assumptions on subjective model updating in Section 6.5.

Table 10: Probability of reporting negative subjective model by experienced and perceived inflation

	(1)	(2)	(3)	(4)	(5)
	Weaker	Weaker	Weaker	Weaker	Weaker
Inflation	0.0510*** (0.00177)	0.0463*** (0.00170)	0.0457*** (0.00165)	0.0292*** (0.00137)	0.0254*** (0.000720)
Inflation measure	CPI	by retirement	by income	by housing	perceived
Controls	All	All	All	All	All
R-squared	0.0303	0.0286	0.0292	0.0237	0.0371
Observations	68269	68269	68269	68269	68269

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the results of estimating a linear probability model of whether a respondent reports that inflation makes the economy weaker in response to Question 1 on various measures of inflation. These are annual CPI inflation, inflation split by whether the respondent is of retirement age, split by whether the respondent has above or below median income, split by the respondent's housing tenure, and finally the respondent's perceived current inflation. Sample begins in 2006 Q1, as this is when the ONS sub-group inflation data is available from. Households not reporting a perceived rate of inflation are dropped in all regressions. All regressions are weighted using the survey weights provided in the IAS.

⁴⁷Finer decompositions of inflation by household characteristics are not reliable, given the data available for the UK (see e.g. ?).

Table 11: Regressions of the proportion of households giving each answer to Question 1 on aggregate variables.

	(1) Proportion	(2) Proportion	(3) Proportion
<i>Stronger</i>			
Inflation	-0.0123*** (0.00193)	-0.0116*** (0.00215)	-0.0108*** (0.00221)
GDP growth		0.00346 (0.00363)	-0.00392 (0.00646)
Constant	0.104*** (0.00431)	0.102*** (0.00550)	0.104*** (0.00638)
<i>No difference</i>			
Inflation	-0.0292*** (0.00303)	-0.0262*** (0.00313)	-0.0257*** (0.00314)
GDP growth		0.0150*** (0.00473)	0.0106 (0.0107)
Constant	0.277*** (0.00772)	0.264*** (0.00883)	0.266*** (0.0103)
<i>Don't know</i>			
Inflation	-0.0154*** (0.00249)	-0.0139*** (0.00262)	-0.0135*** (0.00267)
GDP growth		0.00762* (0.00423)	0.00428 (0.00987)
Constant	0.153*** (0.00687)	0.147*** (0.00757)	0.148*** (0.00884)
Omitted quarters	None	None	2008Q2-2009Q1
Observations	70	70	66

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the results of regressing the proportion of households giving each answer to Question 1 (except 'weaker') on annual CPI inflation and quarter-on-quarter real GDP growth. Proportions are computed using survey weights. The R^2 of the core regressions in column 1 are 0.388 (stronger), 0.534 (no difference), and 0.355 (don't know).

Similar patterns in reverse are observed for the other answers. Table 11 repeats the regressions of Table 9, replacing the dependent variable with the proportion of respondents choosing each of the other answers to Question 1. In all cases, inflation accounts for a large share of the variation in survey answers, and higher inflation is associated with significantly lower proportions giving each

answer. Higher GDP growth is associated with higher proportions on these other answers, but that relationship is not significantly different from zero for any answer when excluding the worst of the Great Recession.

To test if the distribution of beliefs about inflation shifts when the economy reaches the ZLB, I estimate an ordered probit regression of subjective models of inflation in the zero lower bound period, and a variety of controls.⁴⁸ A response that inflation makes the economy stronger is coded as the highest value, and inflation makes the economy weaker is the lowest value (I exclude the ‘don’t know’ answers). A positive coefficient on the zero lower bound period would therefore imply a shift towards believing inflation makes the economy stronger, as we would expect if households follow a standard New Keynesian model. This is not what the results in Table 12 show: there is no significant shift towards a positive view of inflation in the ZLB period.

Table 12: Ordered probit regressions of subjective models of inflation on whether the economy is at the zero lower bound on nominal interest rates.

	(1)	(2)	(3)
	Subjective model	Subjective model	Subjective model
Subjective model			
ZLB	-0.00801 (0.00937)	-0.00785 (0.00962)	-0.00513 (0.00972)
Controls	None	Household	Household + macro
Observations	83526	83526	83526

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the results of an ordered probit regression of answers to Question 1 on an indicator for whether the UK economy was at the zero lower bound, defined as the period from 2009Q2 to the end of 2019 (end of the sample). The ordering is: “stronger”, “no difference”, “weaker”. Those answering “no idea” are omitted. All regressions are weighted using the survey weights provided in the IAS.

D.5 Perceived and expected inflation across households

To account for possible selection bias from missing observations, I estimate a version of Table 2 amended for selection as in Heckman (1979). As in Appendix D.3, I predict observing perceived and expected inflation using the components of the economic literacy indicator in Michelacci and Paciello (2024), this time removing the component concerning whether perceived inflation is reported as this is closely related to the dependent variables of the regressions. The results are in Table 13. The predictors used in the first stage are strongly significant. Qualitatively the second-stage results are unchanged from Table 2, and the quantitative differences are small.

⁴⁸The first column of Table 12 has no controls, the second includes the set of household-level covariates used throughout the paper, and the third adds inflation and GDP growth.

Table 13: Perceived and expected inflation are higher for those with more negative subjective models, with selection correction

	(1)	(2)
	Perceived inflation	Expected inflation
end up stronger	-0.724*** (0.0319)	-0.607*** (0.0301)
make little difference	-0.548*** (0.0213)	-0.478*** (0.0201)
dont know	-0.452*** (0.0292)	-0.407*** (0.0280)
Inverse Mills ratio	0.714*** (0.248)	0.168 (0.191)
<i>Selection stage</i>		
r affects π in 1-2 months	0.133*** (0.0209)	0.192*** (0.0209)
r affects π in 1-2 yrs	0.278*** (0.0205)	0.310*** (0.0205)
Pseudo- R^2 (selection)	0.0477	0.0536
Controls	All	All
Time FE	Yes	Yes
Observations	95339	95339

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The table reports the coefficients from estimating a linear regression of perceived and expected inflation on the responses to Question 1, augmented with the inverse Mills ratio from a first-stage probit regression of whether the dependent variable is observed on measures of economic literacy. The omitted category is the belief that inflation makes the economy weaker. The model is run using the 2-step limited information method in Heckman (1979). Time fixed effects and controls as in footnote 26 are included in both stages.

E New Keynesian model further details

E.1 Proofs

Supply-side derivations. There is a perfectly competitive final goods producer which combines intermediate goods varieties with a CES production function. The demand facing an intermediate goods producer j with price P_t^j is therefore:

$$Y_t^j = \left(\frac{P_t^j}{P_t} \right)^{-\varepsilon} Y_t \quad (180)$$

where ε is the elasticity of substitution between intermediate varieties, Y_t is aggregate production and:

$$P_t = \left(\int (P_t^j)^{1-\varepsilon} dj \right)^{\frac{1}{1-\varepsilon}} \quad (181)$$

is the aggregate price level. Intermediate goods firms are therefore monopolistic. They produce using labor as their only input, according to a linear production function:

$$Y_t^j = A_t N_t^j \quad (182)$$

where A_t is exogenous total factor productivity, common to all firms j .

The profit maximization problem of intermediate goods producer j in period t is therefore:

$$\max_{P_t^j, N_t^j} \mathbb{E}_t \sum_{s=0}^{\infty} \Lambda_{t,t+s} \left[P_{t+s}^j Y_{t+s}^j - W_{t+s} N_{t+s}^j - \frac{\Psi}{2} \left(\frac{P_t^j}{P_{t-1}^j} - 1 \right)^2 P_t Y_t \right] \quad (183)$$

subject to the demand function (180) and the production function (182). Ψ is a positive constant, reflecting the degree of price stickiness. Firms have rational expectations. They are owned by a mutual fund which pays out equally to all households, so the stochastic discount factor is:

$$\Lambda_{t,t+s} = \beta^s \left(\frac{\bar{C}_{t+s}}{\bar{C}_t} \right)^{-\frac{1}{\sigma}} \quad (184)$$

It is well-known that solving this firm problem, noting that all firms are identical so choose the same price, and then log-linearizing about the steady state with $\Pi_t = P_t/P_{t-1} = 1$, gives:⁴⁹

$$\pi_t = \beta E_t^{RE} \pi_{t+1} + \frac{\varepsilon - 1}{\Psi} mc_t \quad (185)$$

where $mc_t = W_t/(A_t P_t)$ is real marginal costs. This corresponds to equation 42, with $\kappa = (\varepsilon - 1)/\Psi$.

The labor packer aggregates varieties from labor unions with a CES function, so the demand facing a labor union k with nominal wage W_t is:

$$N_t^k = \left(\frac{W_t^k}{W_t} \right)^{-\varepsilon_N} N_t \quad (186)$$

⁴⁹See e.g. Rotemberg (1987), Ascari and Rossi (2012).

where ε_N is the elasticity of substitution between labor varieties, N_t is aggregate labor supply, and:

$$W_t = \left(\int (W_t^k)^{1-\varepsilon_N} dk \right)^{\frac{1}{1-\varepsilon_N}} \quad (187)$$

is the aggregate nominal wage.

Labor unions maximize wage revenue, net of an increasing convex disutility from supplying labor of $\nu(N_t^k)$. They are also subject to quadratic costs of adjusting real wages, and are owned by all households, giving the same stochastic discount factor $\Lambda_{t,t+s}$ as firms. The problem of a labor union k in period t is therefore:

$$\max_{W_t^k, N_t^k} E_t^{RE} \sum_{s=0}^{\infty} \Lambda_{t,t+s} \left[W_{t+s}^k N_{t+s}^k - \nu(N_{t+s}^k) - \frac{\Psi_N}{2} \left(\frac{W_t^k}{W_{t-1}^k} \frac{P_{t-1}}{P_t} - 1 \right)^2 W_t N_t \right] \quad (188)$$

I take the real wage rigidity parameter $\Psi_N \rightarrow \infty$, so the the solution to the union problem is to set constant real wages:

$$\frac{W_t^k}{P_t} = \frac{W_{t-1}^k}{P_{t-1}} \quad (189)$$

All unions are therefore identical, and the log-linearized solution for real marginal costs is:

$$mc_t = w_t - p_t - a_t = -a_t \quad (190)$$

Finally, I derive equation 45 with a guess-and-verify approach. Guess that the law of motion for inflation is as in equation 45, where $E_t^{RE} v_{\pi_{t+1}} = 0$. The rational expectation of π_{t+1} is therefore $E_t^{RE} \pi_{t+1} = \rho_\pi \pi_t$. Substituting this and equation 190 into equation 42 gives:

$$\pi_t = -\frac{\kappa}{1 - \beta \rho_\pi} a_t \quad (191)$$

Assume TFP follows the exogenous AR(1) process:

$$a_t = \rho_a a_{t-1} + v_{at} \quad (192)$$

This we have that:

$$\pi_t = \rho_a \pi_{t-1} - \frac{\kappa}{1 - \beta \rho_\pi} v_{at} \quad (193)$$

which verifies the guess and implies $\rho_\pi = \rho_a$.

Forecasts using the subjective model. The subjective model represented by equations 47 - 49 can be written in VAR form as:

$$Y_t = A^i Y_{t-1} + B^i U_t \quad (194)$$

where:

$$Y_t = (\pi_t, y_t, i_t)', \quad A = \begin{bmatrix} \rho_\pi^i & 0 & 0 \\ (\alpha^i + \lambda^i \phi^i) \rho_\pi^i & \rho_y^i & 0 \\ \phi \rho_\pi & 0 & 0 \end{bmatrix} \quad (195)$$

$$U_t = (u_{\pi t}, u_{yt}, u_{it})', \quad B = \begin{bmatrix} 1 & 0 & 0 \\ \alpha^i + \lambda^i \phi^i & 1 & \lambda^i \\ \phi^i & 0 & 1 \end{bmatrix}$$

To form a forecast of future variables, a fully-informed agent would use:

$$E_{it}^* Y_{t+s} = (A^i)^s Y_t \quad (196)$$

That is, their forecasts are optimal given their subjective model. To find $(A^i)^s$, first find diagonal matrix D^i and matrix P^i such that:

$$A^i = P^i D^i (P^i)^{-1} \quad (197)$$

This is satisfied with:

$$P = \begin{bmatrix} 0 & (\phi^i)^{-1} & 0 \\ 0 & \frac{(\alpha^i + \lambda^i \phi^i) \rho_\pi^i}{\phi^i (\rho_\pi^i - \rho_y^i)} & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad D^i = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \rho_\pi^i & 0 \\ 0 & 0 & \rho_y^i \end{bmatrix} \quad (198)$$

We then have that:

$$(A^i)^s = P^i (D^i)^s (P^i)^{-1} = P^i \cdot \begin{bmatrix} 0^s & 0 & 0 \\ 0 & (\rho_\pi^i)^s & 0 \\ 0 & 0 & (\rho_y^i)^s \end{bmatrix} \cdot (P^i)^{-1}$$

$$= \begin{bmatrix} (\rho_\pi^i)^s & 0 & 0 \\ \frac{(\alpha^i + \lambda^i \phi^i) \rho_\pi^i}{\rho_\pi^i - \rho_y^i} ((\rho_\pi^i)^s - (\rho_y^i)^s) & (\rho_y^i)^s & 0 \\ \phi^i (\rho_\pi^i)^s & 0 & 0 \end{bmatrix} \quad (199)$$

This implies equations 52 - 54.

Output law of motion. Aggregating over household consumption functions (55), and using equation 44, we have:

$$y_t = \bar{\Theta}y_t - \sigma\beta r_t + \int \omega_i \tilde{\chi}_i E_{it} \pi_t di \quad (200)$$

where ω_i is a weight on household i such that aggregate consumption $c_t = \int \omega_i c_{it} di$, and:

$$\bar{\Theta} = \int \omega_i \frac{1 - \beta}{1 - \beta \rho_{yi}} di \quad (201)$$

Rearranging, and substituting out for $E_{it} \pi_t$ using equation 50, yields

$$y_t = \alpha \pi_t + \lambda r_t + \rho_y \int \hat{\omega}_i E_{it-1} \pi_{t-1} di \quad (202)$$

with

$$\begin{aligned} \alpha &= \frac{1}{1 - \bar{\Theta}} \int \omega_i \tilde{\chi}_i \tau_i di, & \lambda &= -\frac{\sigma\beta}{1 - \bar{\Theta}}, \\ \rho_y &= \frac{1}{1 - \bar{\Theta}}, & \hat{\omega}_i &= \omega_i \rho_{\pi i} \tilde{\chi}_i (1 - \tau_i) \end{aligned} \quad (203)$$

Using this law of motion, and equation 43, y_{t-1} can be written as:

$$y_{t-1} = \left[\lambda \phi + \rho_y \int \omega_i \tilde{\chi}_i \tau_i di \right] \pi_{t-1} + \rho_y \int \hat{\omega}_i E_{it-2} \pi_{t-2} di \quad (204)$$

Using equation 50, the lagged term in the true law of motion can be written as:

$$\int \hat{\omega}_i E_{it-1} \pi_{t-1} di = \left[\int \hat{\omega}_i \tau_i di \right] \pi_{t-1} + \int \hat{\omega}_i \rho_{i\pi} (1 - \tau_i) E_{it-2} \pi_{t-2} di \quad (205)$$

The lagged terms in the subjective model (204) and the true law of motion (205) for y_t are both therefore determined by realized π_{t-1} and a weighted average of $E_{it-2} \pi_{t-2}$, with the weights increasing in $\hat{\omega}_i$. While not exactly the same, equation 49 therefore provides a tractable approximation to the functional form of 202.

E.2 Microfounding the imposed properties of information and subjective models

First, I show that the subjective model properties defined in Section 6.5 are sufficient for the model to qualitatively match empirical Results 2 and 3. I begin with Result 3. Subjective model property 2 implies that:

$$\frac{\partial \tilde{\chi}_i}{\partial E_{it}\pi_t} = \frac{\partial \hat{\alpha}_{it}}{\partial E_{it}\pi_t} \cdot \frac{\partial \tilde{\chi}_i}{\partial \hat{\alpha}_{it}} = \frac{\partial \hat{\alpha}_{it}}{\partial E_{it}\pi_t} \cdot \frac{\beta(1-\beta)\rho_{\pi i}}{(1-\beta\rho_{\pi i})(1-\beta\rho_{yi})} < 0 \quad (206)$$

where the second equality makes use of equation 55. Higher perceived inflation is therefore associated with more negative beliefs about the impact of inflation on consumption.

For Result 2, note that from equation 50:

$$\frac{\partial \tilde{\chi}_i}{\partial \pi_t} = \tau_i \frac{\partial \tilde{\chi}_i}{\partial E_{it}\pi_t} \quad (207)$$

Combining this with equation 206, and the fact that $\tau_i \geq 0$, we have that:

$$\frac{\partial}{\partial \pi_t} \left[\Pr \left(\tilde{\chi}_i |_{\hat{\alpha}_{it}} < X \right) \right] \geq 0 \quad (208)$$

for any threshold X . The inequality is strict if $\tau_i > 0$, and is an equality otherwise. As realized inflation rises, each household becomes more likely to report that inflation makes the economy weaker.

Next, I show how a model with information costs leads to equation 57.

For this, it is helpful to first derive an expression for the expected utility loss due to imperfect information.

Lemma 2 *Let $E_{i0}^* U_{i0}^*$ denote the expected utility of an otherwise identical household who observes Π_t precisely before choosing C_{it} . Furthermore, let $\hat{U}_{i0}^*$ and $\hat{U}_{i0}$ denote the log-quadratic approximation to the discounted utility of the fully-informed and uninformed households respectively. The expected utility loss from imperfect information about Π_t is:*

$$E_{i0}(\hat{U}_{i0}^* - \hat{U}_{i0}) = \frac{(\bar{C}_i)^{1-\frac{1}{\sigma}}}{2\sigma} E_{i0} \sum_{t=0}^{\infty} \beta^t (c_{it} - c_{it}^*)^2 \quad (209)$$

where lower-case letters are log-deviations of the corresponding variables from steady state, and c_{it}^* denotes the period- t consumption of the fully-informed household.

Proof. The proof is an adaptation of the derivation of expression (34) in [Maćkowiak and Wiederholt \(2015\)](#). First, substitute the budget constraint (40) into the utility function (39) to obtain:

$$E_{i0}U_{i0} = E_{i0}\beta^t \frac{1}{1 - \frac{1}{\sigma}} \left(\frac{R_{t-1}}{\Pi_t} B_{it-1} + Y_t - B_{it} \right)^{1 - \frac{1}{\sigma}} \quad (210)$$

Write this in log-deviations from steady state, where $\bar{X}$ denotes the steady state value of the corresponding variable X_t , and $x_t \equiv \log(X_t/\bar{X})$ is the corresponding log-deviation:

$$E_{i0}U_{i0} = E_{i0}\beta^t \frac{1}{1 - \frac{1}{\sigma}} \left(\bar{R}\bar{B}_i \exp(r_{t-1} - \pi_t + b_{it-1}) + \bar{Y} \exp(y_t) - \bar{B}_i \exp(b_{it}) \right)^{1 - \frac{1}{\sigma}} \quad (211)$$

We then take a quadratic approximation of this with respect to each variable in log-deviation, about the steady state. For this, define $z_t = (r_{t-1}, \pi_t, y_t)'$ as the vector of exogenous variables taken as given by the household in period t . The past asset choice b_{it-1} is also taken as given in period t , and b_{it} is the only choice variable. After the quadratic approximation, expected discounted utility is given by:

$$\begin{aligned} E_{i0}U_{i0} \approx E_{i0}\hat{U}_{i0} = \bar{U}_i + E_{i0} \sum_{t=0}^{\infty} \beta^t & \left[h_b b_{it} + h_z z_t + \frac{1}{2} H_{bb,-1} b_{it} b_{it-1} + \frac{1}{2} H_{bb,0} b_{it}^2 + \frac{1}{2} H_{bb,1} b_{it+1} b_{it} \right. \\ & + \frac{1}{2} b_{it} H_{bz,0} z_t + \frac{1}{2} b_{it} H_{bz,1} z_{t+1} + \frac{1}{2} z_t' H_{zz,0} z_t + \frac{1}{2} z_t' H_{zb,-1} b_{it-1} + \frac{1}{2} z_t' H_{zb,0} b_{it} \Big] \\ & + \beta^{-1} \left(h_{-1} b_{i,-1} + \frac{1}{2} H_{-1} (b_{i,-1})^2 + \frac{1}{2} H_{bb,1} b_{i,-1} b_{i0} + \frac{1}{2} b_{i,-1} H_{bz,1} z_0 \right) \end{aligned} \quad (212)$$

where $\beta^t h_b$ denotes the first derivative of U_{i0} with respect to b_{it} , evaluated at the steady state. Similarly, h_z denotes the vector of first derivatives of U_{i0} with respect to z_t , evaluated at steady state. The matrices $\beta^t H_{jk,\tau}$ denote the second derivatives of U_{i0} with respect to j_t and $k_{t+\tau}$, for $j_t, k_t \in \{b_{it}, z_t\}$, evaluated at steady state. $\beta^{-1} h_{-1}$ and $\beta^{-1} H_{-1}$ are the first and second derivatives of U_{i0} with respect to initial wealth $b_{i,-1}$, evaluated at steady state. As in [Maćkowiak and Wiederholt \(2015\)](#), note that there are no cross-products of b_t and z_{t-1} , because from equation 211 the first derivative of U_{i0} with respect to b_{it} does not depend on any elements of z_{t-1} . Similarly, there are no terms in the interaction of z_t and z_{t-1} or z_{t+1} .

We now simplify this, using several properties of the coefficient vectors and matrices. First, we

have that $z'_t H_{zb,0} b_{it} = b_{it} H_{bz,0} z_t$. Second:

$$\begin{aligned} E_{i0} \sum_{t=0}^{\infty} \beta^t \frac{1}{2} H_{bb,-1} b_{it} b_{it-1} &= E_{i0} \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \beta^{-1} H_{bb,1} b_{it} b_{it-1} \\ &= \frac{1}{2} \beta^{-1} H_{bb,1} b_{i,-1} b_{i0} + E_{i0} \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \beta^{-1} H_{bb,1} b_{it} b_{it+1} \end{aligned} \quad (213)$$

Similarly:

$$\begin{aligned} E_{i0} \sum_{t=0}^{\infty} \beta^t \frac{1}{2} z'_t H_{zb,-1} b_{it-1} &= E_{i0} \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \beta^{-1} b_{it-1} H_{bz,1} z_t \\ &= \frac{1}{2} \beta^{-1} b_{i,-1} H_{bz,1} z_0 + E_{i0} \sum_{t=0}^{\infty} \beta^t \frac{1}{2} b_{it} H_{bz,1} z_{t+1} \end{aligned} \quad (214)$$

Using these, and the fact that $h_b = 0$, the log-quadratic approximation to utility becomes:

$$\begin{aligned} E_{i0} \hat{U}_{i0} &= \bar{U}_i + E_{i0} \sum_{t=0}^{\infty} \beta^t \left[h_z z_t + \frac{1}{2} H_{bb,0} b_{it}^2 + H_{bb,1} b_{it+1} b_{it} \right. \\ &\quad \left. + b_{it} H_{bz,0} z_t + b_{it} H_{bz,1} z_{t+1} + \frac{1}{2} z'_t H_{zz,0} z_t \right] \\ &\quad + \beta^{-1} \left(h_{-1} b_{i,-1} + \frac{1}{2} H_{-1} b_{i,-1}^2 + H_{bb,1} b_{i,-1} b_{i0} + b_{i,-1} H_{bz,1} z_0 \right) \end{aligned} \quad (215)$$

Next, we find b_{it}^* , the optimal asset holdings chosen each period by a fully-informed household. This satisfies the first order condition:

$$E_{i0}^* \left[H_{bb,0} b_{it}^* + H_{bb,1} b_{it+1}^* + \beta^{-1} H_{bb,1} b_{it-1}^* \right] = -E_{i0}^* \left[H_{bz,0} z_t + H_{bz,1} z_{t+1} \right] \quad (216)$$

Define the expected utility of a fully-informed household, $E_{i0}^* \hat{U}_{i0}^*$, as the expected discounted utility if the household chooses this optimal saving behavior. The expected utility loss from deviating from this rule is:

$$\begin{aligned} E_{i0}(\hat{U}_{i0}^* - \hat{U}_{i0}) &= E_{i0} \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2} H_{bb,0} (b_{it}^*)^2 + H_{bb,1} b_{it+1}^* b_{it}^* - \frac{1}{2} H_{bb,0} b_{it}^2 - H_{bb,1} b_{it+1} b_{it} \right] \\ &\quad + E_{i0} \sum_{t=0}^{\infty} \beta^t (b_{it}^* - b_{it}) (H_{bz,0} z_t + H_{bz,1} z_{t+1}) + E_{i0} \beta^{-1} \left(H_{bb,1} b_{i,-1} b_{i0}^* - H_{bb,1} b_{i,-1} b_{i0} \right) \end{aligned} \quad (217)$$

where I have used that $b_{i,-1}^* = b_{i,-1}$.

Substituting in equation 216 we have:

$$E_{i0}(\hat{U}_{i0}^* - \hat{U}_{i0}) = E_{i0} \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2} H_{bb,0} (b_{it}^*)^2 + H_{bb,1} b_{it+1}^* b_{it}^* - \frac{1}{2} H_{bb,0} b_{it}^2 - H_{bb,1} b_{it+1} b_{it} \right] \\ - E_{i0} \sum_{t=0}^{\infty} \beta^t (b_{it}^* - b_{it}) (H_{bb,0} b_{it}^* + H_{bb,1} b_{it+1}^* + \beta^{-1} H_{bb,1} b_{it-1}^*) + E_{i0} \beta^{-1} \left(H_{bb,1} b_{i,-1} b_{i0}^* - H_{bb,1} b_{i,-1} b_{i0} \right) \quad (218)$$

Collecting terms and rearranging:

$$E_{i0}(\hat{U}_{i0}^* - \hat{U}_{i0}) = E_{i0} \sum_{t=0}^{\infty} \beta^t \left[-\frac{1}{2} H_{bb,0} (b_{it}^*)^2 - \frac{1}{2} H_{bb,0} b_{it}^2 + H_{bb,0} b_{it} b_{it}^* + H_{bb,1} b_{it+1}^* b_{it} \right. \\ \left. - H_{bb,1} b_{it+1} b_{it} \right] + E_{i0} \sum_{t=0}^{\infty} \beta^t \beta^{-1} H_{bb,1} b_{it-1}^* (b_{it} - b_{it}^*) + E_{i0} \beta^{-1} H_{bb,1} b_{i,-1} (b_{i0}^* - b_{i0}) \quad (219)$$

The second summation can be written as:

$$E_{i0} \sum_{t=0}^{\infty} \beta^t \beta^{-1} H_{bb,1} b_{it-1}^* (b_{it} - b_{it}^*) = \beta^{-1} H_{bb,1} b_{i,-1} b_{i0} - E_{i0} \beta^{-1} H_{bb,1} b_{i,-1} b_{i0}^* \\ + E_{i0} \sum_{t=0}^{\infty} \beta^t H_{bb,1} b_{it}^* (b_{it+1} - b_{it+1}^*) \quad (220)$$

where I have again used $b_{i,-1}^* = b_{i,-1}$.

Substituting this into the expected utility loss and collecting terms:

$$E_{i0}(\hat{U}_{i0}^* - \hat{U}_{i0}) = E_{i0} \sum_{t=0}^{\infty} \beta^t \left[-\frac{1}{2} H_{bb,0} (b_{it} - b_{it}^*)^2 - H_{bb,1} (b_{it} - b_{it}^*) (b_{it+1} - b_{it+1}^*) \right] \quad (221)$$

Differentiating the instantaneous utility function $U_{p,t}$ twice gives:

$$H_{bb,0} = \frac{\partial^2 U_{ip,t}}{\partial b_{it}^2} \Big|_{ss} = -\frac{1}{\sigma} \bar{C}_i^{-\frac{1}{\sigma}-1} \bar{B}_i^2 (1 + \beta^{-1}), \quad H_{bb,1} = \frac{\partial^2 U_{ip,t}}{\partial b_{it} \partial b_{it+1}} \Big|_{ss} = \frac{1}{\sigma} \bar{C}_i^{-\frac{1}{\sigma}-1} \bar{B}_i^2 \quad (222)$$

Therefore:

$$E_{i0}(\hat{U}_{i0}^* - \hat{U}_{i0}) = -\frac{1}{\sigma} \bar{C}_i^{-\frac{1}{\sigma}-1} \bar{B}_i^2 E_{i0} \sum_{t=0}^{\infty} \beta^t \left[-\frac{1 + \beta^{-1}}{2} (b_{it} - b_{it}^*)^2 + (b_{it} - b_{it}^*) (b_{it+1} - b_{it+1}^*) \right] \quad (223)$$

Next, we transform this into an equation involving consumption choices, rather than asset choices. Log-linearizing the budget constraint (40) gives:

$$\bar{C}_i c_{it} = \beta^{-1} \bar{B}_i (r_{t-1} - \pi_t + b_{it-1}) - \bar{B}_i b_{it} + \bar{Y} y_t \quad (224)$$

Subtracting the equivalent for the fully-informed household:

$$\bar{C}_i (c_{it} - c_{it}^*) = \beta^{-1} \bar{B}_i (b_{it-1} - b_{it-1}^*) - \bar{B}_i (b_{it} - b_{it}^*) \quad (225)$$

We substitute this into equation 223 and rearrange. To see how the rearrangement works, define $\Delta_{it} = \bar{B}_i / \bar{C}_i \cdot (b_{it} - b_{it}^*)$, so that equation 225 becomes:

$$\Delta_{it} = \beta^{-1} \Delta_{it-1} - (c_{it} - c_{it}^*) \quad (226)$$

Substituting out for $(b_{it} - b_{it}^*)$ and $(b_{it+1} - b_{it+1}^*)$ in equation 223 using the definition of Δ_{it} gives:

$$E_{i0}(\hat{U}_{i0}^* - \hat{U}_{i0}) = -\frac{1}{\sigma} \bar{C}_i^{1-\frac{1}{\sigma}} E_{i0} \sum_{t=0}^{\infty} \beta^t \left[-\frac{1+\beta^{-1}}{2} \Delta_{it}^2 + \Delta_{it} \Delta_{it+1} \right] \quad (227)$$

The terms inside the square brackets can be rearranged to:

$$\begin{aligned} -\frac{1}{2} \Delta_{it}^2 - \frac{1}{2\beta} \Delta_{it}^2 + \Delta_{it} \Delta_{it+1} &= -\frac{1}{2} \frac{1}{\beta^2} \Delta_{it-1}^2 + \frac{1}{\beta} \Delta_{it-1} (c_{it} - c_{it}^*) - \frac{1}{2} (c_{it} - c_{it}^*)^2 \\ &\quad - \frac{1}{2\beta} \Delta_{it}^2 + \Delta_{it} \Delta_{it+1} \\ &= -\frac{1}{2} \frac{1}{\beta^2} \Delta_{it-1}^2 + \frac{1}{\beta} \Delta_{it-1} (c_{it} - c_{it}^*) - \frac{1}{2} (c_{it} - c_{it}^*)^2 - \frac{1}{2\beta} \Delta_{it}^2 + \Delta_{it} (\beta^{-1} \Delta_{it} - (c_{it+1} - c_{it+1}^*)) \\ &= -\frac{1}{2} (c_{it} - c_{it}^*)^2 + \frac{1}{2\beta} (\Delta_{it}^2 - \Delta_{it-1}^2) - (\Delta_{it} (c_{it+1} - c_{it+1}^*) - \frac{1}{\beta} \Delta_{it-1} (c_{it} - c_{it}^*)) \end{aligned} \quad (228)$$

where the first and second equalities involve substituting out using equation 226.

Substituting this into equation 227, canceling terms when they appear from multiple periods, and noting that $\Delta_{i,-1} = 0$, we obtain equation 209. ■

Substituting the consumption functions of informed and uninformed households (equation 55) into this expected utility loss from imperfect information gives:

$$E_{i0}(\hat{U}_{i0}^* - \hat{U}_{i0}) = \frac{(\bar{C}_i)^{1-\frac{1}{\sigma}}}{2\sigma} \tilde{\chi}_i^2 E_{i0} \sum_{t=0}^{\infty} \beta^t (\pi_t - E_{it} \pi_t)^2 \quad (229)$$

Following the rational inattention literature I assume that increasing information precision is costly to the household. The utility cost of a signal s_{it} is given by:

$$\mathcal{C}(\{s_{it}\}^t) = \psi \sum_{t=0}^{\infty} \beta^t I(\pi^t; s_{it} | \mathcal{I}_{it-1}) \quad (230)$$

where $\psi > 0$ is a positive constant and $I(\pi^t; s_{it} | \mathcal{I}_{it-1})$ is the Shannon mutual information between priors and posteriors in period t . This cost function is common in the rational inattention literature (Maćkowiak et al., 2023).

To solve for optimal information processing, I make the simplifying assumption that the household chooses information as if they are certain about the parameters of their subjective model. Similarly, they ignore that they will update those parameters after receiving information.

The household information choice problem then has the same form as the firm's rational inattention problem in Maćkowiak and Wiederholt (2009). As is standard in the rational inattention literature, I make the following further assumptions:

Assumption 1: (π_t, s_{it}) has a stationary Gaussian distribution.

Assumption 2: When the household decides on their information strategy in period 0, they receive a long sequence of signals of their chosen form. This implies that $E_{it}(\pi_t^2 | \mathcal{I}_{it})$ is constant over time.

Assumption 3: In period t , households can only process information about variables realized up to period t . They cannot process any information about realizations of inflation in future periods.⁵⁰

Under these assumptions, Maćkowiak and Wiederholt (2009) show that the optimal signal is of the form in equation 46, absent the public noise ε_t , which we have only added for the purpose of creating aggregate variation in information for use in finding the causal effect of exogenous movements in inflation perceptions. Using that signal structure, the utility cost of period- t signal s_{it} is:

$$\begin{aligned} I(\pi^t; s_{it} | \mathcal{I}_{it-1}) &\equiv H(\pi_t | s_i^{t-1}) - H(\pi_t | s_i^t) = \frac{1}{2} \log_2 \left(\frac{\text{Var}(\pi_t | \mathcal{I}_{it-1})}{\text{Var}(\pi_t | \mathcal{I}_{it})} \right) \\ &= \frac{1}{2} \log_2 \left(\frac{1}{1 - \tau_i} \right) \end{aligned} \quad (231)$$

where the final equality uses standard properties of the Kalman filter.

Assumption 2 implies that $E_{i0}(\pi_t - E_{it}\pi_t)^2$ is constant over time. From the properties of the

⁵⁰This ensures that as the cost of information approaches zero the household information set in period t contains realized values of all period t variables, but not realizations of variables in future periods, as in standard full-information models. See Jurado (2021) for a detailed discussion of this assumption.

Kalman filter:

$$E_{i0}(\pi_t - E_{it}\pi_t)^2 = \frac{(1 - \tau_i)\sigma_\pi^2}{1 - \rho_{\pi i}^2(1 - \tau_i)} \quad (232)$$

Using these results, and evaluating the resulting geometric series in the utility losses and costs of information, the household information choice problem reduces to:

$$\min_{\tau_i} \frac{(\bar{C}_i)^{1-\frac{1}{\sigma}}}{2\sigma} \cdot \frac{\tilde{\chi}_i^2(1 - \tau_i)\sigma_\pi^2}{1 - \rho_{\pi i}^2(1 - \tau_i)} + \frac{\psi}{2} \log_2 \left(\frac{1}{1 - \tau_i} \right) \quad (233)$$

subject to $\tau_i \in [0, 1]$. The first order condition for an interior solution is:

$$\frac{1 - \tau_i}{(1 - \rho_{\pi i}^2(1 - \tau_i))^2} = \frac{\delta_i^*}{(1 - \rho_{\pi i}^2)^2} \tilde{\chi}_i^{-2} \quad (234)$$

where:

$$\delta_i^* = \frac{2\psi\sigma(1 - \rho_{\pi i}^2)^2}{(\bar{C}_i)^{1-\frac{1}{\sigma}}\sigma_\pi^2 \ln(2)} \quad (235)$$

Since $\delta_i^* > 0$, the τ_i implied by this first order condition is always strictly less than 1. Finally, we find the region where $\tau_i \geq 0$ binds. Differentiating the left hand side of equation 234 with respect to τ_i gives:

$$\frac{\partial}{\partial \tau_i} \left(\frac{1 - \tau_i}{(1 - \rho_{\pi i}^2(1 - \tau_i))^2} \right) = -\frac{1 + \rho_{\pi i}^2(1 - \tau_i)}{(1 - \rho_{\pi i}^2(1 - \tau_i))^3} < 0 \quad (236)$$

The left hand side of equation 234 is therefore strictly decreasing in τ_i . The constraint therefore binds whenever the right hand side is sufficiently large, that is when:

$$\tilde{\chi}_i^2 < \delta_i^* \quad (237)$$

If, as in the calibration used in our quantitative exercises (see Appendix E.3), all households have the same $\rho_{\pi i}$, then δ_i^* is constant across households. This completes the derivation of equation 57.

Finally, I provide a microfoundation for equation 58. The household faces Knightian uncertainty about the α_i parameter in their subjective model. After observing the realization of s_{it} , the household updates their subjective model to reflect this: following the literature on ambiguity aversion they make decisions using worst-case beliefs (Hansen and Sargent, 2008).

Formally, the household selects beliefs and actions as if they are playing a game with an ‘evil agent’, who distorts α_i to minimize expected utility, while the household simultaneously chooses c_{it} to maximize expected utility. The maximization problem is solved by the consumption function in equation 55 with the updated $\hat{\alpha}_{it}$.

To solve the evil agent problem, we then need to find the indirect expected utility when

households follow this consumption function. Begin with the expected utility of a household who is fully-informed about inflation each period. To simplify the problem, here I assume that $\sigma \rightarrow 1$, so the instantaneous utility from consumption C_{it} is $\log(C_{it})$. The log-quadratic approximation of expected discounted utility, substituting in the consumption function of the informed household, is therefore:

$$E_{i0}^* \hat{U}_{i0}^* = E_{i0}^* \sum_{t=0}^{\infty} \beta^t \left(\frac{1-\beta}{1-\beta\rho_{yi}} y_t - \sigma\beta r_t + \frac{\beta\rho_{\pi i}[(1-\beta)(\alpha_i + \lambda_i\phi_i) - \sigma(\phi_i\beta - 1)(1-\beta\rho_{yi})]}{(1-\beta\rho_{\pi i})(1-\beta\rho_{yi})} \pi_t \right) \quad (238)$$

Substituting out for expected future inflation, interest rates, and real income using the subjective model (equations 52 - 54) gives indirect utility as a function of current observables and subjective model parameters:

$$E_{i0}^* \hat{U}_{i0}^* = \frac{1-\beta}{(1-\beta\rho_{yi})^2} y_0 - \sigma\beta r_0 + \frac{1}{1-\beta\rho_{\pi i}} \left(\frac{\beta\rho_{\pi i}(\alpha_i + \lambda_i\phi_i)}{1-\beta\rho_{yi}} - \sigma\beta^2\phi_i\rho_{\pi i} + \tilde{\chi}_i \right) \pi_0 \quad (239)$$

Finally, use the expression for the expected utility loss from limited information (equation 229) to find the expected indirect utility of the potentially uninformed household:

$$E_{i0} \hat{U}_{i0} = \frac{1-\beta}{(1-\beta\rho_{yi})^2} y_0 - \sigma\beta r_0 + \frac{1}{1-\beta\rho_{\pi i}} \left(\frac{\beta\rho_{\pi i}(\alpha_i + \lambda_i\phi_i)}{1-\beta\rho_{yi}} - \sigma\beta^2\phi_i\rho_{\pi i} + \tilde{\chi}_i \right) E_{i0} \pi_0 - \frac{\log(\bar{C}_i)}{2(1-\beta)} \tilde{\chi}_i^2 \frac{(1-\tau_i)\sigma_{\pi}^2}{1-\rho_{\pi i}^2(1-\tau_i)} \quad (240)$$

where I have used that the expected variance of inflation perception gaps is constant (equation 232). Differentiating the expected indirect utility with respect to α^i gives:

$$\frac{\partial E_{i0} \hat{U}_{i0}}{\partial \alpha_i} = \frac{\beta(2-\beta)\rho_{\pi i}}{(1-\beta\rho_{\pi i})(1-\beta\rho_{yi})} E_{i0} \pi_0 - \frac{\beta\rho_{\pi i} \log(\bar{C}_i)(1-\tau_i)\sigma_{\pi}^2}{(1-\beta\rho_{\pi i})(1-\beta\rho_{yi})(1-\rho_{\pi i}^2(1-\tau_i))} \cdot \tilde{\chi}_i \quad (241)$$

Expected indirect utility is therefore increasing in α_i if and only if perceived inflation is sufficiently high, and otherwise it is decreasing. Intuitively, when perceived inflation is high, the worst case is that high inflation is associated with low real incomes. However, when perceived inflation is lower, the reverse is true. The worst case is then that inflation supports real incomes, and so the ambiguity averse household distorts their subjective model in that direction, with a positive $\hat{\alpha}_{it}$.

Formally, suppose that the evil agent chooses the distorted $\hat{\alpha}_{it}$ to minimize expected utility net of a quadratic cost of distortions from α_{i0} :

$$\hat{\alpha}_{i0} = \arg \min_{\alpha} E_{i0} \hat{U}_{i0} + \frac{\psi_{\alpha}}{2} (\alpha - \alpha_{i0})^2 \quad (242)$$

The first order condition is:

$$\hat{\alpha}_{i0} = \alpha_{i0} + \alpha_{i1} E_{i0} \pi_0 + \alpha_{i2} \tilde{\chi}_i \quad (243)$$

where:

$$\alpha_{i1} = -\frac{\beta(2-\beta)\rho_{\pi i}}{\psi_{\alpha}(1-\beta\rho_{\pi i})(1-\beta\rho_{yi})} < 0 \quad (244)$$

$$\alpha_{i2} = -\frac{\beta\rho_{\pi i} \log(\bar{C}_i)(1-\tau_i)\sigma_{\pi}^2}{\psi_{\alpha}(1-\beta\rho_{\pi i})(1-\beta\rho_{yi})(1-(\rho_{\pi i})^2(1-\tau_i))} \quad (245)$$

In all calibrations used in the paper, $\bar{C}_i = 1$ for all households i , which implies $\alpha_{i2} = 0$. In that case equation 243 gives equation 58.

E.3 Parameters for figures in Sections 6 and 7

Figure ??. All parameters as in Table 14, except α_{i0} distributed such that $\tilde{\chi}_i$ is in the range $[-1, 1]$, and ψ set at 0.2×10^{-3} . This scales all attention so that the change in attention with subjective models is clear in the figure.

Table 3 and Figure 5. The calibrated parameters are set out in Table 14. β and σ are set to standard values, and ϕ is set such that the Taylor principle is just satisfied, as found on average for the UK by Lee et al. (2013). For the remaining parameters of subjective models, including the mean of the α_{i0} distribution, I estimate equations 47-49 using OLS on UK data from 1993-2019. The longer sample than the survey data is to allow for more precise estimation of model parameters. It is not extended further back because of the structural break in many UK macroeconomic time series at the end of 1992 identified by Benati (2006).

For the inflation data, I take the log first difference of quarterly CPI (ONS series MM23). I de-mean and remove seasonal variation by regressing the series on quarter-of-the-year dummies, and taking the residual as my quarterly inflation series. As well as being used in the calibration, this

series is used to generate the simulated paths for perceived inflation and the aggregate consumption elasticity to inflation.

The interest rate data is 3-month money market rates, taken from the OECD Main Economic Indicators. To be consistent with the model equations, I transform this annualized rate into a gross quarterly interest rate, then take logs and de-mean. Following [Harrison and Oomen \(2010\)](#), I allow the mean interest rate to vary with changes in the broad regime of UK monetary policy, which I take to occur in 2009Q1 as interest rates hit the ZLB.

I proxy real income with real wages, since the model is approximated around a steady state with no saving. I begin by summing ONS series ROYH, ROYK, and ROYJ to obtain a measure of total nominal wages. I then divide this by total hours (ONS series YBUS) and working age population (ONS series MGSL) to obtain nominal wages per worker per hour. Finally, I divide by the level of CPI (including the seasonal adjustment carried out in the computation of inflation) to obtain real wages. I then take logs and hp-filter the series to obtain the cyclical component. This is estimated to be reasonably persistent, ($\rho_w = 0.731$), but still this implies a very small amplification from real income changes: $(1 - \bar{\Theta})^{-1} = 1.04$.

For $(\sigma(\alpha_{i0}), \alpha_{i1}, \psi)$ I target three moments from the IAS data. The first is the average ratio of ‘weaker’ to ‘stronger’ answers in response to Question 1. The raw proportions are inappropriate since we do not know how far either side of a true $\tilde{\chi}_i = 0$ is considered ‘little difference’ by the respondents, but the ratio still gives the balance between negative and positive models of the economy. That ratio is on average 7.533.

The second target is the estimated elasticity of the proportion with negative models to inflation, that is the coefficient from regressing $\Pr(\text{‘weaker’})$ on current inflation and a constant. That elasticity is 0.090.

Finally, the third target is an estimate of the average Kalman gain across the population, which helps to identify the information cost parameter ψ . For this, take Equation 50 and average across households to give:

$$E_I(E_{it}\pi_t) = \bar{\tau}\pi_t + (1 - \bar{\tau})\rho_\pi E_I(E_{it-1}\pi_{t-1}) \quad (246)$$

where I have used the fact that all households are calibrated to have the same ρ_π , and in the model information, and so τ_i , is decided before the households update their subjective models, and so is independent of perceived inflation. Denoting $\bar{E}_t\pi_t$ as the average perceived inflation in time t , I therefore estimate:

$$\bar{E}_t\pi_t = \gamma_1\pi_t + \gamma_2\bar{E}_{t-1}\pi_{t-1} \quad (247)$$

by OLS, restricting $\gamma_2 = \rho_\pi(1 - \gamma_1)$, where ρ_π is as in Table 14. The estimated γ_1 therefore gives an estimate of the average Kalman gain across the population. This target is 0.448.

Table 14: Calibration

Parameter	Value	Source	Parameter	Value	Source
β	0.99	standard	σ_π	0.003	estimated model
σ	1	standard	σ_i	0.004	estimated model
ϕ	β^{-1}	Lee et al. (2013)	σ_y	0.008	estimated model
$E_I \alpha_{i0}$	-0.732	estimated model	$\sigma(\alpha_{i0})$	0.613	targets
λ	-0.037	estimated model	α_{i1}	-234	targets
ρ_π	0.329	estimated model	ψ	0.787×10^{-9}	targets
ρ_y	0.731	estimated model			